

DEPARTMENT OF INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Informatics

On the Feasibility of Pixel-Free Visual Odometry Using Video Codec Motion Vectors

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Zur Machbarkeit pixelfreier visueller Odometrie mit Bewegungsvektoren aus Videocodecs

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I confirm that this master's thesis in informatics is my own work and I have documented all sources and material used.

Munich, 23 September 2026 Emir Özer

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Abstract

Wearables and small drones need to understand how they move while operating with limited power and computation. Visual odometry estimates this movement from camera images, usually by tracking image features over time. Yet many camera devices already contain dedicated hardware that searches for matching image regions to compress video. Could the motion information produced by this widely available hardware also support camera tracking?

This thesis explores that possibility using motion vectors from H.264 video compression. These vectors describe how the encoder predicts image blocks; they are designed to save data, so they do not necessarily follow the same physical points over time. The proposed method combines vectors from two cameras with known camera geometry to estimate depth and camera movement. Its tracking and motion estimation use these measurements without consulting image pixels. This approach is compatible with compress-then-analyze: a camera could send the required compressed streams to a companion phone, which reuses their motion information for localisation.

Experiments on an established drone benchmark recover coherent camera trajectories, particularly on easier sequences. More demanding sequences expose limitations, and methods that refine their tracks using image pixels remain more accurate overall. The results nevertheless demonstrate that useful camera tracking from codec motion vectors is feasible. They provide one starting point for turning existing video-processing resources into a source of geometric information. The experiments establish this possibility; live hardware operation and the potential energy savings remain to be evaluated.

Kurzfassung

Wearables und kleine Drohnen müssen ihre eigene Bewegung erfassen können, obwohl ihnen nur begrenzte Energie und Rechenleistung zur Verfügung stehen. Visuelle Odometrie schätzt diese Bewegung aus Kamerabildern, üblicherweise durch die Verfolgung von Bildmerkmalen über die Zeit. Viele Kameraгeräte enthalten jedoch bereits spezialisierte Hardware, die zur Videokompression nach übereinstimmenden Bildregionen sucht. Könnten die dabei gewonnenen Bewegungsinformationen auch zur Schätzung der Kamerabewegung dienen?

Diese Arbeit untersucht diese Möglichkeit anhand von Bewegungsvektoren aus der H.264-Videokompression. Die Vektoren beschreiben, wie der Encoder Bildblöcke vorhersagt. Da sie der Datenreduktion dienen, verfolgen sie nicht zwangsläufig dieselben physischen Punkte über die Zeit. Das vorgeschlagene Verfahren kombiniert Vektoren aus zwei Kameras mit bekannter Kamerageometrie, um Tiefe und Kamerabewegung zu schätzen. Die Verfolgung und Bewegungsschätzung nutzen diese Messungen, ohne auf Bildpixel zuzugreifen. Dieser Ansatz ist mit Compress-then-Analyze vereinbar: Eine Kamera könnte die benötigten komprimierten Videoströme an ein gekoppeltes Smartphone senden, das deren Bewegungsinformationen für die Lokalisierung wiederverwendet.

Experimente auf einem etablierten Drohnen-Benchmark liefern schlüssige Kameratrajektorien, insbesondere auf einfacheren Sequenzen. Anspruchsvollere Sequenzen zeigen Grenzen auf; Verfahren, die ihre Korrespondenzen anhand von Bildpixeln verfeinern, bleiben insgesamt genauer. Dennoch belegen die Ergebnisse, dass eine brauchbare Schätzung der Kamerabewegung allein aus Codec-Bewegungsvektoren möglich ist. Sie bieten einen Ausgangspunkt dafür, vorhandene Ressourcen zur Videoverarbeitung als Quelle geometrischer Informationen zu nutzen. Die Experimente zeigen diese Machbarkeit; der Live-Betrieb auf entsprechender Hardware und mögliche Energieeinsparungen müssen noch untersucht werden.

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1. Introduction

1.1. Motivation

For cameras on wearables, embedded systems, and lightweight drones, useful localisation must fit within a limited power and computation budget. Video encoding offers an attractive resource for this purpose: H.264 has a mature, widely deployed hardware ecosystem, and its encoders already perform motion estimation to compress successive images [1]. A geometric estimator that can use these motion estimates could benefit from an accelerator already available in many camera pipelines.

The appeal of dedicated hardware is concrete. One reported mobile-platform example contrasts an 80 mW power budget for specialized video/audio hardware with 1000 mW for a CPU [2]. These platform-specific budgets motivate treating video encoding as a source of useful visual computation. Once the encoder has searched for matching blocks, its output contains correspondence information that an odometry system could reuse. Such correspondences can connect the two views of a stereo camera or track scene structure across successive images. Reusing them for stereo matching and temporal tracking is therefore a plausible route to reducing the additional work required for localisation.

This opportunity fits naturally into a *compress-then-analyze* (CTA) architecture. A camera device can encode its observations and send the coded streams to a companion phone for analysis [3, 1]. Compression makes that communication practical, but the receiving phone still has to infer motion from what arrives. If codec correspondences themselves support odometry, the receiver can perform geometric estimation without first building a pixel-based tracking frontend. The potential benefit is thus shared across the system: dedicated coding hardware supplies motion information, the link carries a compressed representation, and the receiver reuses that information for localisation.

The representation also offers flexibility in where analysis is performed. A device with spare computation could filter codec correspondences before transmission and send the surviving observations, while a more constrained camera could leave that work to its receiver. This makes the balance between local computation and communication a design choice, as motivated by the trade-off between CTA and *analyze-then-compress* (ATC) [4]. The choice is especially relevant when a receiving phone must share its energy and thermal budget with other tasks.

The geometric feasibility of this idea is the prerequisite examined here: *do codec motion vectors retain enough information for useful camera tracking without pixel-based refinement?* Their compression-oriented block matches need not be accurate or persistent enough for this purpose. Demonstrating a workable odometry construction would establish a reason to explore systems built around this widely available computational resource. This thesis

provides one such construction and studies where its geometric information is sufficient.

1.2. Problem Statement and Scope

We consider calibrated stereo visual odometry using H.264 side information, without decoded image intensities in tracking or geometric estimation, IMU measurements, GPS, or an external pose prior. Rectification and encoding use pixels during offline stream preparation. The frontend and estimator consume motion-vector metadata, calibration, rectification masks, and timestamps. The term *pixel-free* applies to this estimation boundary.

The H.264 encodings use a single reference frame (`ref=1`) and disable B-frames (`bframes=0`). In each temporal stream, inter-frame prediction uses the preceding encoded view; stereo correspondence uses independent I-P-P triples. This short reference structure is relevant to live systems, where observations must be available with limited delay. It does not make the present input preparation a live pipeline: the experiments use two-pass encoding and reverse complete image sequences to obtain reciprocal vectors.

The restriction exposes difficulties that pixel refinement can otherwise resolve. Codec vectors describe block-wise prediction decisions rather than persistent scene points, their spatial support depends on the partition structure, and their values are selected for compression rather than physical motion [5, 1]. This study investigates one construction that recovers stereo depth, validates temporal associations, maintains identities, and integrates block-displacement observations into a point-based estimator.

The experiments establish an analysis interface and its geometric performance. They use prepared temporal, reverse-temporal, and shifted stereo streams; a live hardware encoder, wireless link, and receiving phone are outside the experimental setup. The costs of producing and transporting those inputs must therefore be included when assessing a future distributed deployment.

1.3. Research Questions

1. How much odometric accuracy remains when temporal correspondence tracking uses raw codec motion vectors without pixel-domain refinement?
2. To what extent can block-based codec motion vectors form persistent, geometrically useful stereo-temporal tracks?
3. How are the properties of the resulting tracks and input sequences related to variation in odometric accuracy?

1.4. Contributions

Pixelless VO (PVO) provides a concrete feasibility demonstration through one stereo pipeline. It combines codec-derived depth, persistent geometric association, and a displacement-

based observation model. Evaluation on all eleven sequences of the established EuRoC MAV benchmark [6] relates trajectory accuracy to the availability and persistence of codec observations. The trajectory visualizations in subsection 6.1.1, especially the easy Machine Hall sequences, show the central result: reasonable tracking can be recovered from motion vectors without pixel refinement. The reported PVO errors on EMH01 and EV101 provide further quantitative evidence of this geometric information.

These results support further exploration of pixel-free odometry and flexible placement of codec-based analysis between capturing and receiving devices. The particular stereo construction, filters, and backend are one way to test that possibility. The measured limitations identify where alternative constructions could improve accuracy, robustness, or the cost of preparing the codec inputs.

1.5. Thesis Structure

Chapter 2 introduces the camera geometry, bundle-adjustment, codec, and robust-registration concepts used throughout the thesis. Chapter 3 positions the examined system with respect to compressed-domain motion estimation, codec-aware tracking, and motion-vector reconstruction. Chapter 4 describes the PVO compressed-domain frontend and its integration into the visual-odometry backend. Chapter 5 defines the reported configurations, dataset, metrics, and diagnostic protocol, and Chapter 6 reports the resulting trajectory and frontend measurements. Finally, Chapter 7 interprets the operating envelope and limitations, before Chapter 8 summarizes the findings and identifies directions for future work.

2. Background

2.1. Visual Odometry Foundations

Visual odometry estimates a time-ordered sequence of camera poses from visual observations. A visual frontend supplies correspondences between frames, while a backend uses their geometric constraints to estimate motion. Visual SLAM extends this local estimation problem with a persistent map and mechanisms for global consistency, such as loop closure [7]. This thesis evaluates local trajectory estimation and therefore uses the term visual odometry unless a cited system explicitly provides broader SLAM functionality.

Let $\mathcal{F}_W$ denote the world frame and $\mathcal{F}_{C_t}$ the camera frame at time t. The pose $\mathbf{T}_{WC_t} \in \mathrm{SE}(3)$ maps a point from $\mathcal{F}_{C_t}$ to $\mathcal{F}_W$ and contains a rotation $\mathbf{R}_{WC_t} \in \mathrm{SO}(3)$ and translation $\mathbf{t}_{WC_t} \in \mathbb{R}^3$. A landmark $\mathbf{p}_j^W$ is a scene point expressed in the world frame. Two-dimensional observations assigned to the same landmark form a track. Frames retained for joint pose and landmark refinement are called keyframes; limiting their number bounds the optimization problem.

2.2. Camera and Stereo Geometry

A calibrated central camera is described by a projection $\pi : \mathbb{R}^3 \to \mathbb{R}^2$ and an inverse projection π^{-1} that maps an image coordinate to a viewing ray. For the pinhole model,

$$\pi\Big(\begin{bmatrix} X & Y & Z \end{bmatrix}^\mathsf{T}\Big) = \begin{bmatrix} f_x X/Z + c_x & f_y Y/Z + c_y \end{bmatrix}^\mathsf{T}. \tag{2.1}$$

Native projections may additionally include lens distortion; rectification instead remaps both stereo cameras to a shared virtual, undistorted projection [8].

The calibrated stereo extrinsics fix the relative rotation and translation between the anchor and stereo views, and the translation magnitude defines the baseline. A point observed in one view must lie on the corresponding epipolar line in the other. Rectification makes these lines horizontal, so correspondence reduces to a one-dimensional disparity search. In the ideal parallel pinhole case, depth is $Z = fb/d$ for focal length f, baseline b, and disparity d. More generally, depth is obtained by triangulating the two calibrated rays. A valid reconstruction must satisfy cheirality, placing the point in front of both cameras [8].

2.3. Nonlinear Optimization for Visual Odometry

For an observation $\mathbf{u}_{j,t}$ of landmark j at time t, the conventional reprojection residual uses the predicted-minus-measured convention,

$$\mathbf{e}_{j,t} = \pi\left(\mathbf{T}_{WC_t}^{-1}\mathbf{p}_j^W\right) - \mathbf{u}_{j,t}. \tag{2.2}$$

Bundle adjustment is the joint refinement of scene structure and viewing parameters by minimizing a cost over their prediction errors [9]. For point-based visual odometry, each landmark couples the poses from which it is observed. A longer, correctly associated track can therefore support the same structure from several views, while a two-frame track supplies only a pairwise constraint. This additional support is useful only when the observations are geometrically consistent; an incorrect long association can instead contaminate several states. A robust loss, such as the Huber loss, reduces the influence of observations whose residuals are inconsistent with the assumed noise model [10].

The visual-only Basalt backend used here retains the bounded-window formulation of its visual-inertial estimator [11]. Recent states and selected keyframes remain active in the optimization; variables leaving the window are marginalized and their information is retained as a prior. This standard marginalization remains enabled in the frozen configuration, whereas the separately evaluated pre-marginalization outlier filter and second optimization pass are not retained. Since visual measurements constrain only relative geometry, a reference pose must be fixed to remove the global gauge freedom. The proposed residual changes the visual measurement term but retains this state-management structure; it is defined in chapter 4.

2.4. H.264 Inter Prediction and Motion Vectors

2.4.1. Prediction, Residuals, and Reconstruction

H.264, also called Advanced Video Coding (AVC) or MPEG-4 Part 10, represents video through a standardized compressed syntax and decoding process [12, 13]. H.264 uses a block-based hybrid coding model combining inter-picture prediction with transform coding of the prediction residual [12, Sec. IV]. The following explanation concerns the progressive-frame, 4:2:0 setting used in this thesis; interlaced and multiview extensions are not required.

Prediction reduces the amount of new information needed to describe a block. Intra prediction uses already reconstructed neighbouring samples within the current picture. Inter prediction instead uses a displaced region of a reference picture. A motion vector describes the displacement used for motion compensation [13, Sec. 3.95]. In addition to prediction information, the encoder can transmit a residual: the difference between the input block and its prediction. This residual is transformed, quantized, and entropy coded. The decoder reconstructs the residual and adds it to the prediction; an in-loop deblocking filter can then modify the reconstructed samples. Reference pictures are reconstructed pictures, so encoder and decoder use matching reference content rather than the encoder's uncompressed input images [12, Sec. IV].

Motion vectors alone therefore do not determine the decoded image. Prediction can be imperfect while the coded residual restores image detail. The quantization parameter controls residual-coefficient quantization, not the spatial step of the motion-vector representation. A visually acceptable decoded frame need not have geometrically accurate motion vectors [12, 5].

Crucially, the standard specifies which bitstreams are valid and how they are decoded, not a unique motion-estimation algorithm [13, Sec. 3.52]. Motion search, partition selection, and rate control are encoder decisions. Their purpose is compression efficiency, often expressed by a rate-distortion objective of the form

$$J(c) = D(c) + \lambda R(c), \tag{2.3}$$

where c is a coding choice, D measures reconstruction distortion, R counts its required bits, and λ trades distortion against rate. Choices include the prediction mode, partitioning, reference picture, motion vectors, and residual coding. Practical searches can use approximations and early exits; H.264 compliance does not imply a globally optimal search or identical vectors from different encoders [12]. The rate cost makes a vector that is cheap to describe competitive with one that better matches image appearance, even before geometric interpretation is considered.

2.4.2. Spatial Support and Partitioning

In 4:2:0 video, the luma component represents brightness, while each chroma component is sampled at half the horizontal and vertical luma resolution. The spatial dimensions used below are in luma samples. A macroblock covers 16×16 luma samples. Inter prediction can use that region as one block or partition it into 16×8, 8×16, or 8×8 regions. Each 8×8 region can independently remain whole or be subdivided into 8×4, 4×8, or 4×4 subpartitions [13, Tables 7-13 and 7-17]. The hierarchy is illustrated in Figure 2.1. H.264 therefore does not require a uniform grid of 8×8 motion vectors.

Each inter-predicted partition or subpartition carries a translational motion description, with one vector for a unidirectionally predicted block. Larger partitions reduce the motion information to encode; smaller ones can represent different displacements within the same macroblock at greater signalling cost. These regions are not feature detections or object segments. Their selection can change between pictures without preserving any identity [12, Sec. IV-H].

Prediction support must also be distinguished from residual transform size. For example, the 4×4 transform operates on prediction-error samples; it does not imply that every inter-prediction vector belongs to a 4×4 region. High-profile coding additionally permits an 8×8 transform [13, Secs. 8.5 and A.2.4]. The MV block dimensions describe where a displacement is applied, not the precision with which a point inside that region has been localized.

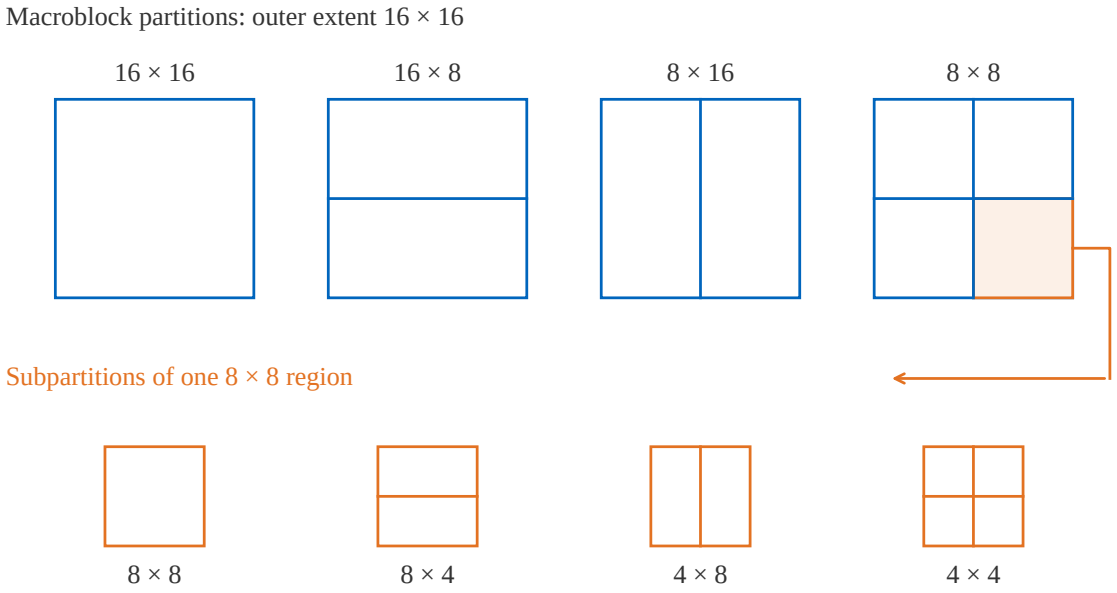


Figure 2.1.: H.264 inter-prediction partition choices in luma samples. Top: partitioning of a 16×16 macroblock. Bottom: possible subdivisions of one 8×8 region; the four regions of a partitioned macroblock can select these independently. The drawing uses a common spatial scale and shows prediction support, not transform blocks. Based on [13, Tables 7-13 and 7-17].

2.4.3. Reference Pictures and Temporal Direction

Pictures are divided into slices, whose types constrain the available coding modes. An I slice uses intra coding. A P slice permits intra coding and inter prediction from one reference list; a B slice additionally permits a prediction combining two reference lists. P and B slices can therefore contain intra-coded regions with no inter-picture motion vector [13, Table 7-10]. The common terms I-, P-, and B-frame are shorthand; a P label alone does not imply a temporal vector at every location. An instantaneous decoding refresh (IDR) picture also resets reference-picture dependencies. An ordinary intra-coded picture does not necessarily prevent later pictures from referring to pictures preceding it [13, Sec. 3.69]. A pattern such as I-P-P specifies coding types, but does not by itself specify every reference relationship [13, Secs. 7.4.5 and 8.2.4].

A vector is meaningful only together with its reference-picture identity. H.264 can retain multiple reconstructed references, and decoding order need not equal display order. In B prediction, either reference list can contain pictures from either temporal direction; a list index is not by itself a timestamp or a direction of motion [12, Secs. IV-H.1 and IV-H.2].

The direction convention matters. Under a current-to-reference lookup convention, a vector $\mathbf{m}$ pairs a location $\mathbf{q}$ in the current coding block with $\mathbf{q} + \mathbf{m}$ in the reference image [13, Sec. 3.95]. If the reference is the preceding frame, forward-time propagation instead maps a point $\mathbf{p}$ in that reference support to $\mathbf{p} - \mathbf{m}$. Thus an encoder reference arrow and a forward-time motion arrow have opposite directions in this case. For example, a horizontal lookup displacement of -2 pixels describes a forward-time displacement of $+2$ pixels between those supports. It is a displacement over the reference interval, not a velocity.

The experiments restrict reference count and prediction structure to simplify this interpretation (chapter 5). Encoding the same images in reverse order is nevertheless a new prediction problem, not a way to obtain the exact inverse of the original vectors. This independently encoded reverse stream is also distinct from the two reference lists of a single B slice. Neither codec syntax nor rate-distortion optimization enforces reciprocal agreement between the two encodings.

2.4.4. Subpixel Precision and Decoded Motion Information

H.264 represents luma motion in quarter-sample units. An integer vector component k therefore represents $k/4$ luma pixels, allowing displacements between recorded sample locations. Motion compensation obtains these fractional reference samples by interpolation: half-sample luma positions use a six-tap filter, and quarter-sample positions use rounded averages of suitable integer- and half-sample values [13, Secs. 8.4.2.2 and 8.4.2.2.1]. In 4:2:0 chroma, the same displacement corresponds to eighth-chroma-sample units because that grid has half the luma resolution; this does not provide eighth-pixel precision on the luma image [13, Sec. 8.4.1.4].

Fractional precision specifies representable displacements, not a bound on error relative to physical image motion. The chosen vector is the result of prediction search and coding decisions, not necessarily a correctly estimated continuous motion rounded to its nearest quarter

pixel. Consequently, 0.25-pixel representation alone justifies neither subpixel correspondence accuracy nor a particular statistical uncertainty model.

The bitstream usually codes a motion-vector difference relative to a predicted vector. For each component, reconstruction has the form

$$\mathbf{m} = \mathbf{m}_{\text{pred}} + \Delta\mathbf{m}, \tag{2.4}$$

where the predictor depends on neighbouring motion information, partition shape, and reference indices [13, Sec. 8.4.1, Eqs. (8-172) and (8-173)]. A zero coded difference is therefore not necessarily a zero displacement. The codec's neighbour-based prediction, which includes median-based cases, saves bits; it is separate from the motion-vector outlier filter introduced in chapter 4.

Skip and direct modes provide another important distinction. A skipped P macroblock has inferred motion information despite carrying no explicit motion-vector difference or residual. Its derived vector need not be zero. B direct and skip modes likewise derive motion from other decoded information [13, Secs. 8.4.1.1 and 8.4.1.2]. This differs from intra coding, which provides no inter-picture vector. Extracting motion therefore requires syntax parsing and reconstruction of motion information, including inferred modes and reference identities, not simply reading explicit vector differences. The standard separates this derivation from the subsequent use of reference intensities for motion compensation [13, Secs. 8.4.1 and 8.4.2]. The estimator in this thesis uses the resulting metadata, not the reconstructed intensity samples; this does not mean that no compressed syntax is decoded.

2.4.5. Motion Vectors versus Optical Flow and Feature Tracks

Optical flow describes apparent image motion between images. A common local model seeks a displacement $\mathbf{f}$ satisfying brightness constancy,

$$I_{t+1}(\mathbf{u} + \mathbf{f}(\mathbf{u})) \approx I_t(\mathbf{u}), \tag{2.5}$$

with spatial assumptions or patch support resolving otherwise ambiguous matches. KLT uses such image information to refine a point displacement; texture is needed to constrain that displacement in both directions [14, 5]. Occlusion, changing illumination, repeated patterns, and the aperture problem still limit optical-flow estimates: an isolated straight edge does not constrain displacement along its own direction. Optical flow is therefore not synonymous with ground-truth scene correspondence.

Codec motion and optical flow are related because both exploit displaced image appearance. Their objectives and outputs differ. Codec prediction trades distortion against coded bits, may use a non-adjacent reference, and assigns one translation to an entire prediction region. It can leave mismatched appearance to the coded residual. Optical-flow estimation instead seeks an image-motion field, without a cost for encoding its vectors in an H.264 stream. Repeating a block vector at every covered pixel produces a dense array, but adds no independent motion measurements or localized features. A useful prediction of the block's appearance can therefore differ from the motion of a particular corner or edge inside it [5, 1].

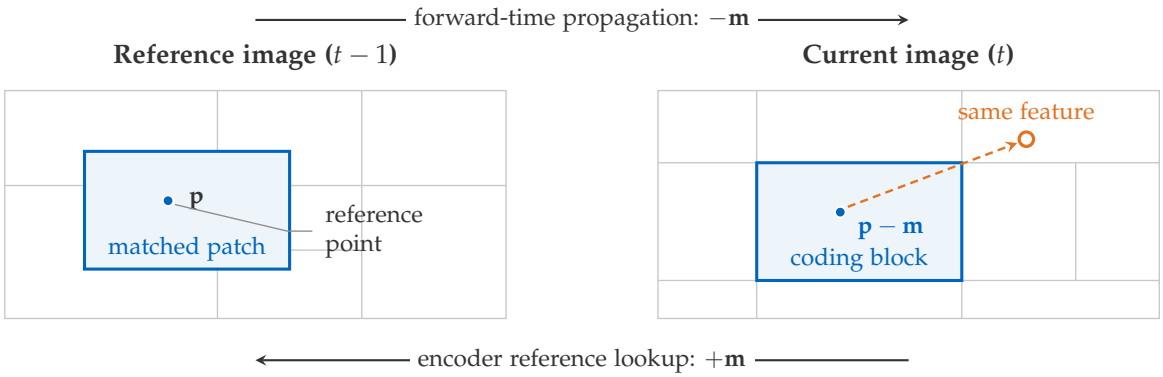


Figure 2.2.: Conceptual block prediction, not measured data. Blue rectangles mark a matched reference patch and its current coding block. Applying their displacement to the reference point produces the filled blue point; the same physical feature may appear elsewhere (orange ring). Gray lines show independently chosen coding partitions. The arrows distinguish encoder reference lookup from chronological propagation for an earlier reference.

As illustrated in Figure 2.2, translating a point with its containing block does not establish feature identity. A matched reference patch need not coincide with a coding partition in the reference image. Although coding partitions cover the current picture, their displaced reference supports can overlap and leave gaps. Repeatedly selecting a block that covers a propagated point can consequently switch the image structure being followed. Conversely, a visible structure may have no usable vector: its region may be intra-coded, or no available source support may cover the queried point. Missing support supplies no motion measurement and does not imply that the point is stationary. Measured examples of departure from a KLT track and abrupt loss of local support are given in Figure 4.7 and Figure 4.8.

Even the presence of a syntactically valid vector does not guarantee a valid geometric observation. H.264 permits reference lookup outside the image by extending boundary samples [12, Sec. IV-H.1]; those extended samples do not constitute additional observed scene content. This motivates separating codec validity from the image bounds and rectification validity required for the geometry used here.

2.4.6. Texture and the Motion-Matching Error Surface

The reliability of a block displacement also depends on the shape of its local matching-error surface. Zhang et al. [15] characterize this surface from the nine integer-pixel displacements centred on an integer-pixel optimum. For an image-matching distortion expressed as a sum of squared differences, the local surface is

$$E(\Delta x, \Delta y) = \sum_{(x,y)\in W} \left[s(x,y) - c(x + d_{0x} + \Delta x, y + d_{0y} + \Delta y) \right]^2, \tag{2.6}$$

where s is the current block, c is the reference image, W is the block support, and $\mathbf{d}_0$ is the selected integer-pixel displacement. The perturbation $(\Delta x, \Delta y)$ is measured around that displacement. With the first-order brightness-constancy approximation

$$c(x + d_{0x} + \Delta x, y + d_{0y} + \Delta y) \approx c(x + d_{0x}, y + d_{0y}) + I_x \Delta x + I_y \Delta y, \tag{2.7}$$

where I_x and I_y are evaluated at the corresponding reference location for each pixel in W, the displacement Hessian is

$$\mathbf{H}_E \approx 2 \sum_{(x,y)\in W} \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} = 2\mathbf{M}_W. \tag{2.8}$$

Thus the local quadratic curvature is, up to a factor of two, the same second-moment matrix used by Shi-Tomasi corner selection [14]. A corner-like region supplies curvature in two displacement directions; an edge leaves one direction weakly constrained, and a low-texture region leaves both weakly constrained. Zhang et al. use the Hessian condition and deviation from flatness to characterize the accuracy available to sub-pixel motion estimation. This local SSD model explains matching ambiguity; it is not a claim that every H.264 encoder minimizes SSD or uses this quadratic fit. Under the rate-distortion tradeoff in Equation 2.3, a flat image-matching surface gives the rate term greater opportunity to affect which displacement is selected. This is an interpretation of the local model, not a guarantee that the chosen displacement represents physical point motion. Empirical comparisons likewise show that codec vectors can deviate substantially from refined optical flow in regions with little texture, repetitive patterns, or strong motion blur [1].

This relationship motivates using corner coincidence as an offline diagnostic of whether a codec block contains structure capable of localizing translation. It is not a correctness test for an individual vector: the actual codec surface depends on two frames, interpolation, partitioning, and rate-distortion search, and even a textured block can be assigned to the wrong physical surface.

This distinction accounts for the principal measurement difficulties considered later: block support is spatially quantized, several scene structures can contribute to one block, different blocks can describe nearly indistinguishable observations, and useful temporal information can be absent at intra-coded regions.

2.5. Robust Point-Cloud Registration

Rigid registration estimates a rotation $\mathbf{R} \in \mathrm{SO}(3)$ and translation $\mathbf{t} \in \mathbb{R}^3$ from putative 3D-3D correspondences $(\mathbf{a}_i, \mathbf{b}_i)$. Incorrect associations make ordinary least squares brittle. TEASER formulates robust registration with a truncated least-squares objective,

$$\min_{\mathbf{R},\mathbf{t}} \sum_i \min\left(\frac{\|\mathbf{b}_i - (\mathbf{R}\mathbf{a}_i + \mathbf{t})\|^2}{\beta_i^2}, \bar{c}^2 \right), \tag{2.9}$$

where β_i bounds the expected inlier noise, $\bar{c}$ is the dimensionless truncation threshold, and the scale is fixed for metric point clouds [16]. Large residuals saturate instead of dominating

the estimate. TEASER++ combines invariant measurements, graph-based outlier pruning, and graduated non-convexity to obtain a robust rigid transform and an inlier set.

CLIPPER addresses the related problem of selecting mutually consistent associations. It represents putative matches as vertices of a consistency graph and uses pairwise geometric compatibility to recover a dense inlier subgraph [17]. This illustrates the distinction between proposing correspondences and validating them geometrically. The implementation in this thesis uses TEASER++, not CLIPPER: its pairwise transform filters stereo-temporal associations and supports identity propagation, but is not used as the reported camera trajectory.

2.6. Trajectory Evaluation

Let $\mathbf{T}_i$ and $\widehat{\mathbf{T}}_i$ be synchronized reference and estimated camera poses. Before evaluation, the estimate is brought into the reference frame with a rigid $\mathrm{SE}(3)$ alignment $\mathbf{S}$ obtained from corresponding positions [18]. The absolute pose error is

$$\mathbf{E}_i = \mathbf{T}_i^{-1}\mathbf{S}\widehat{\mathbf{T}}_i, \tag{2.10}$$

and absolute trajectory error (ATE) is the root mean square of $\|\mathrm{trans}(\mathbf{E}_i)\|$ over the trajectory [19]. It measures global disagreement after alignment and is therefore sensitive to accumulated drift.

Relative translation error compares motion over a fixed interval Δ. Its pose error is

$$\mathbf{F}_i^{\Delta} = \left(\mathbf{T}_i^{-1}\mathbf{T}_{i+\Delta}\right)^{-1}\left(\widehat{\mathbf{T}}_i^{-1}\widehat{\mathbf{T}}_{i+\Delta}\right), \tag{2.11}$$

with the translational statistic computed from $\mathrm{trans}(\mathbf{F}_i^{\Delta})$ [19]. It describes local drift over the chosen interval rather than global trajectory agreement, so it need not rank methods in the same order as ATE. The alignment and interval used in the experiments are specified in chapter 5.

3. Related Work

3.1. Compression and the Placement of Visual Analysis

Compress-then-analyze and analyze-then-compress allocate different work to a capturing device and its receiver. Hofer et al. [3] investigate H.264 transmission for edge-assisted visual SLAM, evaluating encoding, transmission, decoding, and feature-analysis latency on a Raspberry Pi and an Intel NUC. Their software OpenH264 experiments motivate transmitting compressed video when capture-side feature extraction is costly; they evaluate the coding pipeline independently of a complete SLAM system and do not measure energy consumption.

Redondi et al. [4] compare JPEG-based CTA and BRISK-feature ATC for image recognition in visual sensor networks. Their measurements and network evaluation relate recognition accuracy, transmission rate, and camera-node energy. The outcome depends on the hardware, feature workload, and link constraints. These studies motivate the allocation of visual computation; the present thesis investigates what the receiver can infer directly from codec motion without a pixel-based correspondence stage.

3.2. Motion-Vector Refinement into Optical Flow

Young, Girod, and Taubman [5] recover a dense optical-flow field from block-based coded motion using confidence-weighted filtering and inpainting. Prediction-unit dimensions inform the confidence model, while decoded image colours guide edge-preserving filtering; optional variational refinement additionally uses image gradients, temporal intensity differences, and warping. This work characterizes how codec motion can initialize a more accurate flow estimate, but its recovered flow is a pixel-guided product rather than a codec-only geometric observation.

3.3. Codec-Aware Tracking

VOCA [1] uses H.264 motion vectors as initialization priors for KLT tracking. The codec prediction moves the initial estimate closer to the subsequent photometric solution, while KLT refinement and conventional tracking modes retain access to decoded image intensities. VOCA provides the implementation context and the reference configurations used in this thesis, but does not impose the same pixel-access restriction.

3.4. Motion Vectors for Camera Motion and Mapping

Li and Song [20] estimate monocular camera motion and planar scene structure from MPEG motion vectors. Their formulation explicitly assumes a structured environment dominated by planes, such as building facades and paved roads. It merges direct and bidirectional-frame observations, rejects insufficient-parallax motion, groups correspondences with homography RANSAC, and jointly tracks planes and camera motion in an extended Kalman filter. It thus establishes an early codec-only navigation result within this planar setting, rather than a general point-based reconstruction. Its B-frame observations also depend on subsequent anchor images: they cannot be formed as soon as each intermediate image is captured, introducing buffering and latency for a live camera. The paper does not establish end-to-end real-time operation. These constraints concern both the required scene structure and the availability of the measurements, independently of the estimator's computational cost.

MoV-SLAM [21] uses H.264 macroblocks and their motion vectors to build long-range correspondence tracks within a complete monocular and stereo SLAM system. Its EXPRESS detector evaluates block intensities to select distinctive regions, while Lucas-Kanade tracking handles missing motion-vector coverage and stereo association. It consequently demonstrates how codec motion can reduce frontend cost without enforcing the pixel-access boundary studied here.

3.5. Motion Vectors for Correspondence and Reconstruction

Zouein et al. [22] convert AV1 block centres and motion vectors into point correspondences, propagate them into multi-frame tracks, and reject directionally inconsistent tracks using cosine similarity. Their evaluation focuses on correspondence density, pairwise geometry, and a small COLMAP structure-from-motion demonstration rather than a dedicated visual-odometry estimator.

The subsequent Gaussian-splatting work [23] reuses this correspondence pipeline to produce dense COLMAP point clouds for initializing 3D Gaussian splatting. It evaluates the effect of denser reconstruction on rendering quality and training time, without introducing calibrated stereo depth or a motion-vector-specific visual-odometry backend.

3.6. Motion-Vector Stereo

Passthrough+ [2] repurposes video-encoding hardware to estimate correspondences between rectified views of a stereo pair. Shifted copies of the stereo images reduce the displacement presented to the encoder and offset the codec block grid, after which consistency checks and calibrated triangulation produce sparse 3D points. This per-frame stereo stage uses codec motion, but the complete scene proxy is not motion-vector-only: the system assumes a high-precision headset pose, typically supplied by an independent SLAM system, and uses pose to propagate the previous depth map into the current view. That pose-assisted temporal model is unavailable under the information restriction of this thesis. The present work adapts

the sparse stereo-reconstruction principle; the differences in preprocessing and downstream use are specified in chapter 4.

3.7. Spatial Feature Management

Feature-based estimators generally regulate both the number and spatial distribution of their observations. VINS-Mono maintains a target population of corner tracks and enforces a minimum separation between newly detected features to obtain a more uniform distribution [24]. Similarly, the low-latency VIO pipeline of Kühne et al. suppresses nearby detector candidates so that the same local feature is not represented repeatedly [25]. Both systems make these decisions from image-derived features. Their common design objective motivates the pixel-free analogue developed here: codec observations that are indistinguishable at the available spatial and depth resolution should not provide duplicated geometric support.

3.8. Positioning of This Work

The compressed-domain systems above span the distinctions central to this thesis: whether codec motion is refined using pixels, whether recovered structure is planar or point-based, whether metric scale is supplied by stereo, and whether correspondences persist beyond a frame pair. Work that merely accelerates video decoding or generic optical flow without using codec side information is outside this comparison.

Prior codec-only navigation establishes that motion vectors can support camera-motion estimation in structured, plane-dominated environments [20]. Other work uses codec motion to initialize pixel-domain tracking or provide correspondences for reconstruction. The present work investigates one way to use these measurements in a calibrated, point-based stereo visual-odometry system without requiring a planar scene model. The evaluated design combines metric motion-vector stereo, persistent 3D association, suppression of geometrically indistinguishable observations, and displacement-based optimization without pixel refinement. Its reverse-temporal streams are nevertheless prepared offline, so this study also leaves causal, end-to-end real-time operation unverified.

No row alone defines the contribution. In particular, codec-only navigation predates this thesis, motion-vector tracks have already been used for reconstruction, and motion-vector stereo originates in Passthrough+. The present study tests how these ingredients behave together under a pixel-free estimation boundary and within a point-based visual-odometry estimator.

Table 3.1.: Comparison of directly related compressed-domain systems. "Pixels" refers to image-intensity access in correspondence generation or geometric estimation, not to the image input required by an encoder.

Work	Input and representation	Scale and pixel access	Track persistence and validation	Estimator or output
Young et al. [5]	Coded MVs Dense 2D flow	No scale Pixels used	*Persistence:* none *Validation:* pairwise confidence	Pixel-guided flow
Li and Song [20]	MPEG MVs Planes and pose	Monocular No pixels	*Persistence:* plane tracks *Validation:* homography RANSAC	Extended Kalman filter
MoV-SLAM [21]	H.264 MVs and LK Point tracks	Mono/stereo Pixels used	*Persistence:* point tracks *Validation:* LK fallback	SLAM backend
Zouein et al. [22]	AV1 MVs 2D point tracks	Monocular No pixels	*Persistence:* MV tracks *Validation:* directional check	Matching and SfM study
Zouein et al. (2026) [23]	AV1 MVs Dense 3D points	Monocular SfM Pixels downstream	*Persistence:* MV tracks *Validation:* geometric checks	COLMAP and 3DGS
VOCA [1]	H.264-seeded KLT Point tracks	Calibrated VO Pixels used	*Persistence:* KLT tracks *Validation:* fallback modes	Sliding window BA
Passthrough+ [2]	Stereo encoder MVs and SLAM pose Sparse depth and dense proxy	Metric stereo Pixels used	*Persistence:* temporal depth *Validation:* stereo consistency	View synthesis
Pixelless VO (this thesis)	H.264 temporal and stereo MVs Persistent 3D points	Metric stereo No estimator pixels	*Persistence:* landmark IDs *Validation:* reciprocal + TEASER++	Sliding window BA

4. Compressed-Domain Visual Odometry

4.1. System Overview

The examined pipeline separates image-dependent preparation from compressed-domain estimation, as summarized in Figure 4.1. Offline preparation encodes native images into temporal H.264 streams and rectified images into stereo streams. The extraction software decodes these streams to obtain their motion-vector metadata. Only codec side information, calibration, masks, and timing metadata enter the tracking frontend and pose estimator. Temporal vectors propose motion, shifted stereo encodings provide metric depth, and robust registration validates the resulting 3D-3D associations. Confirmed identities are converted into observations for a windowed visual-odometry backend.

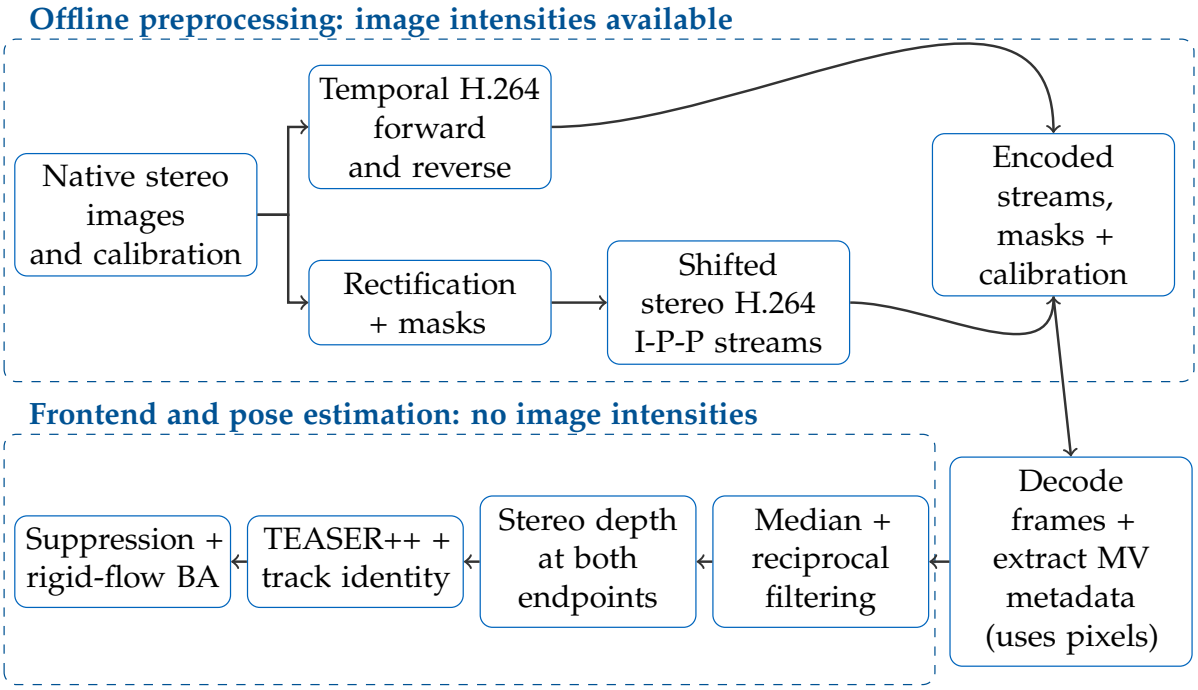


Figure 4.1.: Processing pipeline and pixel-access boundary. Rectification and both H.264 paths operate on images before evaluation. Native images feed the forward/reverse temporal encodings, while rectified images feed the shifted stereo encodings. The extractor reconstructs image pixels, which are not passed to the frontend. The pixel-free boundary encloses correspondence filtering, stereo lookup, association, and pose estimation; these stages consume only side information, masks, timing data, and calibrated geometry.

4.2. Input and Pixel-Access Boundary

For each timestamp, the estimator receives temporal motion vectors from the forward and reverse encodings, stereo motion vectors from the shifted I-P-P encodings, the width and height of every prediction block, reference direction, fractional displacement and frame type. It additionally receives the two-camera calibration, synchronized timestamps, the virtual rectification model, and the two rectification validity masks. These quantities are sufficient to express codec supports in native camera coordinates and to form calibrated rays.

In compressed-domain operation, image loading is disabled before observations enter the frontend. The frontend neither constructs image pyramids nor runs FAST, KLT, patch alignment, descriptor extraction, or a photometric test. Image intensities are used to generate the H.264 streams during offline preprocessing, and the extraction software reconstructs images while reading codec side information. Neither the tracking frontend nor pose estimation consumes those intensities. The Shi-Tomasi and blur measurements reported later are likewise post-hoc diagnostics and are not available to pose estimation.

4.3. Stereo Rectification and Video Preprocessing

Rectification is constructed separately for each supplied stereo calibration. The camera supplying temporal track candidates is designated the anchor camera A, and the other view is the stereo camera S. Let ${}^{A}\mathbf{t}_{S}$ be the position of the stereo camera expressed in the anchor-camera frame. The horizontal axis of the virtual camera is $\mathbf{e}_x = {}^{A}\mathbf{t}_{S} / \|{}^{A}\mathbf{t}_{S}\|$. The sum of the two native optical axes, expressed in the anchor-camera frame, is projected onto the plane orthogonal to $\mathbf{e}_x$ and normalized to form $\mathbf{e}_z$; $\mathbf{e}_y = \mathbf{e}_z \times \mathbf{e}_x$ completes the orientation. This produces a common view direction while making the calibrated baseline horizontal.

Both cameras are remapped into the same undistorted pinhole projection. Its resolution equals the native EuRoC resolution, its principal point is the image centre, and its common focal length is the smallest of the four native horizontal and vertical focal parameters. Each virtual ray is rotated into the respective native camera and projected with that camera's calibrated distortion model. Bilinear sampling produces the offline rectified images. Pixels for which this mapping is invalid are set to zero, and a binary validity mask for each camera is stored with the virtual intrinsics and rotations. During estimation, an entire codec block is rejected if either of its rectified supports intersects an invalid region.

Camera streams are paired by their recorded nanosecond timestamps. If one camera contains an unmatched terminal image, only that unpaired timestamp is removed from the longer metadata stream; both the temporal video and its reverse counterpart are generated from the resulting ordered timestamp list. EuRoC has two cameras, so its calibrated pair is used directly. For a dataset with more than two cameras, one calibrated pair would have to be designated as the anchor and stereo views; additional cameras would remain outside the Passthrough+-derived depth stage.

4.4. Codec-Domain Stereo Correspondences

Only the sparse encoder-stereo stage of Passthrough+ [2] is adapted. The shifted views are represented as independent full-frame encodings rather than subframes of one hardware-encoder mosaic. Temporal motion-search and partition settings are retained, while the reference structure, rate control, and encoded frame rate follow the stereo configuration in chapter 5. Each EuRoC camera pair is remapped into a calibration-specific common virtual projection before encoding. During estimation, stereo motion vectors are queried only at temporal candidate locations; their endpoints are converted back to native camera rays and triangulated with the calibrated baseline.

The original hemispherical depth grid, Laplacian densification, stereoscopic rendering, and pixel-based subpixel refinement are not used. Nor is its temporal depth-map propagation: Passthrough+ projects the previous depth map using a headset pose obtained from an independent SLAM system [2], whereas no external pose estimate is available here. Sparse stereo depths instead serve as geometric support for temporal association and visual odometry.

Writing L for the anchor view and R for the stereo view, three horizontal pre-shifts, 0, 8, and 32 pixels, are combined with four codec grid phases, $(0,0)$, $(4,0)$, $(0,4)$, and $(4,4)$ pixels. The resulting twelve independent streams contain repeated triples

$$I(L_t^{s,g}), \quad P(R_t^g), \quad P(L_t^{s,g}), \tag{4.1}$$

where s denotes the disparity pre-shift and g the grid phase. With one reference frame, the first P-frame supplies left-to-right predictions and the second supplies right-to-left predictions. The known artificial shifts are removed from the reported block coordinates. Left-to-right and right-to-left blocks are paired one-to-one when their supports overlap and their componentwise cycle error is at most 1.0 pixel. The intersection of the two supports becomes the validated rectangle.

A native anchor-camera query $\mathbf{u}$ is first unprojected, rotated into the virtual frame, and projected onto the rectified grid. Every variable-sized codec block whose source rectangle contains this location is considered; blocks are not expanded into per-pixel points. Applying that block's fractional motion vector gives a candidate in the rectified right view. Both endpoints must lie in valid rectification support, and their vertical displacement in the common grid must not exceed one pixel. The encoded right endpoint is then converted to a native stereo-camera ray. The frontend triangulates distances along unit native-camera bearings. Candidates are rejected when their calibrated epipolar error exceeds 0.005, triangulation is degenerate, either depth is at most 0.333 m, the anchor-camera depth exceeds 30 m, or reprojection into the stereo camera differs from the candidate by more than 1.5 pixels.

When several encodings or overlapping variable-size blocks cover the query, one stereo observation is returned. Its ranking cost is

$$C = e_{\text{repr}} + e_{\text{epi}} + 10^{-4}wh + 10^{-3}d_c, \tag{4.2}$$

Here e_{repr} is the stereo reprojection distance in pixels, and $e_{\text{epi}} = |\hat{\mathbf{b}}_L^\top \mathbf{E} \hat{\mathbf{b}}_R|$ is a dimensionless epipolar residual. The bearings are unit vectors in their native camera frames, and $\mathbf{E}$ is the

essential matrix formed from the calibrated relative rotation and unit baseline direction. The product wh is block area and d_c is the squared, block-size-normalized distance from the query to the block centre. This preference resolves duplicate codec hypotheses locally; the later suppression stage handles geometrically indistinguishable temporal measurements.

4.4.1. Grid Sampling and Disparity Pre-Shifts

The grid phases and horizontal pre-shifts address two different aspects of stereo correspondence. Grid phases increase spatial sampling density. Applying the same phase g to both views moves the codec's sampling grid relative to the scene while preserving disparity. In the eight-pixel sampling example used by Passthrough+ [2], the phases $(0,0)$, $(4,0)$, $(0,4)$, and $(4,4)$ interleave into a grid with four-pixel spacing in both directions, as shown in Figure 4.2. The combined samples refine spatial coverage. This spacing describes the sampling construction; the encoded prediction blocks and their validated intersections retain variable sizes.

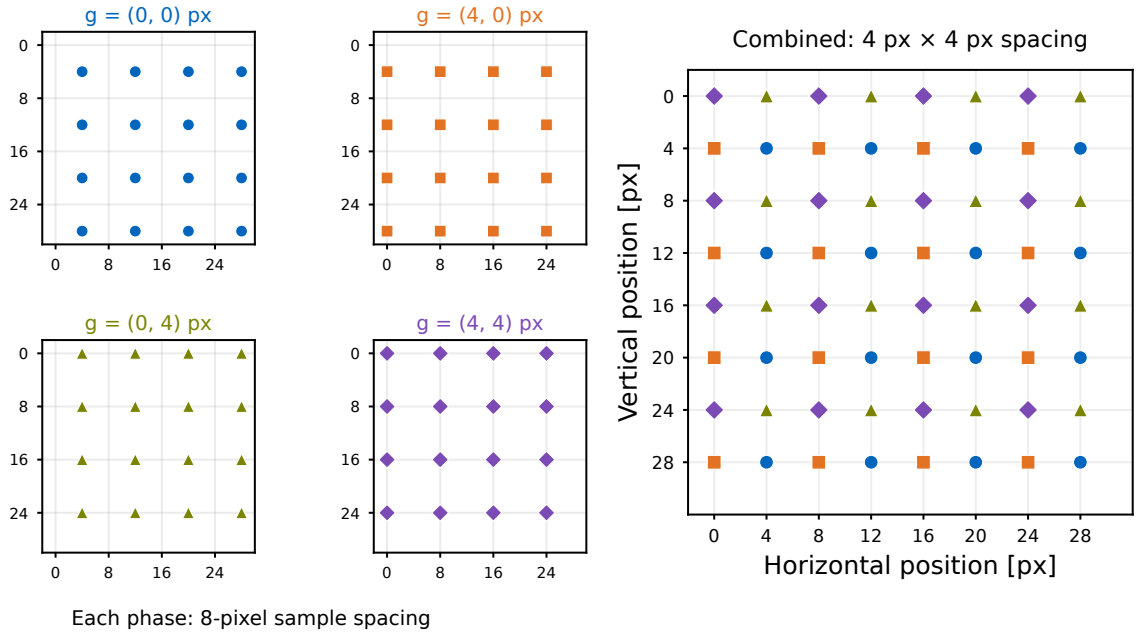


Figure 4.2.: Grid-phase interleaving, illustrated for an eight-pixel base grid. The four phases sample different positions in the same image coordinates. Their union has four-pixel spacing horizontally and vertically. Colour and marker shape identify the originating phase. The same phase is applied to both stereo views, so it changes sampling position without changing disparity.

Horizontal pre-shifts change the displacement presented to the encoder. Passthrough+ reports that encoders favour short motion vectors, making large disparities from nearby surfaces harder to recover [2]. For disparity $d = u_L - u_R$, shifting the left image by s pixels

presents $d - s$ to the encoder. The known shift is restored before triangulation. With focal length f_x and baseline B, rectified axial depth satisfies

$$Z = \frac{f_x B}{d}, \qquad |d - s| = \left| \frac{f_x B}{Z} - s \right|. \tag{4.3}$$

Here Z is axial depth in the virtual rectified camera, used to interpret the pre-shift examples and depth visualizations. The tracking frontend instead triangulates distance along native-camera unit bearings, denoted λ below.

Figure 4.3 shows two locations whose depth is recovered by different pre-shifts. At the EMH01 location at 135 s, a disparity of 29.25 pixels gives $Z \approx 1.31$ m. The 32-pixel pre-shift reduces the required displacement to 2.75 pixels and provides an accepted match. The displayed 0- and 8-pixel proposals instead have a vertical error of 1.75 pixels and fail the one-pixel check. At the EMH02 location at 95 s, disparity is 8.50 pixels and $Z \approx 4.52$ m; the 8-pixel pre-shift reduces the required displacement to 0.50 pixels and supplies the depth.

For each location, all four grid phases are checked: neither of the other pre-shifts returns an accepted block covering that query. Thus, a pre-shift can determine whether a particular scene location supplies depth, and the useful pre-shift changes with disparity. Local appearance and codec decisions also affect availability, so useful depth ranges can overlap. The illustrated matches pass rectification-mask, vertical-displacement, reciprocal-cycle, and one-to-one overlap checks; they precede temporal tracking and backend landmark selection. An overview of the correspondences from all twelve transforms on a later EMH02 frame is given in Figure B.8.

Figure 4.4 and Figure 4.5 show depth renderings in three scenes: EV101 at 60 s, EMH01 at 115 s, and EMH02 at 135 s. A separate grayscale depth inset is placed inside each rectified scene image. Stereo-block support surviving the rectification and bidirectional checks is sampled on a four-pixel grid. Each supported sample displays calibrated axial depth, using the median where several blocks cover the same location. Brighter values indicate nearer surfaces; black marks locations without a displayed sample. Unsupported regions are left unfilled. These post-hoc renderings make the codec-derived geometry inspectable; grayscale intensity is not passed to the estimator.

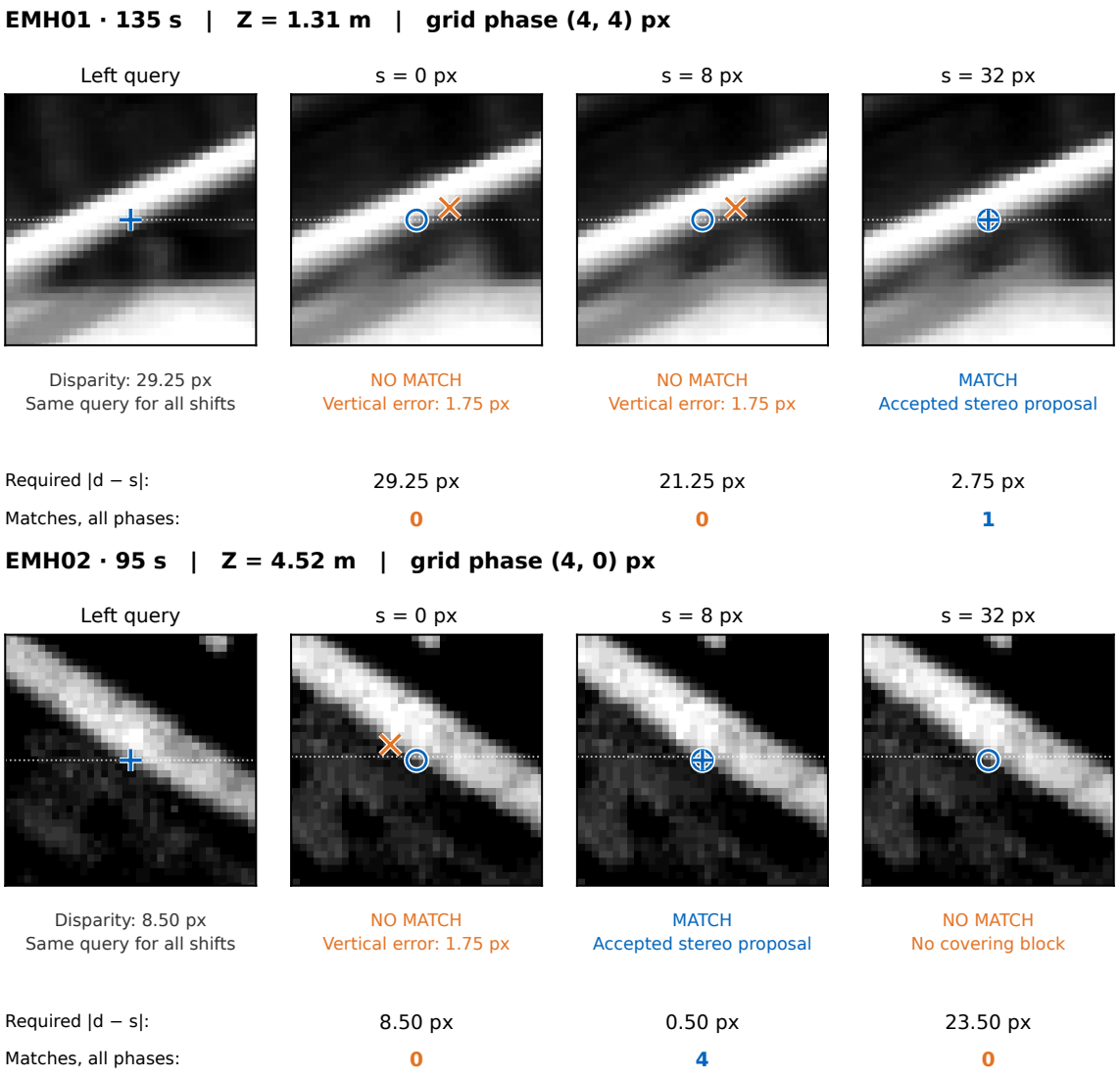


Figure 4.3.: Pre-shift-dependent depth recovery at two later scene locations. Each row holds the left query and displayed grid phase fixed. Blue circles repeat the right-image match supplied by the successful pre-shift; orange crosses mark rejected proposals. The dotted line is the query's epipolar row. The required residual $|d - s|$ is computed from the recovered disparity; the counts below include all four grid phases. Left and right crops within a row share one grayscale display scale.

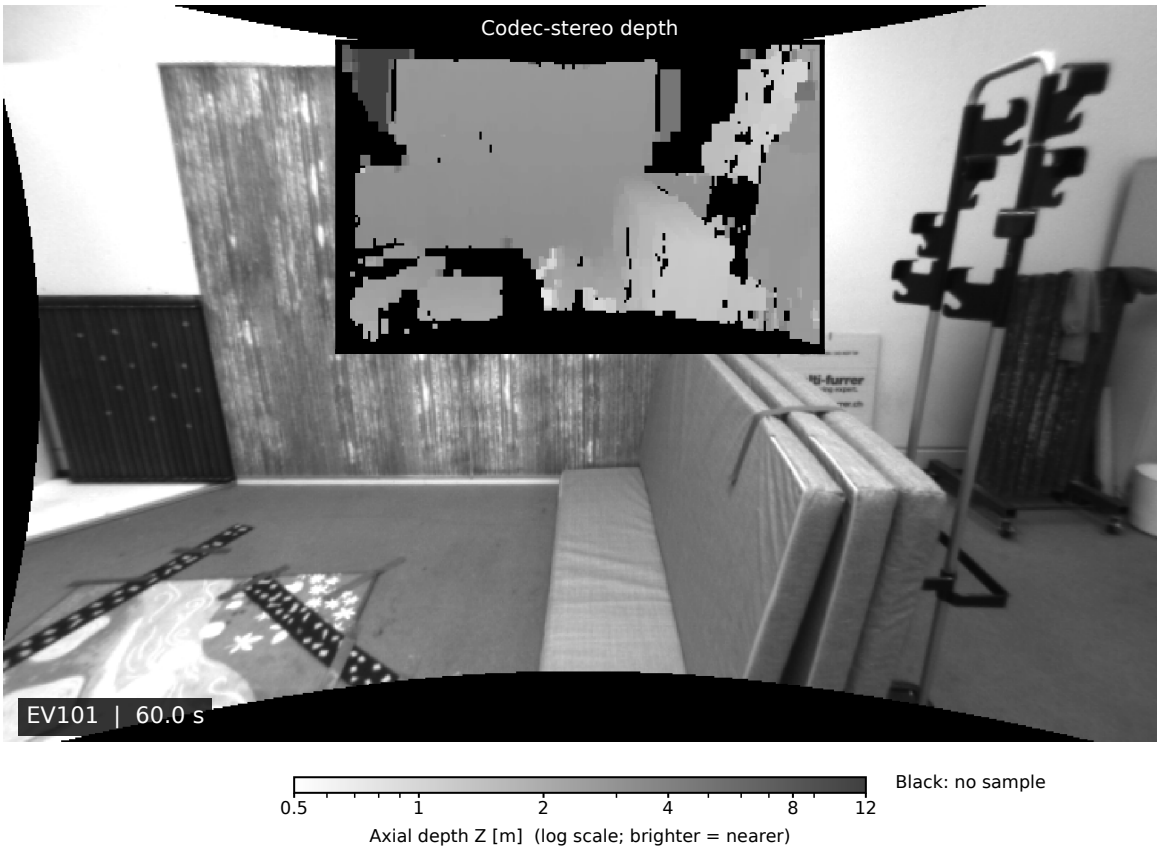


Figure 4.4.: Codec-stereo depth for EV101 at 60 s. The inset shows the full field of view on a four-pixel sampling grid, with nearer surfaces brighter on the logarithmic depth scale. Black marks unsupported samples; depths outside the displayed 0.5 m to 12 m range are clipped only for grayscale visualization.

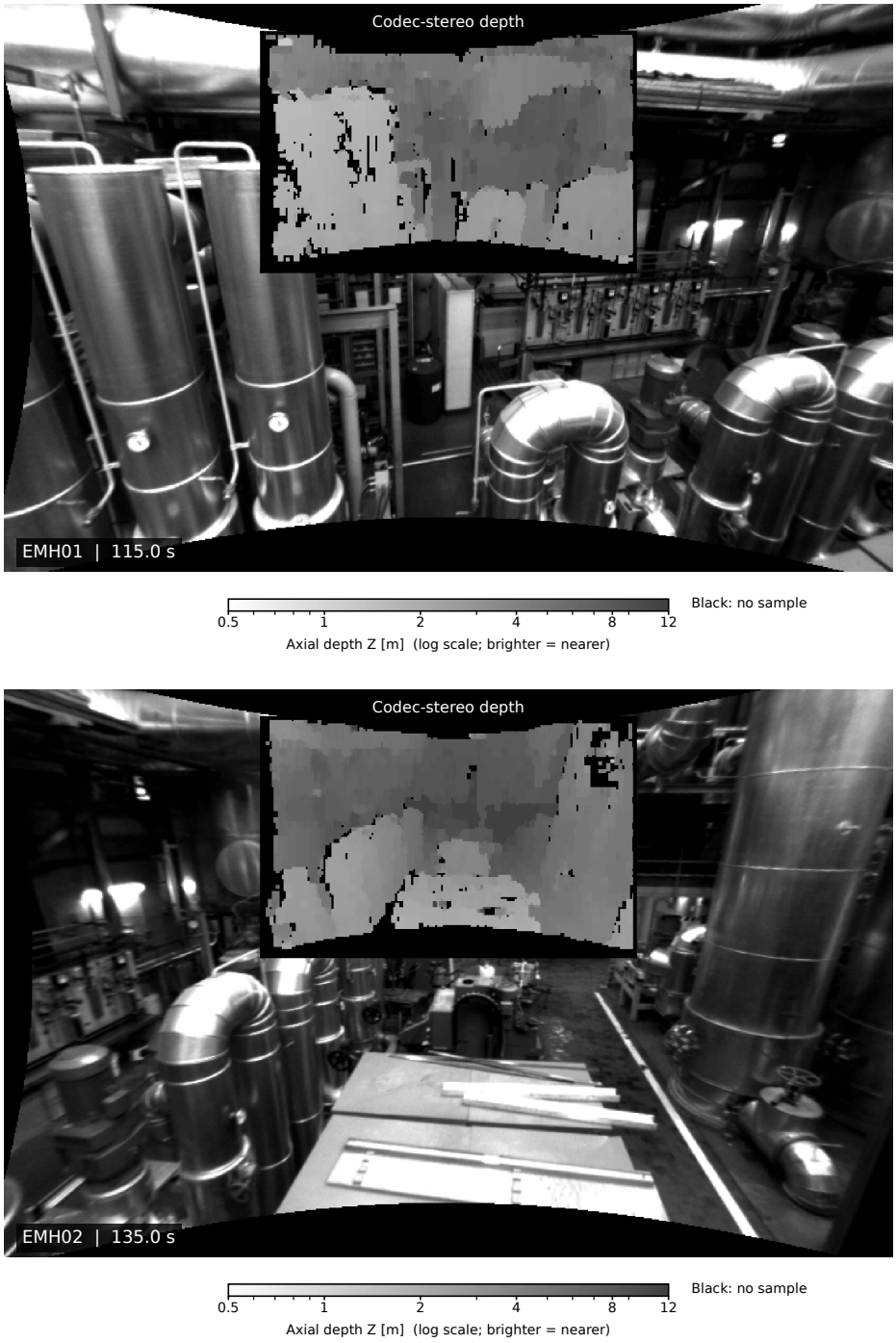


Figure 4.5.: Additional codec-stereo depth maps in later Machine Hall scenes: EMH01 at 115 s (top) and EMH02 at 135 s (bottom). Both use the grayscale scale and sampling convention of Figure 4.4. The separate depth insets distinguish nearby equipment from more distant structure while retaining unsupported regions in black.

4.5. Temporal Motion-Vector Candidates

Motion vectors are read from the H.264 side data with their reference direction, block dimensions, integer numerator, and scale. Only vectors referencing the preceding encoded frame are used. Dividing the numerator by the scale preserves the fractional current-to-reference displacement $\mathbf{v}$. In the forward encoding, a coding-block centre $\mathbf{q}_t$ corresponds to the reference-support centre $\mathbf{q}_{t-1} = \mathbf{q}_t + \mathbf{v}$. The temporal candidate runs from $\mathbf{q}_{t-1}$ to $\mathbf{q}_t$, with forward displacement $\mathbf{m} = -\mathbf{v}$. Later continuation may apply a covering block's displacement to an established track location; it does not repeatedly snap that location back to the block centre.

The compression-artifact filter tests local agreement without consulting the image. Motion vectors are accumulated into a 16×16-pixel grid and their two displacement components are averaged independently in each occupied cell. For a cell q, let $\bar{\mathbf{v}}_q$ denote this mean and let $\widetilde{\mathbf{v}}_q$ be the componentwise median of the occupied cells in its 3×3 neighbourhood. Provided that at least three neighbouring cells are occupied, every vector assigned to q receives the score

$$d_{\mathrm{med}}(q) = \left\| \bar{\mathbf{v}}_q - \widetilde{\mathbf{v}}_q \right\|_2 . \tag{4.4}$$

It is retained when $d_{\mathrm{med}} \leq 5$ pixels. Border cells, cells without sufficient neighbourhood support, out-of-image endpoints, and endpoints in masked-out image regions are rejected. The result is illustrated in Figure 4.6: the large magenta vectors disagree with the coherent downward motion that dominates their neighbourhoods.

The forward encoding supplies $t - 1 \rightarrow t$ motion. A second video encoded in reverse image order supplies the same interval from the opposite temporal direction; its vectors and timestamps are inverted back into forward-time coordinates. The fixed single-reference encoder setting prevents a temporal vector from silently spanning more than one frame. At an I-frame boundary, the available direction is used if exactly one encoding supplies inter prediction. If neither direction supplies a candidate, the affected tracks terminate rather than receiving a copied or pixel-derived displacement.

A motion vector describes a block prediction, not the identity of a scene point inside that block. Naively assigning a persistent ID to whichever block covers the previous sample can therefore fail in two distinct ways. First, the selected block can continue to produce plausible motion after its displacement has ceased to follow the same localized structure. Figure 4.7 shows this effect over fourteen transitions on a strong-gradient railing edge. The codec path and a post-hoc KLT track initially agree, then separate progressively while raw codec vectors remain available.

Second, local inter-prediction support can disappear although the tracked image structure remains visible. In Figure 4.8, raw motion vectors remain present in the destination image at the first unsupported transition, but none has a source block covering the previously propagated point. The KLT diagnostic remains stable over the subsequent observations. The trace alone does not identify whether this local absence was caused by an intra-coded block, a partition change, or another encoder decision. It nevertheless demonstrates why an absent vector cannot be equated with an absent scene point.

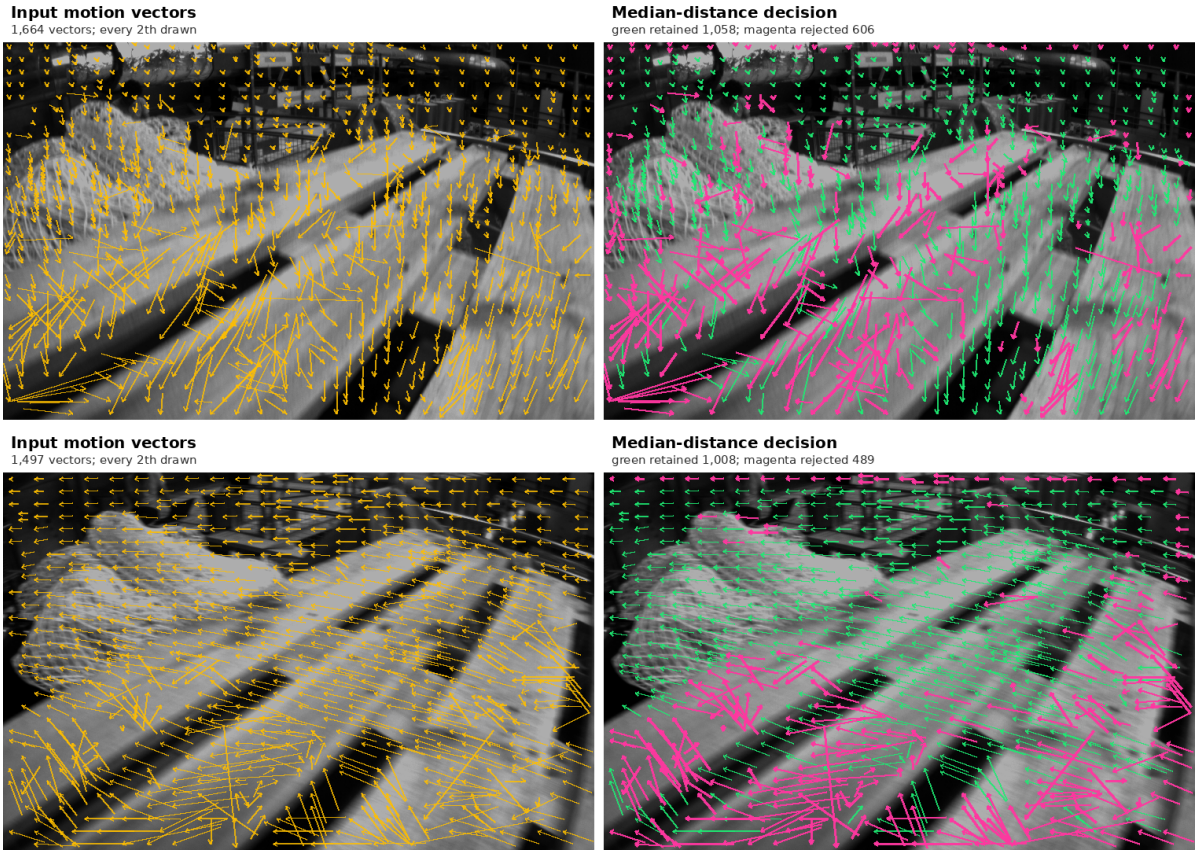


Figure 4.6.: Examples of median-filter decisions on two EMH01 frames. In each row the left panel shows the input field; on the right, retained vectors are green and rejected vectors are magenta. Arrows use the measured displacements; dense populations are deterministically subsampled only for legibility.

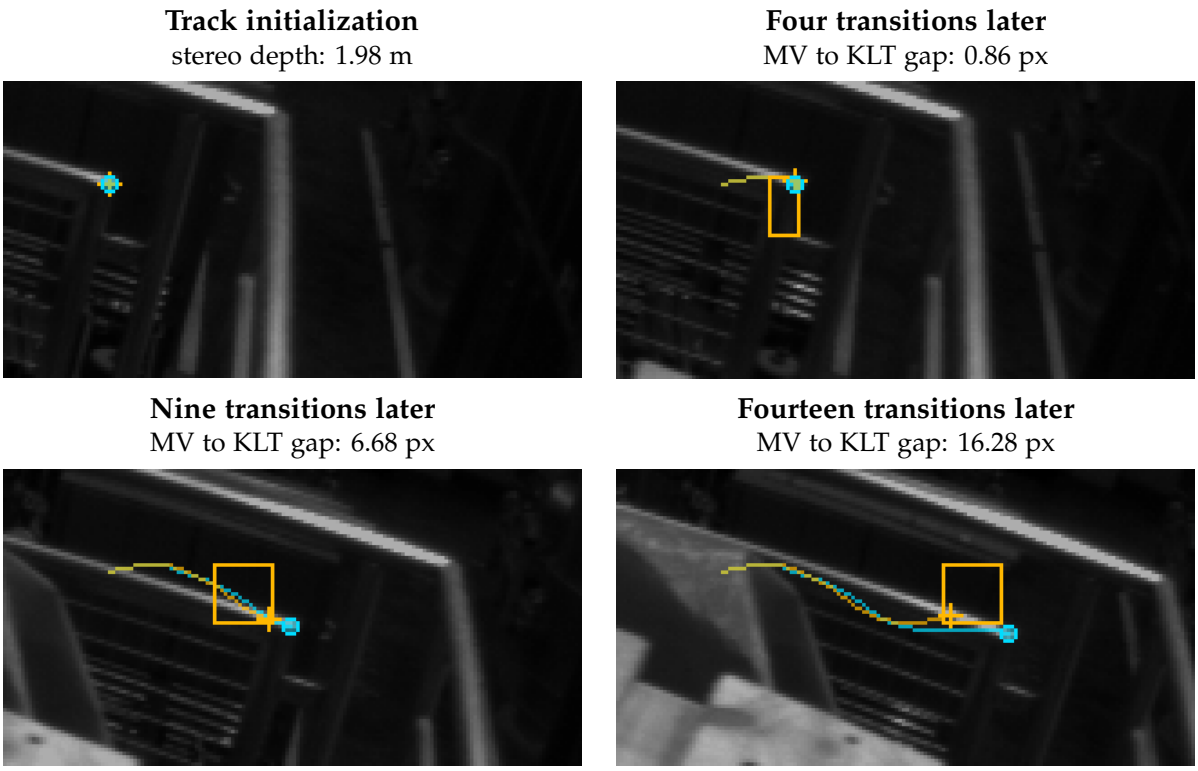


Figure 4.7.: Failure of naive ID propagation on EMH01. Both diagnostic tracks start at the same selected image feature. Orange repeatedly applies the raw, unfiltered H.264 vector whose source block covers the preceding point; its rectangle shows that step's destination block support. Cyan is an OpenCV pyramidal KLT track computed after the experiment and retained only while its per-step forward-backward error is at most one pixel. It is diagnostic and is not consumed by the estimator. The nearest accepted compressed-stereo sample is 3.43 pixels from the birth location and estimates a depth of 1.98 metres. The trails show the full histories up to each displayed stage.

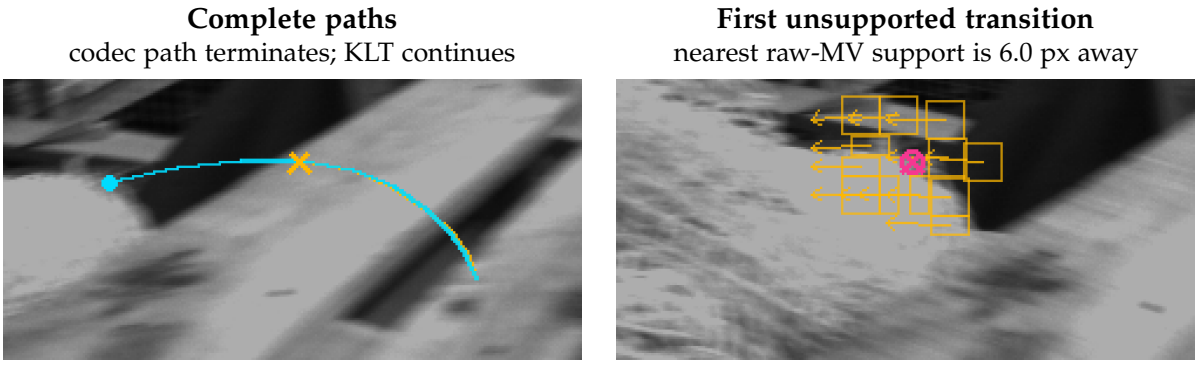


Figure 4.8.: Abrupt local loss of codec support on EMH01. Orange and cyan have the same meanings as in Figure 4.7. The raw MV path follows the KLT diagnostic for six transitions and then terminates; the left panel places both complete paths on one image. The right panel shows a nearby subsample of the raw H.264 source supports and displacements used for the next transition. Other vectors are present locally, but none covers the propagated query marked in magenta. Decoded intensities are used only to construct this explanatory diagnostic.

These two observations motivate reciprocal and geometric validation; neither post-hoc KLT track is an additional pixel-domain gate in the method.

4.6. Forward-Backward Consistency

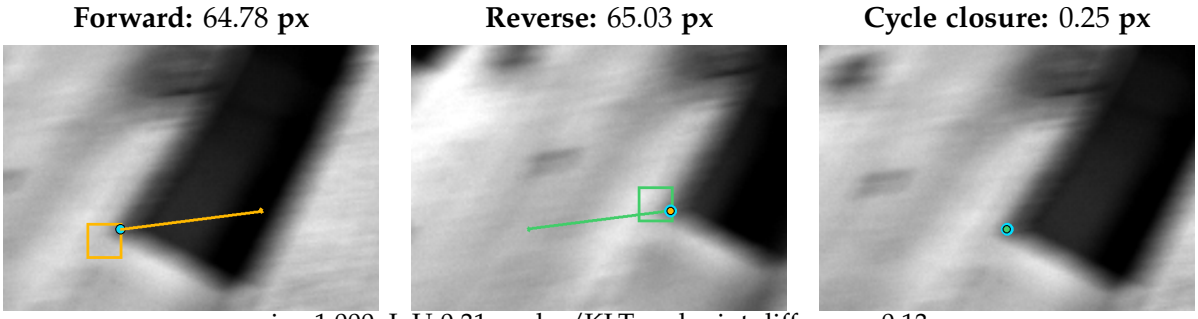
Tracking a point forward and then backward should return to its starting location when both estimates describe the same correspondence. Kalal et al. use this time-reversal principle to detect tracking failures from forward-backward error [26]. Zouein et al. apply a related directional-consistency test to tracks assembled from AV1 motion vectors [22]. Here the principle is adapted to H.264 block supports rather than image-tracked points.

Let $\mathbf{v}_f$ be a forward motion vector and let $\mathbf{v}_b$ be a vector from the reverse encoding, expressed in the same temporal direction. A candidate pair is retained only if its source and destination support rectangles overlap in both directions and

$$\frac{\mathbf{v}_f^{\mathsf{T}}\mathbf{v}_b}{\|\mathbf{v}_f\|\|\mathbf{v}_b\|} \geq 0.98, \qquad \|\mathbf{v}_f - \mathbf{v}_b\|_2 \leq 1.5 \text{ px}. \tag{4.5}$$

The minimum intersection-over-union of the corresponding source and destination supports must additionally be at least 0.15. Two zero-displacement vectors are treated as directionally consistent. The validated support is the intersection that remains after translating both block supports into a common frame.

The retained hybrid policy uses reciprocal candidates for consensus estimation and continuation of established tracks. The larger best-direction pool may propose new tracks, but these remain provisional until a reciprocal continuation confirms a third observation. If only one encoding direction is available at an I-frame boundary, that direction is used as a fallback.



cosine 1.000, IoU 0.31; codec/KLT endpoint difference 0.13 px

Figure 4.9.: An accepted large-displacement reciprocal match on EMH01. The cyan marker identifies a post-hoc KLT feature, the orange arrow is the forward codec motion, and the green arrow is the reverse-encoding motion. In the middle panel, the cyan KLT ring encloses the orange codec endpoint; in the final panel, the green return point lies within 0.25 px of the cyan source point. The KLT observation also passes its own forward-backward check with 0.06 px error. Image intensities are used only for this explanatory validation and are not inputs to reciprocal correspondence selection.

4.7. Robust Association and Track Identity

For a temporal candidate $(\mathbf{u}_{t-1}, \mathbf{u}_t)$, compressed-domain stereo supplies the distance λ along the native anchor-camera bearing at each endpoint. Let $\hat{\mathbf{b}}_A(\mathbf{u})$ denote the corresponding unit bearing. These ranges and bearings form a putative 3D-3D association

$$\mathbf{a}_i = \lambda_{t-1,i}\,\hat{\mathbf{b}}_A(\mathbf{u}_{t-1,i}), \qquad \mathbf{b}_i = \lambda_{t,i}\,\hat{\mathbf{b}}_A(\mathbf{u}_{t,i}). \tag{4.6}$$

Under the static-scene assumption, the inlier pairs share one rigid transformation. TEASER++ estimates this consensus with fixed scale and a 0.1 m noise bound [16]. Each candidate is then tested against the estimated transform. For points beyond 5 m, a further 1.5 px reprojection gate rejects associations whose metric residual is small only because of weak depth geometry. If registration fails, no temporal association is accepted for that frame.

An accepted continuation inherits the existing landmark ID. A new association receives a provisional ID and is withheld from the backend until the following frame confirms it. TEASER++ therefore acts as an association filter; its frame-to-frame transform is not inserted as a camera-pose measurement.

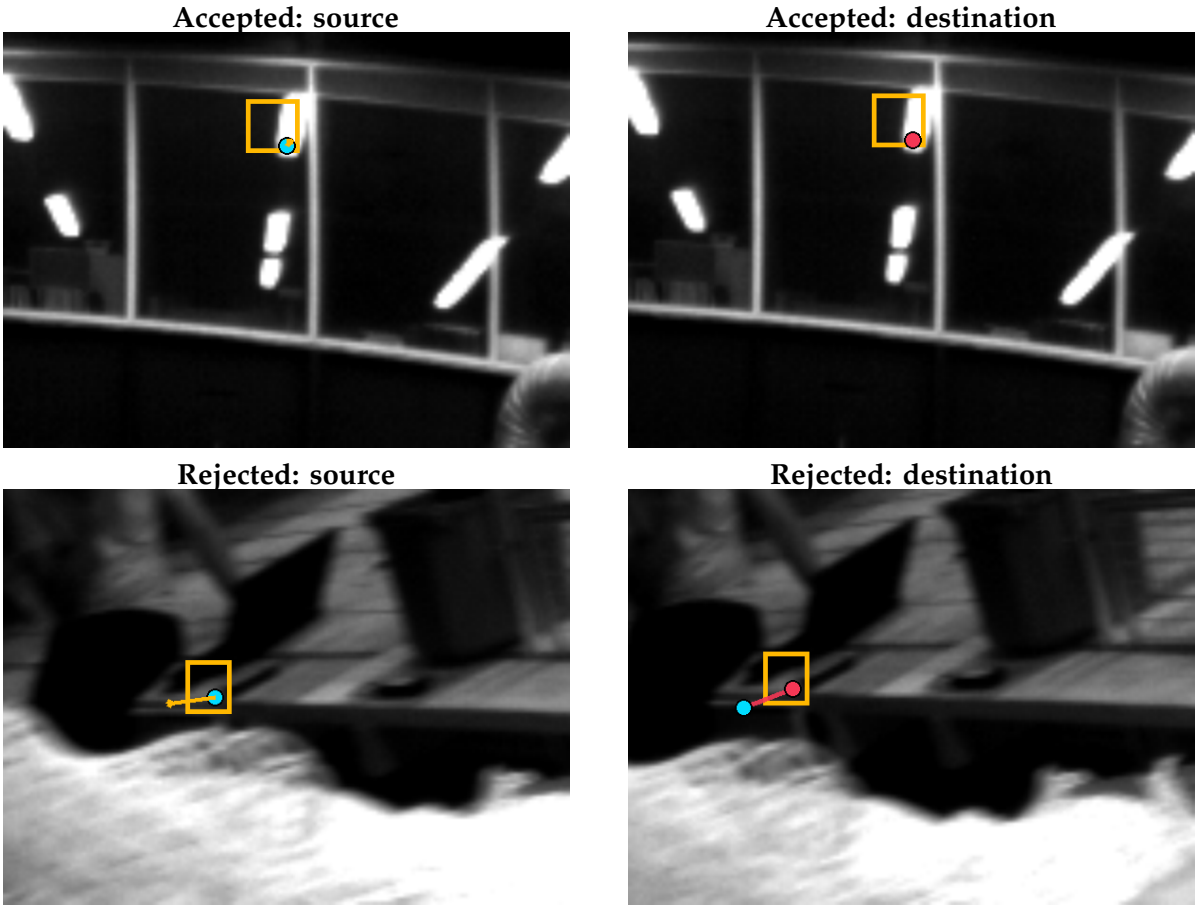


Figure 4.10.: Individual robust-association decisions on EMH01. Orange outlines show the exact codec-block support, cyan marks a post-hoc KLT endpoint, and magenta marks the codec-motion endpoint. The accepted example has a 0.12 px endpoint difference. In the rejected example, the distinctly localized corner moves to the cyan position while the codec candidate ends 13.8 px away. Both KLT observations pass a one-pixel forward-backward check. KLT is used only to explain the decisions after estimation; TEASER++ receives the stereo 3D-3D candidates and no image data.

4.8. Suppression of Indistinguishable Measurements

Conventional feature frontends suppress nearby detections to avoid repeatedly representing the same local structure and to preserve spatially useful support [24, 25]. Codec blocks have no image-derived detector response on which to perform conventional non-maximum suppression. The proposed frontend instead suppresses measurements that cannot be distinguished at its available geometric resolution.

H.264 represents motion vectors at quarter-pixel precision and assigns each vector to an extended prediction block. This finite spatial resolution motivates treating sufficiently similar codec-derived observations as duplicate support. Two accepted measurements are considered duplicates when their positions at both temporal endpoints differ by at most 1.125 px in the L_∞ norm and their depth intervals overlap at both endpoints.

The interval model assigns a single-endpoint tolerance of 0.5625 px, half the selected suppression distance, giving a stereo-disparity tolerance of 1.125 px. It forms a disparity surrogate $f_x B/\lambda$ from the triangulated camera range, perturbs it by the disparity tolerance, and maps the limits back through the pinhole depth relation. A nonpositive lower disparity limit gives an unbounded upper range limit. This is a pinhole approximation for comparing depth support. The tolerances were selected in the parameter study; quarter-pixel representation alone does not establish a bound on physical correspondence error.

Candidates are processed in age order: established IDs are considered before new births, and monotonically assigned IDs order older tracks first. The oldest eligible representative is retained and suppresses younger indistinguishable candidates. This favors continuity without inspecting image intensities.

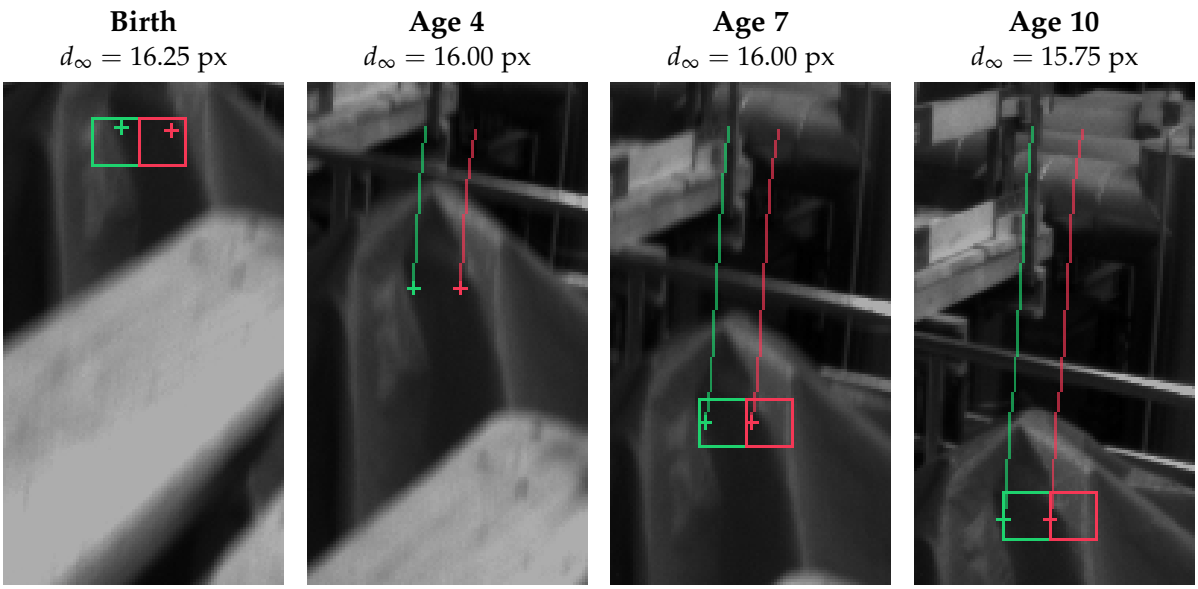


Figure 4.11.: Two nearby but distinguishable EMH01 tracks over their complete ten-frame shared lifetime. Green and magenta identify the two retained IDs; the trails show their preceding image-plane observations. The centers are 16.25 pixels apart at birth and remain between 15.75 and 16.25 pixels apart while moving approximately 135 pixels through the image. Both IDs therefore survive age-prioritized suppression. This illustrates that the method does not enforce generic spatial sparsity: it removes only pairs that meet the 1.125 pixel and depth-overlap criteria. Rectangles show the exact H.264 block dimensions rather than enlarged display regions.

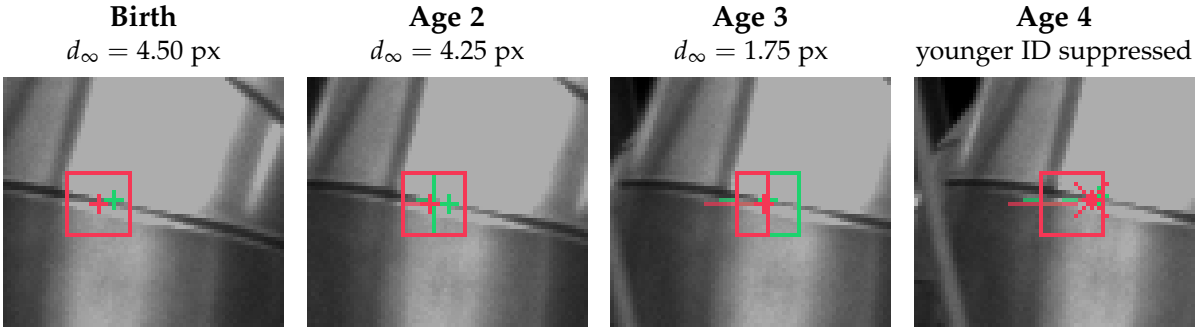


Figure 4.12.: An age-prioritized suppression event recorded at the original 2.25-pixel component-ablation setting on EMH01. The two accepted tracks are distinguishable at birth, approach one another, and end 1.75 pixels apart for two consecutive observations. The older green ID is retained; the crossed magenta ID is omitted once the endpoint and depth-overlap tests both identify the measurements as indistinguishable. The selected PVO setting applies the same rule with a 1.125-pixel bound.

4.9. Rigid-Flow Bundle Adjustment

4.9.1. Observation Model

Conventional bundle adjustment treats an image coordinate $\mathbf{u}_{j,t}^{c}$ as a noisy observation of a projected landmark,

$$\mathbf{e}_{j,t}^{c} = \widehat{\mathbf{u}}_{j,t}^{c} - \mathbf{u}_{j,t}^{c}, \qquad \widehat{\mathbf{u}}_{j,t}^{c} = \pi_c\left(\mathbf{T}_{WC_t}^{-1}\mathbf{p}_j^{W}\right). \tag{4.7}$$

A codec block instead supplies a displacement over an extended prediction region, without identifying a particular scene point within that region. For an anchor-camera track propagated from an initial sample, the stored coordinate after several accepted displacements is

$$\mathbf{u}_{j,t}^{c} = \mathbf{u}_{j,0}^{c} + \sum_{k=1}^{t} \mathbf{m}_{j,k}^{c}. \tag{4.8}$$

Absolute reprojection assigns each resulting coordinate a point-observation residual. An earlier block-selection or displacement error can therefore affect several later residuals. A fixed per-observation variance does not represent this temporal correlation; it does not, however, imply that the coordinates are assumed noise-free.

The evaluated alternative replaces eligible absolute factors with displacement factors. Let s be the predecessor timestamp explicitly recorded for observation (j, t, c). In the evaluated frontend this is the preceding processed frame when the same ID has a stored observation in that camera. It is not reassigned to an older retained keyframe if that predecessor is removed. The shared landmark and two poses predict

$$\Delta\widehat{\mathbf{u}}_{j,t}^{c} = \widehat{\mathbf{u}}_{j,t}^{c} - \widehat{\mathbf{u}}_{j,s}^{c}, \tag{4.9}$$

while the recorded measurement is

$$\mathbf{m}_{j,t}^{c} = \mathbf{u}_{j,t}^{c} - \mathbf{u}_{j,s}^{c}. \tag{4.10}$$

The rigid-flow residual is their difference,

$$\mathbf{r}_{j,t}^{c} = \Delta\widehat{\mathbf{u}}_{j,t}^{c} - \mathbf{m}_{j,t}^{c}. \tag{4.11}$$

Here *rigid flow* means the image displacement predicted by reprojecting one static landmark through two camera poses. It is not an additional measured 3D motion. The factor constrains a difference of projections without attaching either endpoint to an absolute codec sample. Other factors still provide absolute constraints, as detailed below. Shared poses and one persistent landmark state couple the available temporal factors across a track. This construction changes the observation model; it neither makes codec correspondences exact nor removes the possibility of accumulated pose drift.

In the evaluated implementation, the frontend records the predecessor and endpoint difference separately for each camera whenever both same-ID samples are present. Anchor-camera samples are propagated using the selected temporal motion vector. Stereo-camera

samples are obtained from codec-stereo correspondences and expressed in native stereo-camera pixels. Their stored endpoint difference supplies $\mathbf{m}_{j,t}^{S}$; it is not a separately queried temporal vector from the stereo camera. When both camera factors are available, they share a landmark and pose pair but use their respective calibrated projections. Missing flow metadata, including for some backfilled birth observations, leaves an absolute observation instead. Figure 4.13 illustrates the two observation models.

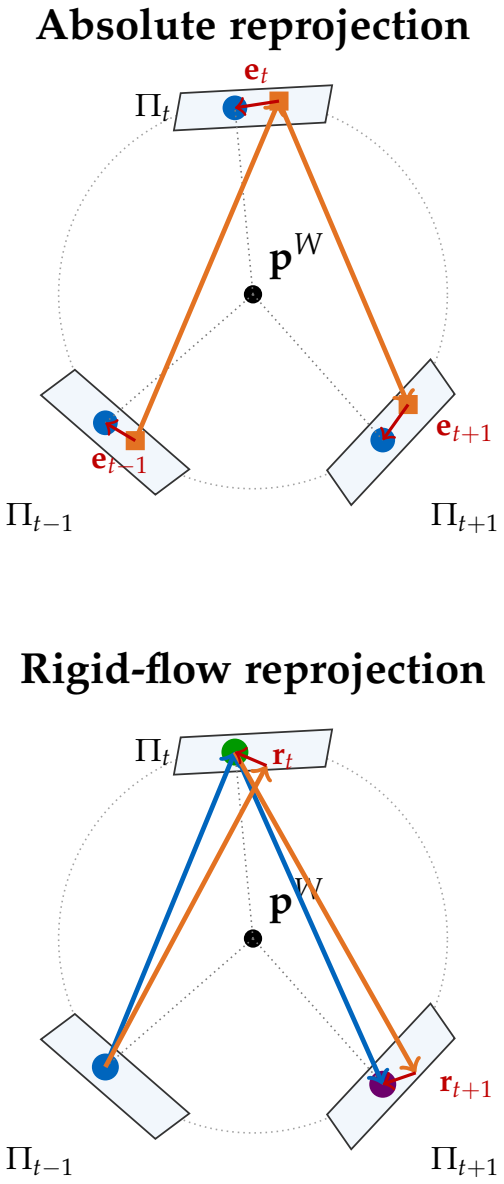


Figure 4.13.: Schematic comparison for one static landmark observed at three consecutive times. Top: absolute factors compare propagated codec samples (orange squares) with individual projections. Bottom: rigid-flow factors compare predicted displacement (blue) with recorded displacement (orange). Red arrows follow the predicted-minus-measured residual convention. Connections between the drawn image planes indicate temporal coupling; displacement subtraction is performed in image-pixel coordinates, not in the page-space geometry of this illustration.

4.9.2. Shared State and Jacobian Assembly

The implementation stores each landmark in its host camera, using two bearing parameters and inverse distance, $\boldsymbol{\ell}_j = (\alpha_j, \beta_j, \rho_j)^{\mathsf{T}}$. The unit bearing $\mathbf{b}(\alpha_j, \beta_j)$ and ρ_j form the homogeneous point

$$\overline{\mathbf{p}}_{h,j}(\boldsymbol{\ell}_j) = \begin{bmatrix} \mathbf{b}(\alpha_j, \beta_j) \\ \rho_j \end{bmatrix}, \qquad \widehat{\mathbf{u}}^c_{j,k} = \pi_c\Big(\mathbf{T}_{kh}\overline{\mathbf{p}}_{h,j}(\boldsymbol{\ell}_j)\Big). \tag{4.12}$$

Here $\mathbf{T}_{kh}$ maps the landmark host camera to camera c at time k; it combines the host and target body poses with the fixed camera extrinsics. Both projections use the same $\boldsymbol{\ell}_j$, rather than independently parameterized points at s and t.

The two predictions are functions of one joint optimization state containing the host pose, predecessor pose, current pose, and shared landmark. Therefore

$$\frac{\partial \mathbf{r}}{\partial \mathbf{x}} = \frac{\partial \widehat{\mathbf{u}}^c_{j,t}}{\partial \mathbf{x}} - \frac{\partial \widehat{\mathbf{u}}^c_{j,s}}{\partial \mathbf{x}}. \tag{4.13}$$

This subtraction requires identical state columns, with zeros for variables absent from a projection. It does not identify the predecessor and current poses as one variable.

Basalt first differentiates a projection with respect to its local host-to-target pose perturbation and the landmark parameters [11]. Denote these derivatives by $\mathbf{D}_k \in \mathbb{R}^{2\times 6}$ and $\mathbf{L}_k \in \mathbb{R}^{2\times 3}$. Let $\mathbf{A}_{kh}$ and $\mathbf{A}_{kk}$ map the absolute host and target body-pose perturbations into that local relative-pose perturbation. For distinct host, predecessor, and current poses, the residual blocks are

$$\begin{aligned} \mathbf{J}_h &= \mathbf{D}_t\mathbf{A}_{th} - \mathbf{D}_s\mathbf{A}_{sh}, & \mathbf{J}_s &= -\mathbf{D}_s\mathbf{A}_{ss}, \\ \mathbf{J}_t &= \mathbf{D}_t\mathbf{A}_{tt}, & \mathbf{J}_\ell &= \mathbf{L}_t - \mathbf{L}_s. \end{aligned} \tag{4.14}$$

Current and predecessor pose terms are placed in separate columns. Subtraction within one block occurs for the shared landmark and, after the chain rule, the shared host pose. If a host and target use the same body pose, their contributions accumulate in the same block. When both timestamp and camera are identical, the relative transform is identity and its pose derivatives are zero.

The backend retains Basalt's pose-linearization convention: the $\mathbf{A}$ maps are evaluated at stored pose linearization points, while the projection transform is updated from the current estimates when those points are held fixed. The local projection derivatives are evaluated using that transform and the current landmark. Thus the block assembly above also applies when marginalization has fixed the pose reference points; it does not imply that every derivative is recomputed at the latest pose estimate.

4.9.3. Factor Allocation and Weighting

Both cameras are enabled for rigid-flow factors in the evaluated configuration. A current observation receives a flow factor only if it is not the landmark's host-camera observation, its finite flow measurement identifies an earlier observation of the same ID in the same camera, and both poses are in the active state ordering. Otherwise, an active observation receives

an absolute factor. The allocation is exclusive: one observation is never represented by both factors in the same joint problem. If a selected flow factor has an invalid projection at either endpoint, it contributes no residual at that linearization; this projection failure does not trigger an absolute fallback. The nonlinear objective and the linear system implement the same residual definition and structural selection rule in separate routines.

For joint-window visual factors, the Huber weight is computed from the two-dimensional residual norm in pixels before scaling by the observation standard deviation. Holding this weight fixed for each linearization gives the weighted residual and Jacobian rows used in the linear system:

$$w_H = \begin{cases} 1, & \|\mathbf{r}\| \le \delta, \\ \delta/\|\mathbf{r}\|, & \|\mathbf{r}\| > \delta, \end{cases} \qquad \widetilde{\mathbf{r}} = \frac{\sqrt{w_H}}{\sigma}\mathbf{r}, \qquad \widetilde{\mathbf{J}} = \frac{\sqrt{w_H}}{\sigma}\mathbf{J}. \tag{4.15}$$

The selected settings are $\sigma = 4$ pixels and $\delta = 0.5$ pixels. Absolute and displacement factors use this same scalar weighting; the model does not introduce a separate flow covariance or temporal error-correlation matrix. The residual ablation changes only this backend factor allocation in joint optimization and marginalization, holding the frontend and other backend settings fixed.

4.9.4. Optimization and Marginalization

After the incoming pose and its observations have been inserted, a pose-only initialization step uses absolute reprojection of already connected landmarks. It performs two Gauss–Newton updates with small damping; its normal equations use unweighted valid projection residuals. The robust weighting described above applies to the subsequent joint optimization and marginalization, not to these initialization updates. All of these coordinates come from codec metadata and calibrated geometry, without consulting image intensities.

A keyframe is requested when the fraction of currently observed anchor-camera IDs connected to landmarks falls below 0.8, provided more than one frame has elapsed since the previous keyframe. New landmarks are triangulated from multi-camera or multi-frame observations. Joint optimization uses absolute square-root QR linearization to eliminate landmark increments, then solves the damped reduced pose system. This is one coupled window problem, not an independent pose fit for each flow measurement.

After optimization, old non-keyframe poses and their observations are removed before the marginalization system is constructed. The most recent pose is retained, so a preceding non-keyframe can still participate in the next joint solve before its removal. After marginalization, the retained state contains at most seven keyframes plus the current pose if it is not already a keyframe; the joint solve can temporarily contain additional poses. The configured three-state limit belongs to the inertial estimator and is inactive here.

Square-root marginalization forms a prior from the existing prior and selected landmarks hosted by departing keyframes or classified as lost. Those landmarks use the same factor-allocation rule on the observations and poses that remain at this stage. Surviving eligible flow factors can therefore enter the prior, but the procedure does not preserve every historical

flow factor: removal of a predecessor can make an observation use absolute reprojection, and there is no replacement flow edge to another retained frame. The separately explored pre-marginalization outlier-removal step and second optimization pass are disabled in the evaluated method.

4.10. Integration and Frame Processing

The retained hybrid policy separates continuation from birth. Established IDs may continue only through reciprocal forward-backward candidates. The larger union of forward and inverted-reverse candidates may propose births, allowing coverage where one direction is preferable, but a new ID remains provisional until a reciprocal observation confirms it in the following frame. The frontend therefore buffers two frames before publishing them to the backend. Once confirmed, the withheld observations for the first two frames and the third observation are emitted with one ID. A provisional track that fails confirmation is discarded. At shutdown, buffered frames are flushed in timestamp order, but unconfirmed births are not promoted.

The per-frame procedure is summarized below.

Algorithm 1: Pixel-free processing of frame t.

Input: temporal and stereo MVs, calibration, masks, tracks, and backend state.
Output: confirmed observations with IDs and updated state.
Normalize MVs; apply source, mask, bounds, and median checks.
Form best-direction birth and reciprocal continuation pools.
Obtain stereo at both endpoints; fit fixed-scale TEASER++ to spatially subsampled 3D pairs.
Retain candidates passing stereo, rigid-consensus, and far-depth gates.
for active IDs j, oldest first, **do**
    Select the reciprocal successor with minimum 3D residual.
    Continue j and record both camera displacements if suppression retains it.
end
Allocate provisional IDs from unused birth candidates surviving suppression.
Confirm provisional IDs after a reciprocal third observation.
Release confirmed observations after the two-frame delay.
For released frames, optimize the current pose and update keyframes and landmarks.
Optimize the active window and marginalize as required.

5. Experimental Methodology

5.1. Research Design

The evaluation separates three forms of evidence. First, Pixelless VO (PVO) is compared with a stationary baseline and two pixel-refined reference configurations. Second, a component ablation holds the stereo frontend, TEASER++ registration, rigid-flow residual, backend settings, and encoded input fixed while enabling reciprocal motion-vector agreement and spatial suppression individually and jointly. Third, offline diagnostics measure track persistence, stereo support, image-structure coincidence, and the conditions associated with large local error. The latter measurements explain the operating envelope but are not inputs to the estimator.

All trajectory tables use full-sequence runs. The reported PVO trajectory uses the operating point selected by the bounded parameter study described below; short smoke tests, abandoned variants, and results from non-equivalent diagnostic builds are excluded. Every reported configuration completed all eleven sequences; a failed or incomplete run would be reported as missing rather than replaced by a numeric value.

5.2. Dataset

Experiments use all eleven sequences of the EuRoC MAV dataset, an established benchmark for visual odometry and SLAM, with its supplied stereo calibration and ground truth [6]. The evaluated sequence aliases, difficulty labels, and frame counts are listed in Table 5.1.

Other relevant benchmarks include TUM-VI [27] and the Monado SLAM Dataset [28]. The choice of EuRoC for the final evaluation is primarily motivated by two practical requirements of the present construction: broad usable coverage after stereo rectification and the availability of depth at both endpoints of temporal correspondences. The encoder-stereo stage adapted from Passthrough+ operates on rectified views [2] and can recover depth only where both cameras observe the scene. EuRoC's nearly parallel camera pair is well suited to the common pinhole projection used here. TUM-VI's wide-angle images and Monado's varied headset camera arrangements require more adaptation to retain useful image support in that projection.

A reanalysis of the archived pilot calibrations quantifies the coverage restriction. Of the calibrated, safe-radius-valid native left-image pixel centres, 97.0 % project inside the chosen rectified image on EuRoC, compared with 51.5 % on TUM-VI and 58.4 % to 78.7 % across the tested Monado camera pairs. These fractions describe native-to-rectified image coverage. Temporal candidates outside this domain cannot obtain stereo depth through the implemented

lookup, irrespective of the availability of temporal motion vectors. Raw image resolution alone does not explain the dataset choice: the Monado Index trials used 960×960 images, exceeding EuRoC's 752×480 resolution.

The preliminary runs also show substantially less usable stereo-temporal support. After the stereo geometric checks, approximately 74.1 % to 74.8 % of temporal candidates had depth at both endpoints on the two easy Machine Hall pilot sequences, compared with about 43 % on TUM-VI room2 and room5 and 6.4 % to 17.3 % on the four tested Monado sequences. Such missing depth prevents candidates from forming the 3D–3D correspondences required by this frontend. These coverage and support limitations motivated concentrating the final feasibility evaluation on EuRoC; further adaptation of rectification and codec-domain stereo to the other camera configurations lies outside its scope. The pilot observations concern the tested configurations and do not establish a general limitation of those datasets or of Passthrough+.

Table 5.1.: EuRoC sequences used by the full-sequence evaluation.

Alias	EuRoC sequence	Difficulty	Frames	Camera rate
EMH01	MH_01_easy	Easy	3682	20 Hz
EMH02	MH_02_easy	Easy	3040	20 Hz
EMH03	MH_03_medium	Medium	2700	20 Hz
EMH04	MH_04_difficult	Difficult	2032	20 Hz
EMH05	MH_05_difficult	Difficult	2273	20 Hz
EV101	V1_01_easy	Easy	2912	20 Hz
EV102	V1_02_medium	Medium	1710	20 Hz
EV103	V1_03_difficult	Difficult	2149	20 Hz
EV201	V2_01_easy	Easy	2280	20 Hz
EV202	V2_02_medium	Medium	2348	20 Hz
EV203	V2_03_difficult	Difficult	1921	20 Hz

Encoded stereo and reverse-temporal streams are aligned by timestamp rather than row number. Where preprocessing produced a one-frame terminal mismatch, the unpaired terminal timestamp was omitted from the longer stream; no timestamp shift or hard common-length truncation was applied.

5.3. Compared Configurations

Stationary Baseline that repeats the initial pose.

TEASER++ only Motion-vector method with TEASER++ and rigid-flow optimization, without forward-backward agreement or non-maximum suppression.

Forward-backward TEASER++-only method with reciprocal agreement and delayed confirmation.

NMS TEASER++-only method with age-prioritized suppression of indistinguishable measurements.

Pixelless VO (PVO) Proposed codec-only method using reciprocal checks, age-prioritized suppression, TEASER++ registration, and rigid-flow optimization.

VOCA MV-first Frozen reference [1]. Codec motion vectors initialize the first KLT pass; tracks not recovered in that pass are retried with unseeded optical flow.

VOCA OF-first Frozen reference. Ordinary KLT optical flow is attempted first; lost tracks are retried with motion-vector initialization.

The stationary baseline and the TEASER++-only estimator are distinct configurations. The VOCA MV-first and OF-first values are taken from their frozen full-sequence evaluations; diagnostic runs are not substituted for those trajectories.

5.4. Encoding and Preprocessing Configuration

All streams are encoded with FFmpeg 4.4.2 and `libx264` in the H.264 High profile with `yuv420p` output. The enabled partitions are `p8x8`, `p4x4`, and `i8x8`. For the temporal streams, the remaining x264 options are `keyint=1000`, `me=umh`, `merange=64`, `subme=6`, `bframes=0`, and `ref=1`. The encoder uses two-pass rate control at $500\,\mathrm{kbit\,s^{-1}}$. Images listed by each camera's timestamp CSV are supplied through FFmpeg's concat demuxer and encoded at 30 frames per second. This encoded rate controls the container timestamps but does not replace the EuRoC nanosecond timestamps used by the estimator and evaluation. H.264 fractional motion is preserved using each vector's reported numerator and scale; no integer rounding is applied.

For every camera, a reverse-temporal stream is made by reversing the same ordered image list and applying the identical temporal encoder settings. The extracted reverse vectors are subsequently inverted and remapped to the corresponding forward-time EuRoC timestamps. Forward and reverse files are required to contain the same number of frames as the synchronized dataset metadata. The single-reference configuration limits prediction dependencies within each encoded stream, but two-pass rate control and reversal of complete sequences make this preparation offline. No end-to-end real-time claim follows from `ref=1` or the absence of B-frames.

Stereo preprocessing first applies the calibration-dependent common projection described in chapter 4. It then encodes twelve independent full-frame stereo streams: disparity shifts 0, 8, and 32 pixels crossed with grid phases $(0,0)$, $(4,0)$, $(0,4)$, and $(4,4)$ pixels. Each source stereo pair is expanded to an I-P-P triple and encoded at 90 frames per second with CRF 23. The temporal motion-search and partition settings are retained, while `keyint=3`, `min-keyint=3`, `scenecut=0`, `bframes=0`, and `ref=1` enforce independent triples. The I-P-P reference structure and shift strategy are adapted from Passthrough+ [2]; the mosaic layout, hardware encoder, and subsequent depth processing are not reproduced.

Preprocessing verifies the frame type of every triple, equal timestamp order for the stereo pair, exact forward/reverse frame counts, the presence of all twelve shift combinations, and

non-empty validity masks. The full command templates are given in Appendix A; the immutable evaluation manifest also records the exact FFmpeg version and hashes of the configurations used.

5.5. Estimator Configuration

All evaluated estimators are visual-only: IMU input and real-time enforcement are disabled. The proposed method uses a calibrated anchor/stereo pair, an absolute square-root QR linearization, square-root marginalization, at most seven retained keyframes plus the current pose after marginalization, and a keyframe connection-ratio threshold of 0.8. With the minimum-frames-after-keyframe parameter set to 1 and a strict counter test, a new keyframe can be requested only after more than one frame has elapsed. The Levenberg–Marquardt iteration-limit parameter is set to 7, with an initial damping value of 10^{-4}. Joint-window visual factors use an observation standard deviation of 4 pixels and a Huber threshold of 0.5 pixels, applied to the residual norm before normalization. Rigid-flow factors are enabled for eligible observations in both the anchor and stereo cameras; host and ineligible observations retain absolute factors. Standard marginalization and lost-landmark removal remain active; the additional pre-marginalization filtering and second optimization pass evaluated in earlier experiments remain disabled.

Table 5.2 lists the selected PVO frontend thresholds. Before TEASER++ estimation, at most one stereo-temporal pair per 40×40-pixel cell is retained using the stereo matching cost. TEASER++ uses fixed scale, a 0.1 m noise bound, GNC-TLS rotation estimation, and heuristic maximum-clique inlier selection [16]. PVO enables the hybrid reciprocal-confirmation policy and indistinguishable-measurement suppression. At the original operating point, the four component-ablation configurations differ only in these two switches; their joint use is labelled "Joint frontend." PVO uses the same complete method with selected frontend thresholds and backend weights.

The numerical values above define the selected PVO operating point. A bounded twelve-candidate study varied reciprocal-agreement gates, suppression distance, Huber threshold, and observation standard deviation on all eleven sequences. The selected point gave the strongest typical-sequence trade-off: it had the lowest median ATE in the original sweep and remained preferable to the local variants around it. One local variant reduced median RTE-6 slightly but greatly enlarged the EV103 failure. Since selection and reporting use the same benchmark, the result is an in-sample operating point rather than a held-out estimate of generalization. The evaluated executable and configuration are identified by SHA-256 hashes in Appendix A.

Table 5.2.: Selected PVO compressed-domain frontend parameters.

Parameter	Value
Median-filter grid / distance threshold	16 px / 5 px
Stereo reprojection gate	1.5 px
Stereo epipolar gate	0.005
Valid stereo depth interval	$(0.333, 30]$ m
TEASER++ noise bound	0.1 m
Far-depth threshold / reprojection gate	5 m / 1.5 px
Reciprocal cosine / minimum IoU	0.98 / 0.15
Reciprocal cycle-error gate	1.5 px
Single-endpoint tolerance in suppression model	0.5625 px
Suppression endpoint distance	1.125 px in L_∞
Confirmation age	3 observations

5.6. Metrics and Aggregation

Absolute trajectory error (ATE) is reported in metres after a single rigid $\mathrm{SE}(3)$ alignment between the estimated and reference trajectories [18, 19]. Relative translation error (RTE-6) is the translation error in metres over non-overlapping six-frame intervals [19]. At the nominal camera rate this corresponds to 0.3 s. Both per-sequence values are root mean square errors (RMSE), computed over associated positions for ATE and relative-pose intervals for RTE-6. ATE measures global trajectory consistency, whereas RTE-6 measures short-horizon drift; a trajectory may therefore have small local errors that accumulate into a larger aligned ATE.

The main tables report every sequence. Across-sequence summaries use the macro mean and median so that long sequences do not receive additional weight. The empirical cumulative distribution and the number of sequences completed are reported alongside these summaries because one failed sequence can dominate an arithmetic mean.

Qualitative trajectory plots use the same single three-dimensional alignment as ATE, with scale fixed to one. They show the saved estimator-origin positions in two orthogonal projections, with equal metric axes and no separate fitting of the projections. Every saved estimate is retained, including boundary poses without a valid ground-truth association; available ground truth is shown over the saved time span. These views reveal path geometry but not the timing or cause of a failure, which requires the episode analysis below.

5.7. Ablation Studies

The component study contains four configurations at the original operating point: TEASER++ only, reciprocal forward-backward agreement only, age-prioritized non-maximum suppression only, and their joint use. Their encoded inputs, stereo processing, TEASER++ configuration, rigid-flow residual, backend settings, and evaluated timestamps are identical. “Joint frontend”

denotes the complete method at the original parameter settings. "PVO" denotes the same method after selecting reciprocal-agreement thresholds, suppression distance, observation standard deviation, and Huber threshold. PVO is reported alongside the four controlled configurations as a parameter-selected variant of the complete method, rather than an additional component-ablation cell.

A separate paired experiment isolates the backend observation model at the selected PVO operating point. It keeps the frontend, encoded inputs, stereo associations, robust registration, window, and loss parameters fixed, and changes only whether eligible temporal observations use conventional absolute reprojection factors or the two-camera rigid-flow factors of Equation 4.11. This comparison is reported separately from the frontend component ablation because it evaluates a different interface in the estimator.

5.8. Track Characteristics

Track length is the number of frames in which one confirmed landmark ID is observed. Because admission is delayed until a third observation, provisional two-frame candidates that subsequently fail are counted separately rather than entered into the confirmed-track histogram. Results report the full distribution and survival proportions in addition to the mean, since a small long-lived tail can be hidden by a large number of short tracks. Track length is kept separate from trajectory error so that multiview support remains visible even when aggregate pose metrics appear similar.

Spatial support is evaluated offline and is never supplied to the estimator. Shi-Tomasi corners are detected with quality level 0.01, minimum distance 7, and block size 7. An observation is marked as structurally coincident when its motion-vector support rectangle contains at least one detected corner. The analysis reports both the fraction of codec regions containing a corner and the fraction of corners covered by their union. Raw temporal vectors, candidates with valid stereo depth at both endpoints, reciprocal candidates, TEASER++ survivors, and suppression-retained candidates are reported separately. The last row also includes newly selected best-direction births, which remain provisional until reciprocal confirmation; it is therefore not a strict subset of the preceding reciprocal-consensus row. Quality levels 0.05 and 0.10 test whether the result is retained for stronger corner responses. These measurements are post-hoc diagnostics of the selected PVO run and do not affect its trajectory.

Frontend counters are normalized by the evaluated frame count. Stereo support is reported as the fraction of queries that return a valid match, together with the availability of depth at consecutive track endpoints. The reported PVO track and depth distributions are computed from the selected PVO runs. Track statistics from reference frontends are used only where their trajectory reproduces the corresponding frozen reference result; non-equivalent diagnostic replays are excluded.

5.9. Failure Analysis

The saved full-sequence trajectories are inspected for large pose updates, sustained local error, stationary outputs during camera motion, and recovery. Each selected episode is examined with its preceding and subsequent frames. Interval boundaries follow the event being examined; they do not divide the sequences into equal time or distance bins. The selection is retrospective and illustrative. Contrasting intervals are included to test whether a proposed explanation also fits successful motion or fails in a different way.

Local comparisons retain the non-overlapping six-frame intervals used by RTE-6. Their actual timestamps are used, since missing pose samples can change the duration of a six-frame interval. Adjacent-pose translation and rotation are also inspected to separate a single discontinuity from sustained error. Accumulated error after aligning the first associated pose illustrates what remains after local recovery; it is explicitly distinguished from the whole-trajectory alignment used for ATE. The archived absolute-residual ablation and VOCA OF-first reference are compared over identical timestamps after checking their complete trajectories against the frozen metrics.

Timestamped diagnostics provide raw and selected candidate counts, emitted frontend IDs, IDs connected to existing landmarks, and optimization iterations. The connected count precedes creation of new landmarks. A stereo-valid candidate has passed the preceding filters and has depth at both temporal endpoints; a zero count at this stage does not isolate which preceding check failed. Eligibility for a rigid-flow factor likewise does not measure the number of independent geometric constraints. Full-sequence track lifetimes, query-success rates, and mean consecutive depth changes remain sequence-level summaries and are not attributed to individual events without timestamped evidence.

Raw images, Shi-Tomasi corners, and an inverse-Laplacian sharpness proxy are used only for offline inspection. The sharpness proxy depends on texture and illumination as well as blur, so it is not treated as a direct measurement of motion blur. Stream frame types are checked where an interruption of temporal MVs is observed. These measurements help locate changes in the input and processing chain, while the interpretation separates observed events from possible causes that the recorded data cannot resolve.

5.10. Reproducibility

The PVO implementation is based on commit `29d09aa0b14e`. The selected evaluation executable and selected configuration are identified by hashes because the suppression distance was exposed as an evaluation-time parameter after that base revision. The evaluation manifest records these hashes, the FFmpeg and Python versions, the eleven canonical dataset identifiers, and the commands used to regenerate the tables and plots. Intermediate data are retained outside the thesis source; the PDF figures included here are immutable exports of that bundle. Missing values are not imputed, and results from binaries that fail the selected-trajectory consistency check are omitted.

6. Results

6.1. Overall EuRoC Results

All reported configurations completed all eleven EuRoC sequences. The full-sequence ATE and RTE-6 results are given in Table 6.1 and Table 6.2. The VOCA columns use their frozen evaluations rather than later instrumented diagnostic runs.

PVO improves on the stationary trajectory in both ATE and RTE-6 on nine of eleven sequences. It therefore recovers camera motion over most of the benchmark, but EV103 and EV203 show that the method is not consistently better than predicting no motion. Its macro ATE is 3.051 m and its median ATE is 1.082 m; its macro RTE-6 is 0.2934 m and its median RTE-6 is 0.0818 m. The gap between mean and median is dominated by the EV103 outlier.

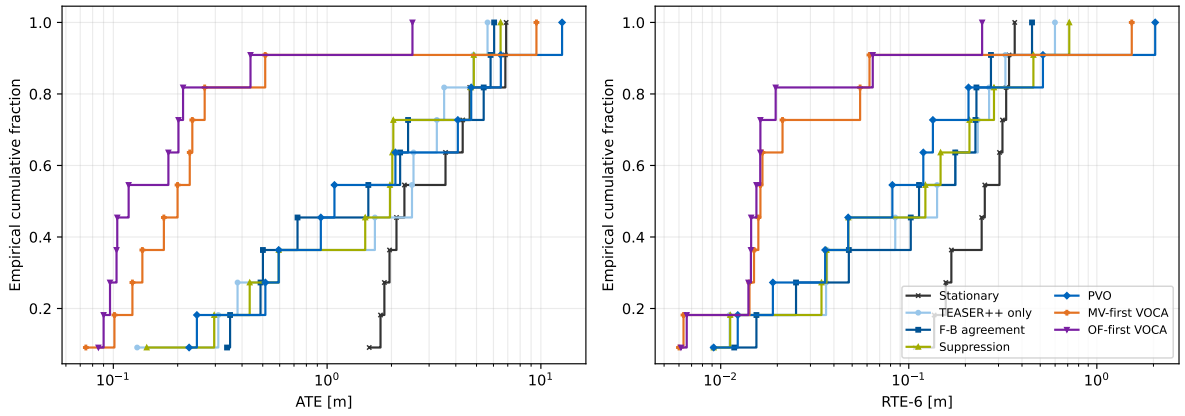


Figure 6.1.: Empirical cumulative distributions over the eleven full-sequence ATE values (left) and RTE-6 values (right). The logarithmic horizontal axes retain the effect of both low-error sequences and large failures.

Table 6.1.: Aligned absolute trajectory error (ATE) in metres on all EuRoC sequences. Lower is better; bold denotes the lowest value in a row.

Sequence	Stationary	PVO	VOCA MV-first	VOCA OF-first
EMH01	4.308	0.246	0.172	**0.118**
EMH02	4.646	0.594	0.123	**0.104**
EMH03	3.580	1.082	**0.200**	0.212
EMH04	6.821	4.722	0.234	**0.202**
EMH05	6.879	4.089	0.228	**0.181**
EV101	1.853	0.226	0.101	**0.096**
EV102	1.778	0.514	**0.074**	0.085
EV103	1.574	12.565	0.513	**0.438**
EV201	2.299	0.935	0.267	**0.090**
EV202	2.111	2.087	0.137	**0.104**
EV203	**1.959**	6.497	9.532	2.508

Table 6.2.: Relative translation error over non-overlapping six-frame intervals (RTE-6), in metres. Lower is better; bold denotes the lowest value in a row.

Sequence	Stationary	PVO	VOCA MV-first	VOCA OF-first
EMH01	0.1573	0.0092	**0.0063**	0.0066
EMH02	0.1685	0.0123	**0.0060**	0.0061
EMH03	0.3659	0.0476	0.0150	**0.0145**
EMH04	0.3414	0.1344	0.0213	**0.0196**
EMH05	0.3149	0.1195	**0.0159**	0.0162
EV101	0.1365	0.0189	0.0142	**0.0140**
EV102	0.3037	0.0359	0.0163	**0.0155**
EV103	0.2536	2.0434	**0.0618**	0.0643
EV201	0.1084	0.0818	0.0551	**0.0163**
EV202	0.2441	0.2078	0.0167	**0.0145**
EV203	0.3296	0.5169	1.5336	**0.2455**

6.1.1. Trajectory Geometry

The trajectory overlays illustrate the central feasibility result: motion vectors alone can support reasonable camera-motion tracking. The easy Machine Hall sequences give especially clear examples: Figure 6.2 and Figure 6.3 show close agreement with the reference paths on EMH01 and EMH02. The medium EMH03 sequence and EV201 follow in Figure 6.4 and Figure 6.5. Their codec-only estimates recover the broad shape of the reference route, with visible deviations in position and vertical motion. Each method is shown separately against ground truth, with common axes for comparison within a sequence.

These results demonstrate one way to recover useful trajectory geometry from codec motion vectors and motivate further investigation of pixel-free VO. Full-sequence PVO overlays for all eleven sequences, including failure cases, are provided in section B.2; the temporal evidence for failures is examined in section 6.8.

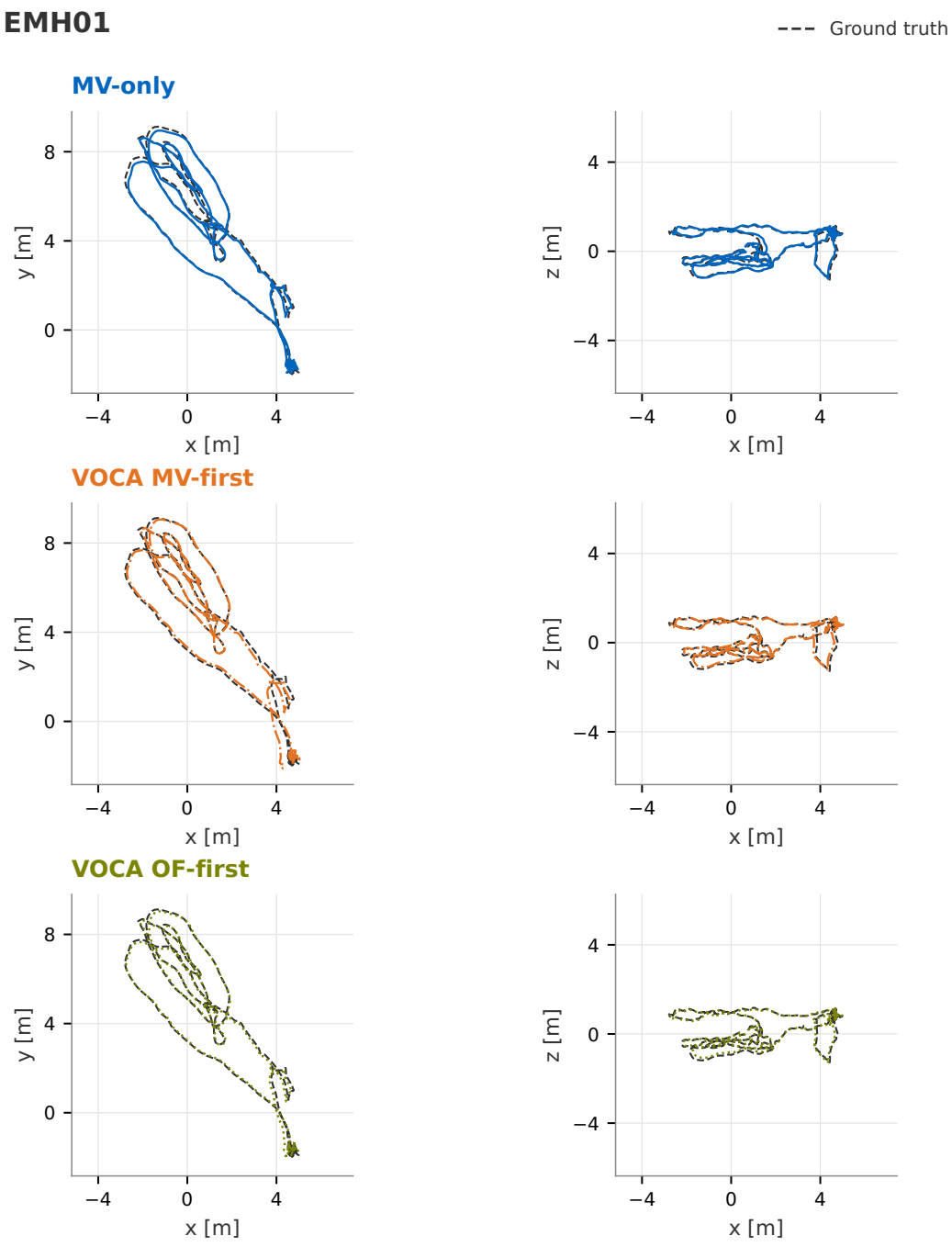


Figure 6.2.: Full-sequence EMH01 trajectories. Top to bottom: MV-only, VOCA MV-first, and VOCA OF-first, each with ground truth. Left: *xy*; right: *xz*. All rows share the same spatial limits and metric scale. Each complete estimate uses its rigid alignment for ATE, shared between projections, without scale correction. All saved estimates are shown.

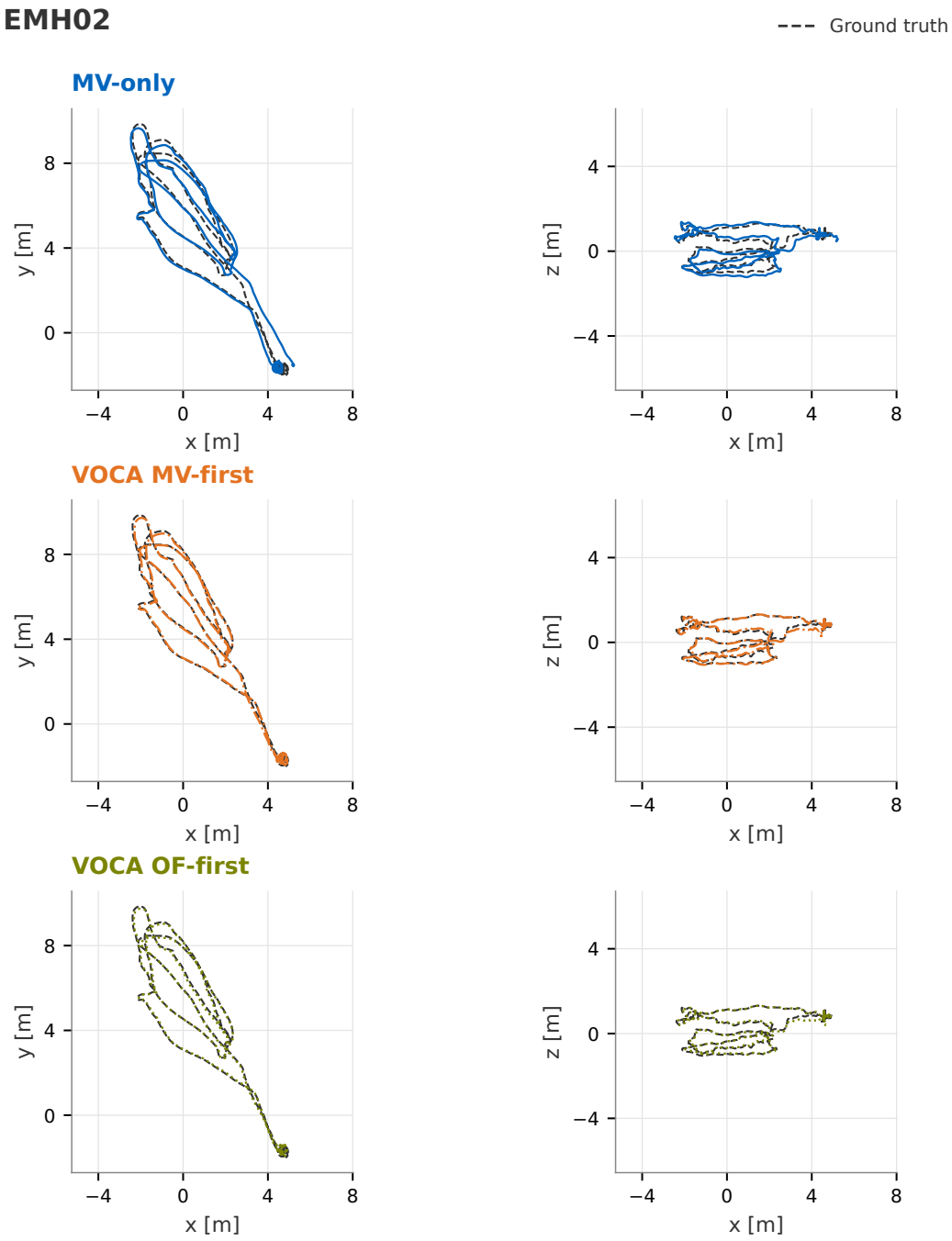


Figure 6.3.: Full-sequence EMH02 trajectories. Top to bottom: MV-only, VOCA MV-first, and VOCA OF-first, each with ground truth. Left: xy; right: xz. Alignment and plotting conventions follow Figure 6.2.

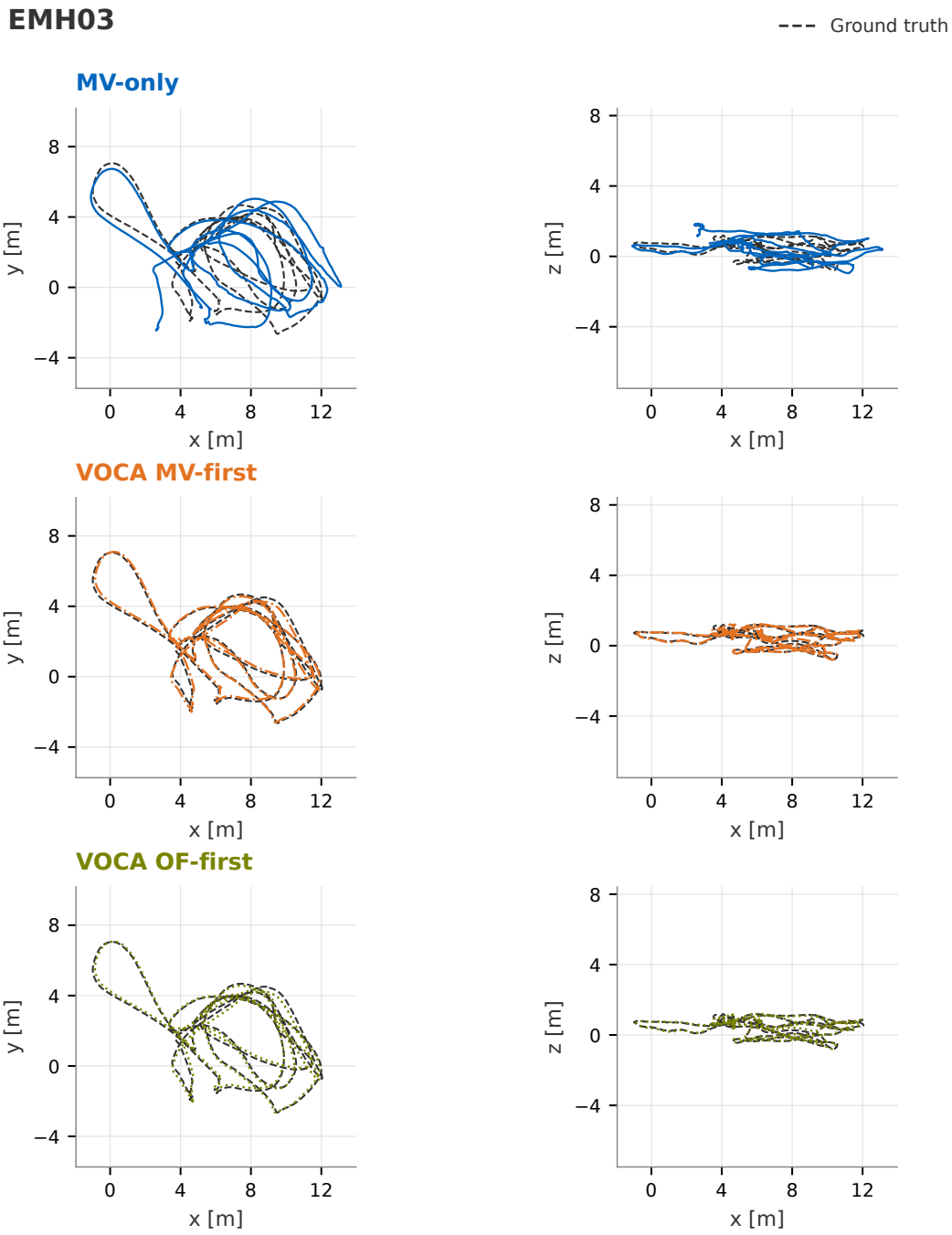


Figure 6.4.: Full-sequence EMH03 trajectories. Top to bottom: MV-only, VOCA MV-first, and VOCA OF-first, each with ground truth. Left: xy; right: xz. Alignment and plotting conventions follow Figure 6.2.

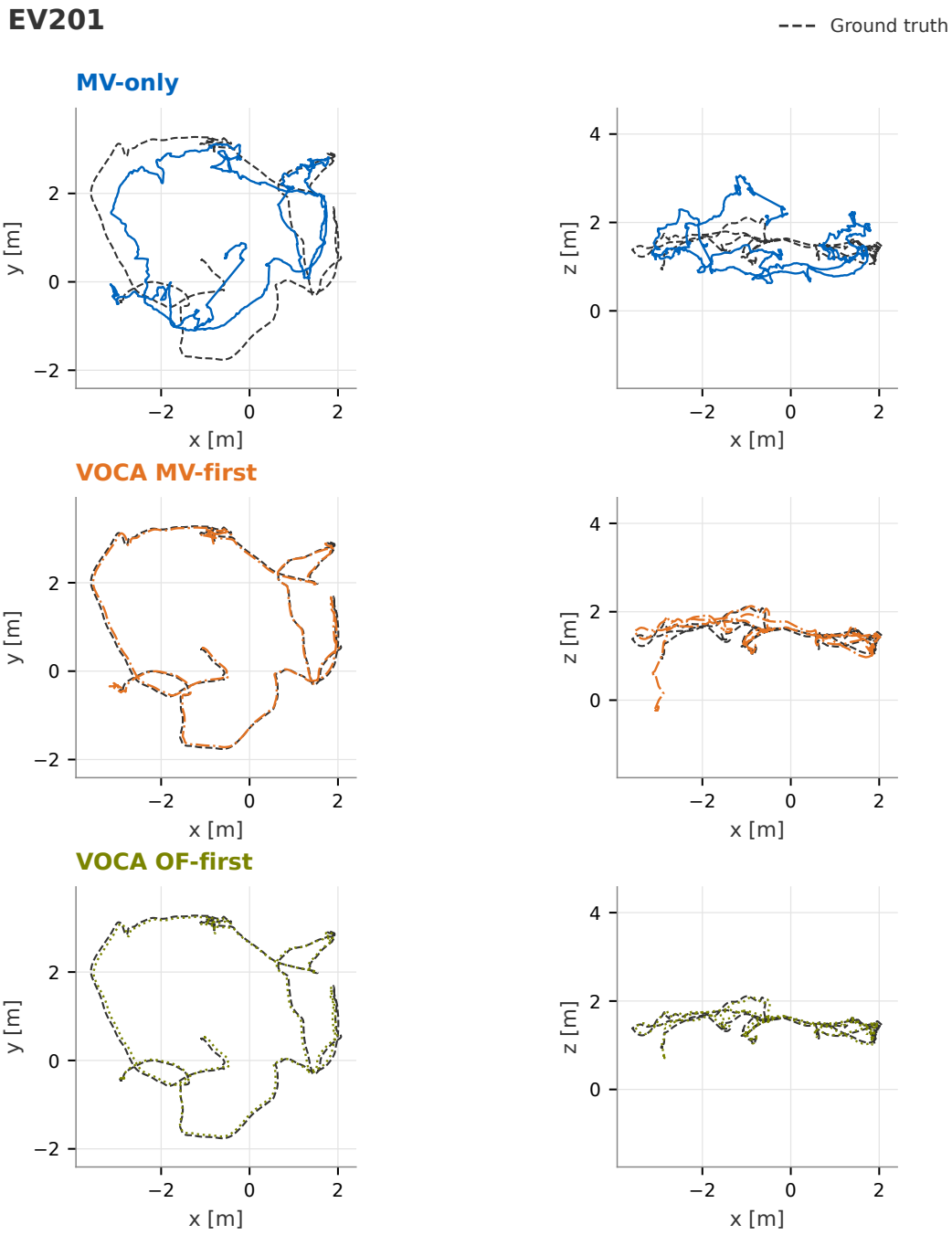


Figure 6.5.: Full-sequence EV201 trajectories. Top to bottom: MV-only, VOCA MV-first, and VOCA OF-first, each with ground truth. Left: *xy*; right: *xz*. Alignment and plotting conventions follow Figure 6.2.

6.2. Results by Sequence Difficulty

For PVO, the easy sequences have a mean ATE of 0.500 m and a median of 0.420 m. The corresponding values are 1.228 m and 1.082 m for the medium sequences, and 6.968 m and 5.609 m for the difficult sequences. Mean RTE-6 rises from 0.0306 m on easy sequences to 0.0971 m on medium and 0.7035 m on difficult sequences.

These labels are descriptive groups, not uniformly spaced levels of numeric difficulty. EV201, for example, is labelled easy but its codec-only ATE is higher than that of the other three easy sequences. Conversely, local RTE-6 remains relatively small on some trajectories whose global ATE is poor.

Table 6.3.: Trajectory and track statistics of PVO grouped by the EuRoC difficulty labels. Observations/frame counts confirmed anchor-camera observations pooled over input frames, not new identities or connected backend landmarks. Lifetimes are pooled over tracks; trajectory means and medians weight sequences equally.

	ATE [m]		RTE-6 [m]			Track lifetime		
Difficulty	Mean	Median	Mean	Median	Obs./frame	Mean [frames]	≥ 5 [%]	Maximum
Easy	0.500	0.420	0.0306	0.0156	998.21	3.521	11.04	204
Medium	1.228	1.082	0.0971	0.0476	475.11	3.403	8.11	77
Difficult	6.968	5.609	0.7035	0.3256	370.99	3.415	7.67	82

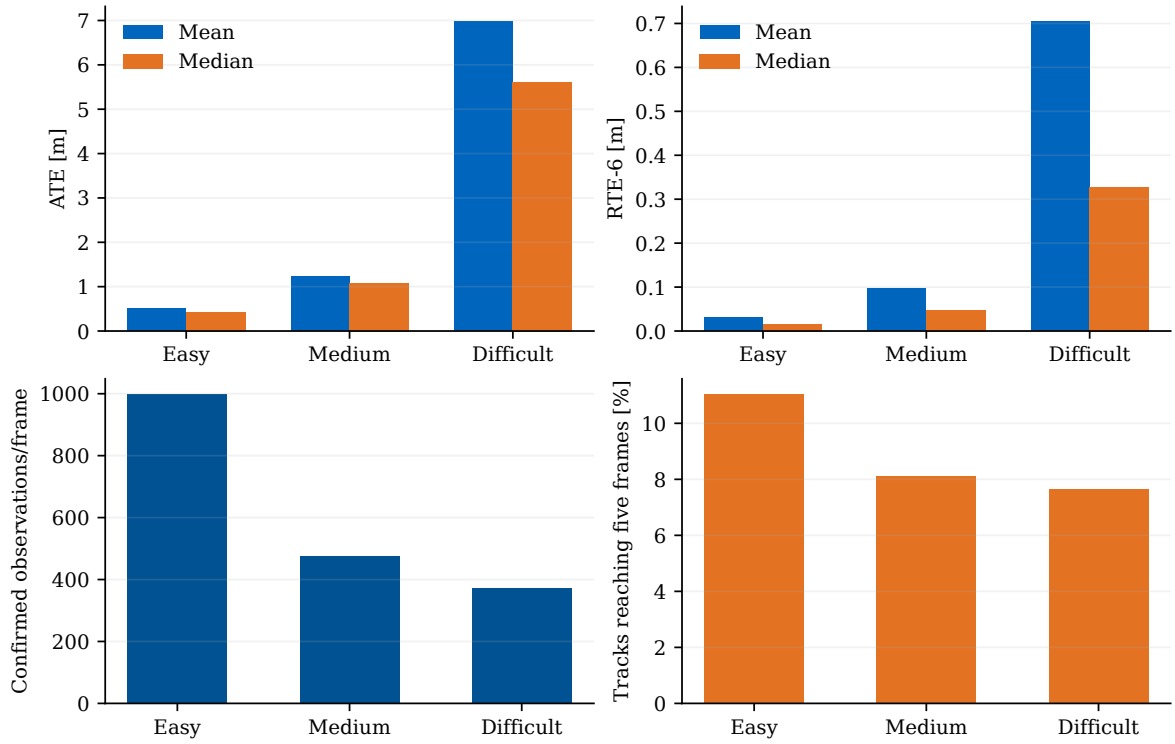


Figure 6.6.: PVO trajectory errors and track support separated by EuRoC difficulty. Confirmed observation density decreases across the groups, while all three remain dominated by short tracks. The lifetime panel reports the proportion reaching at least five frames.

6.3. Component Ablation

Table 6.4 and Table 6.5 give the full-sequence component results at the original operating point and the selected PVO result. Bold marks the lowest codec-only value in each row. "Joint frontend" uses the complete method at the original settings; PVO uses the same components with selected reciprocal gates, suppression distance, observation standard deviation, and Huber threshold. PVO is included for comparison but is not another cell of the controlled component ablation.

Table 6.4.: Full-sequence ATE for the codec-only component ablation, in metres.

Sequence	TEASER++ only	F-B agreement	Suppression	Joint frontend	PVO
EMH01	**0.129**	0.488	0.143	0.232	0.246
EMH02	0.310	0.352	0.296	**0.203**	0.594
EMH03	3.261	1.559	1.508	1.569	**1.082**
EMH04	**3.527**	6.045	4.856	5.672	4.722
EMH05	5.629	5.401	4.706	5.318	**4.089**
EV101	0.381	0.340	0.434	**0.223**	0.226
EV102	0.596	**0.501**	0.594	0.622	0.514
EV103	2.536	2.196	**2.019**	3.406	12.565
EV201	1.673	**0.727**	1.968	0.899	0.935
EV202	2.495	2.392	**2.040**	2.671	2.087
EV203	**4.842**	5.817	6.479	7.635	6.497
Macro mean	2.307	2.347	**2.277**	2.586	3.051
Median	2.495	1.559	1.968	1.569	**1.082**

Table 6.5.: Full-sequence RTE-6 for the codec-only component ablation, in metres.

Sequence	TEASER++ only	F-B agreement	Suppression	Joint frontend	PVO
EMH01	**0.0091**	0.0118	0.0092	0.0115	0.0092
EMH02	**0.0112**	0.0155	0.0112	0.0159	0.0123
EMH03	0.0848	0.1025	0.0477	0.0978	**0.0476**
EMH04	0.1779	0.1767	0.1476	0.1488	**0.1344**
EMH05	0.1414	0.2290	0.1226	0.1365	**0.1195**
EV101	0.0363	0.0252	0.0343	0.0190	**0.0189**
EV102	0.0368	0.0480	0.0365	0.0494	**0.0359**
EV103	0.3264	**0.2262**	0.2839	0.2585	2.0434
EV201	0.2674	0.1134	0.4599	0.1005	**0.0818**
EV202	0.2328	0.2735	0.2102	0.2564	**0.2078**
EV203	0.5989	0.4523	0.7135	**0.4365**	0.5169
Macro mean	0.1748	0.1522	0.1888	**0.1392**	0.2934
Median	0.1414	0.1134	0.1226	0.1005	**0.0818**

PVO has the lowest median ATE and RTE-6 and the lowest RTE-6 on seven sequences. Suppression at the original operating point retains the lowest macro mean ATE, while the joint frontend at that point retains the lowest macro mean RTE-6. The PVO macro means are dominated by the EV103 failure. Relative to TEASER++ only, PVO lowers ATE on six sequences and RTE-6 on eight. The sequence rows make the non-monotonic effect explicit: the selected parameters improve typical local drift but do not eliminate large failures.

6.4. Observation-Model Ablation

The paired comparison in Table 6.6 uses the complete selected PVO frontend and changes only the backend observation factor. The rigid-flow residual has lower ATE on eight of eleven sequences and lower RTE-6 on seven. It also lowers both medians. The macro means favour absolute reprojection because the rigid-flow run fails strongly on EV103, and absolute reprojection is also better on EV203. The displacement formulation therefore improves the typical sequence in this experiment but does not remove the method's sequence-level failure modes.

Table 6.6.: Full-sequence ablation of the observation model at the selected PVO operating point. All encoded inputs, frontend decisions, robust associations, loss parameters, and window settings are shared. Lower is better; bold marks the lower value within each metric.

	ATE [m]		RTE-6 [m]	
Sequence	Absolute	Rigid flow	Absolute	Rigid flow
EMH01	0.253	**0.246**	**0.0089**	0.0092
EMH02	**0.482**	0.594	0.0207	**0.0123**
EMH03	1.215	**1.082**	0.0488	**0.0476**
EMH04	4.960	**4.722**	**0.1294**	0.1344
EMH05	4.266	**4.089**	0.1279	**0.1195**
EV101	0.263	**0.226**	0.0219	**0.0189**
EV102	0.531	**0.514**	0.0393	**0.0359**
EV103	**2.538**	12.565	**0.2910**	2.0434
EV201	1.152	**0.935**	0.0824	**0.0818**
EV202	2.281	**2.087**	0.2621	**0.2078**
EV203	**4.565**	6.497	**0.4264**	0.5169
Macro mean	**2.046**	3.051	**0.1326**	0.2934
Median	1.215	**1.082**	0.0824	**0.0818**

6.5. Track Length and Geometric Support

Across all eleven sequences, PVO produces 5,230,916 confirmed tracks. Their mean length is 3.48 frames, the median is 3, the 90th percentile is 4, the 95th percentile is 5, and the longest track spans 204 frames. Only 29.85% survive for at least four frames, 9.92% for at least five, 0.25% for at least ten, and 0.05% for at least twenty. The method therefore creates persistent identities, but the typical constraint has little multiview support.

No quantitative track-length comparison with the VOCA MV-first and OF-first references is reported. The necessary lifetime instrumentation was evaluated in a separate diagnostic build that did not reproduce the frozen reference trajectories: its ATE on the first easy Machine Hall sequence differed from both reference values. Retaining those track statistics would therefore mix different operating points. The trajectory comparison in Table 6.1 and Table 6.2 remains valid because it uses the archived frozen reference results directly.

The stereo-query yield also varies substantially by sequence. In PVO it is 13.48% on EMH01, 13.56% on EMH02, and 19.88% on EV101, but falls to 1.99%, 1.14%, and 0.71% on EV201 to EV203. Among observations with depth at consecutive endpoints, the mean absolute depth change ranges from 0.0262 m to 0.0424 m. This conditional statistic does not measure the larger population for which one or both depths are absent.

Figure 6.7 retains the individual sequence measurements alongside the signed RTE-6 difference, PVO minus VOCA MV-first. The plot describes which support measurements accompany each complete trajectory; it does not locate a failure or identify its cause. In particular, similar full-sequence track lifetimes can conceal a brief interruption or an extended period of incorrect motion. section 6.8 examines those events in their temporal and visual context.

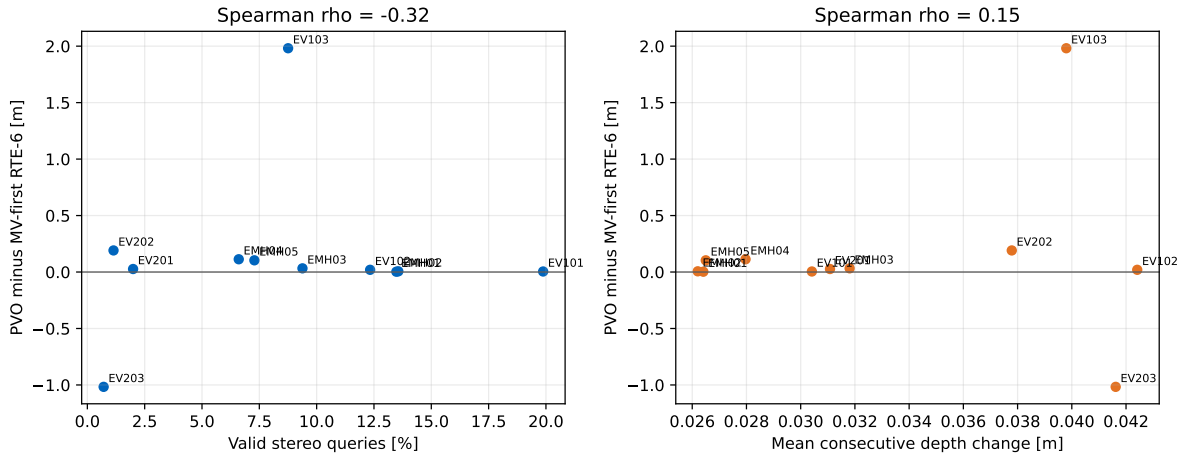


Figure 6.7.: Valid-stereo rate (left) and mean absolute consecutive depth change (right) against the signed RTE-6 difference, PVO minus VOCA MV-first. Positive vertical values indicate greater PVO error. Labels identify the eleven sequence aggregates.

6.6. Coincidence with Conventional Image Structure

The error-surface analysis in Equation 2.8 provides the reason for measuring this coincidence. Corner-like structure gives a block matching cost curvature in two displacement directions, whereas an edge or weakly textured support leaves at least one direction ambiguous. The following test therefore measures whether filtering concentrates motion-vector support near image structure that can condition translational matching; it does not label a motion vector as geometrically correct.

Decoded images are used only in this offline diagnostic. The estimator itself does not receive the Shi-Tomasi responses reported in Table 6.7. A motion-vector observation is counted as structurally coincident when its exact prediction-block rectangle contains at least one Shi-Tomasi corner. The rows give an ordered processing summary: raw anchor-camera temporal vectors, vectors with valid stereo depth at both temporal endpoints, reciprocal forward-backward candidates, TEASER++ survivors, and candidates retained after age-prioritized suppression. The retained row also includes new best-direction births awaiting reciprocal confirmation, so it branches from the strict reciprocal path rather than forming its final subset.

Table 6.7.: Pooled full-sequence Shi-Tomasi coincidence across all eleven EuRoC sequences. Region support is the fraction of codec regions containing a corner; feature coverage is the fraction of detected corners lying in the union of those regions.

Stage	Regions	Regions with corner	Corner coverage
Raw temporal MVs	84,675,992	30.59%	98.58%
Valid stereo depth	41,353,155	37.45%	59.88%
Forward-backward candidates	13,899,680	34.38%	32.48%
TEASER++ survivors	6,856,688	39.57%	18.70%
Suppression-retained candidates	23,057,616	44.54%	43.03%

Requiring stereo depth raises the region-level corner ratio from 30.59% to 37.45%. Contrary to a monotonic-enrichment hypothesis, reciprocal agreement then lowers it to 34.38%. TEASER++ raises it to 39.57%, while the branched suppression-retained pool reaches 44.54%. The first four candidate counts fall at every stage, so TEASER++ is both selective and more concentrated near Shi-Tomasi structure than its reciprocal input. The feature-coverage column is reported separately because the dense raw block union covers nearly every corner even when individual regions are poorly localized.

Stricter Shi-Tomasi quality levels test whether the coincidence is limited to weak corner responses. The same final codec supports are held fixed while only the offline diagnostic threshold changes.

At the default quality level, 44.54% of retained regions contain a corner and their union covers 43.03% of the detected corners. As the diagnostic retains only stronger corners, region coincidence falls because fewer responses exist, but coverage of those responses rises to 51.75%. This does not imply that a retained codec region is a uniquely localized feature or that the estimator observes the corner response.

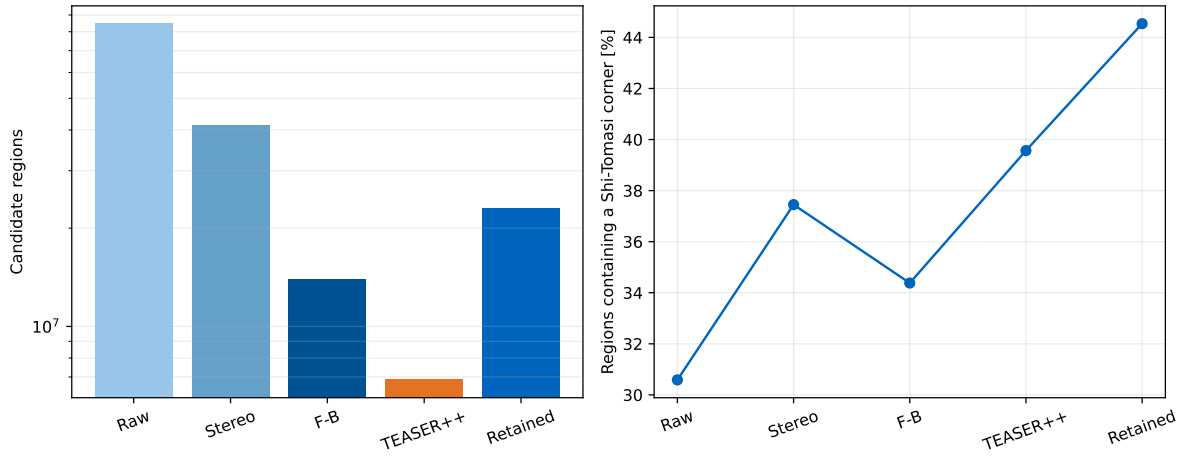


Figure 6.8.: Candidate count and structural coincidence through the codec-only selection stages, pooled over all eleven sequences.

Table 6.8.: Pooled coincidence of final codec supports with Shi-Tomasi corners at increasingly selective quality levels.

Quality level	Corners	Regions with corner	Corner coverage
0.01	18,921,761	44.54%	43.03%
0.05	8,365,939	25.51%	48.39%
0.10	5,076,456	17.64%	51.75%

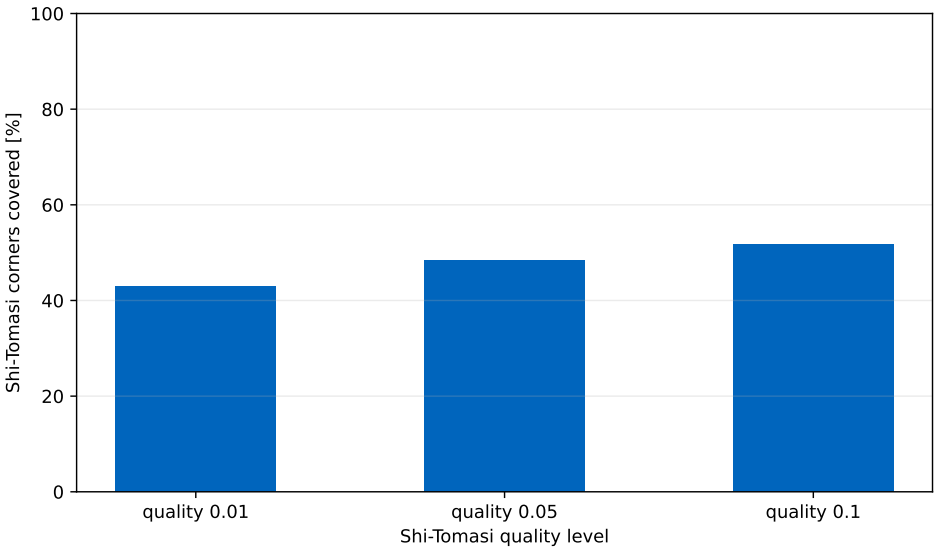


Figure 6.9.: Pooled Shi-Tomasi corner coverage by the fixed final codec supports at three detector quality levels.

6.7. Reference Configurations

The MV-first and OF-first VOCA configurations have access to image intensities that are deliberately unavailable to the proposed estimator. They therefore serve as higher-information operating points rather than objectives that the PVO is expected to surpass. Their median ATE values are 0.200 m and 0.118 m, respectively, compared with 1.082 m for PVO. Median RTE-6 is 0.0163 m for VOCA MV-first, 0.0155 m for VOCA OF-first, and 0.0818 m for PVO.

The macro mean alone obscures reference failures: VOCA MV-first reaches 9.532 m ATE and 1.534 m RTE-6 on EV203. The per-sequence tables and empirical distributions are therefore retained rather than reducing the benchmark to a single ranking. The central result is that codec side information alone supports non-stationary estimates on a substantial part of the benchmark, while a large and sequence-dependent accuracy gap remains.

6.8. Failure Episodes

The full-sequence errors can arise through markedly different events. The following cases follow the observations and estimated motion before, during, and after an error. They use the selected PVO runs, with the paired absolute residual and frozen VOCA OF-first results where indicated. The intervals were selected by inspecting the complete saved trajectories; they are illustrative episodes, not a partition of the benchmark or a model that predicts failure. Times in the figures are relative to the first ground-truth-associated pose of the sequence. Exact intervals and sources are retained with the analysis assets.

6.8.1. Sustained Drift in a Dark, Sparsely Detailed View

On EMH04, the view changes from machinery and pipework to a broad cabin roof and the dark hall behind it. Figure 6.10 follows this transition. During a 7.8 s interval of sustained error, the images contain an average of 124 Shi-Tomasi corners, compared with 751 in a later interval with more visible machinery. RTE-6 falls from 0.3584 m during the degraded interval to 0.0209 m in the later interval, even though mean ground-truth path speed increases from 1.19 $\mathrm{m\,s^{-1}}$ to 1.37 $\mathrm{m\,s^{-1}}$. Faster translation therefore does not account for the difference between these two views.

Codec candidates remain abundant. The degraded interval contains approximately 2229 temporal candidates per frame with valid stereo at both endpoints, and 217 current IDs per frame are connected to existing backend landmarks. These counts do not collapse to zero. The estimated motion nevertheless becomes incorrect in both magnitude and direction, and the first-pose-aligned orientation error grows to approximately 67°. Once the view improves, local translation error becomes small again, but much of the orientation error remains. Local recovery therefore does not imply recovery of the accumulated trajectory.

The images and support measurements are consistent with less informative codec predictions over broad surfaces and dark regions. The connection to block-matching ambiguity follows the error-surface argument in Equation 2.8 [15, 14]. However, the recorded counts do not measure disparity accuracy or the spatial rank of the retained point set. They cannot

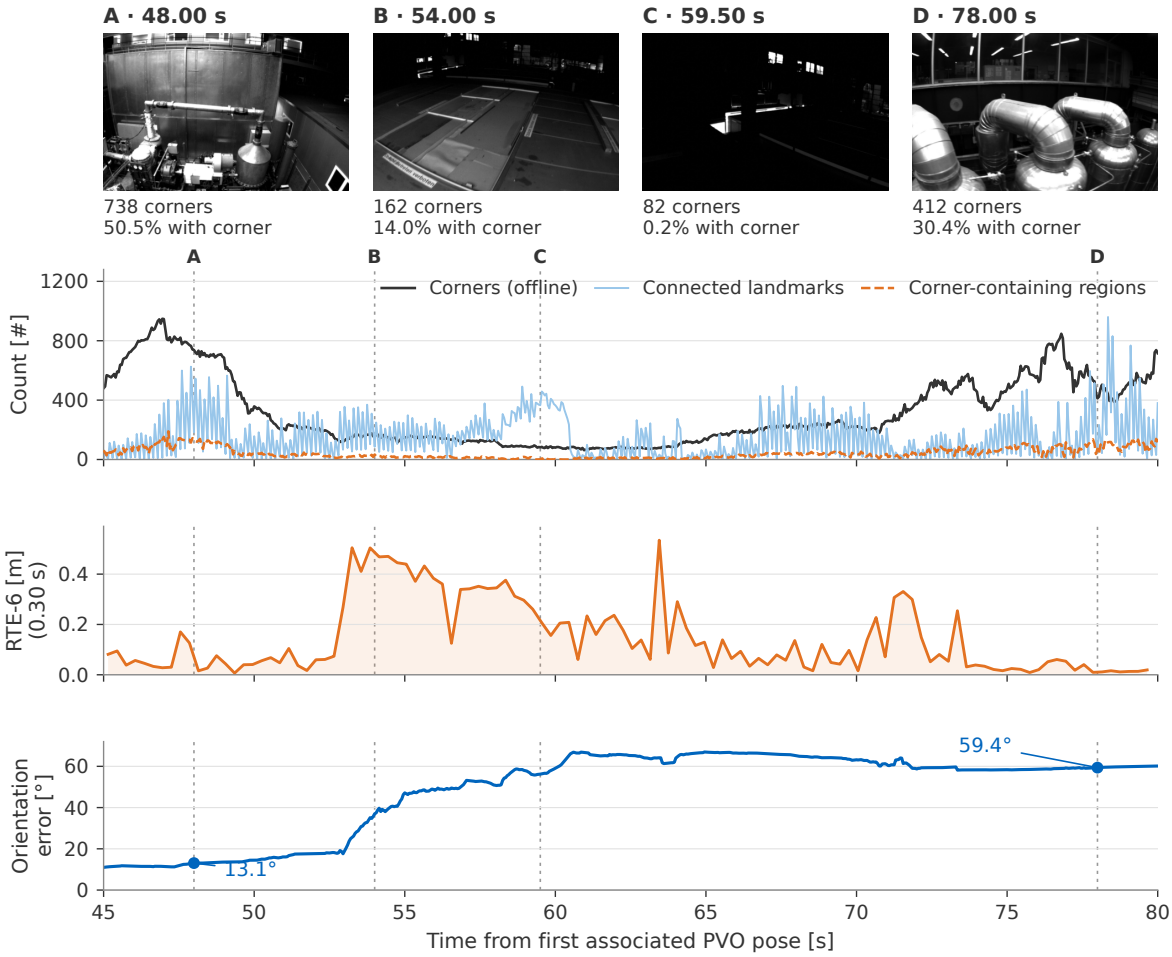


Figure 6.10.: EMH04 before, during, and after sustained drift. Raw anchor-camera images are shown only for diagnosis. The synchronized traces distinguish changing image structure, local translation error, and persistent orientation error. Corner-containing regions are TEASER-stage MV destination blocks containing at least one offline Shi-Tomasi corner; the percentages divide their count by all logged TEASER-stage regions in each sampled frame. Connected IDs instead count observations linked to existing backend landmarks. Orientation error is aligned at the first pose, not over the full trajectory.

distinguish incorrect stereo depth, temporal displacement bias, and unfavorable point geometry as the main source of this episode. The fact that the paired absolute-residual run also degrades in this interval provides an additional limit on its interpretation.

6.8.2. A Brief Loss of Connections During Rapid Motion

EMH03 contains a shorter event in which a six-frame interval reaches 0.4423 m RTE-6. It has an average of only 12.6 current IDs connected to existing landmarks. A separate rapid-motion interval in the same sequence has slightly greater translation and angular speed, yet its RTE-6 is only 0.0265 m. That interval retains 186.4 connected IDs on average. The two image intervals contain similar numbers of detected corners, 656 and 670 per frame, respectively.

This comparison separates the availability of visible structure from its conversion into persistent observations. The failed interval is not simply the faster or less corner-rich of the two. Its small connected set coincides with a large local error, although the archived counters do not identify which individual candidate losses produced that set. The absolute-residual run has almost the same error in the burst, 0.4397 m, which again indicates that replacing rigid flow alone does not remove the event.

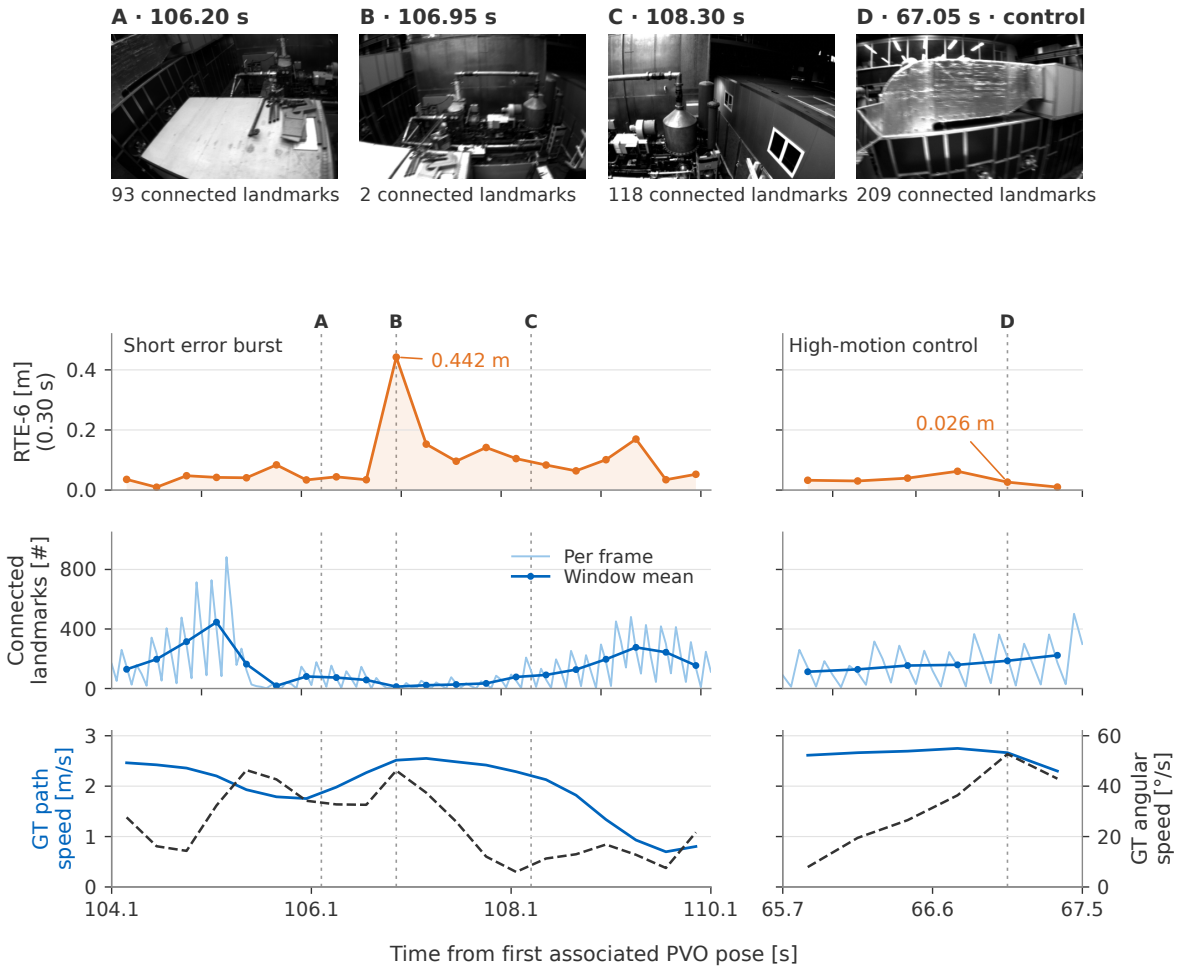


Figure 6.11.: A short EMH03 error burst and a rapid-motion comparison interval. Connected IDs are current observations linked to existing backend landmarks. Similar image corner counts do not imply similar surviving temporal support; the faster comparison interval retains substantially more connections.

6.8.3. Missing Temporal Observations, a Pose Jump, and a Stall

EV103 illustrates a discontinuity rather than gradual drift. The view approaches a bright window beside a broad dark partition. Visible corners and emitted track IDs decrease. Inspection of the encoded streams also identifies an I-frame in each direction at the endpoints of one temporal transition, leaving no ordinary temporal MVs for that connection. Three frames later, the estimated pose jumps by 37.716 m and rotates by 174.42° in one frame. Ground truth changes by only 0.050 m and 0.77°. At the jump, only two current IDs are connected to existing landmarks; the optimizer records eight iterations, six of them rejected.

A subsequent transition again lacks temporal MVs in both directions. The frontend then emits zero points for three consecutive frames and the pose remains exactly unchanged, although ground truth moves approximately 0.04 m per frame. Image corners have already returned to at least 350 per frame. Raw MV candidates return after the missing transition, but no candidate survives the logged stage requiring filtering and valid stereo at both endpoints for the next two frames. These records locate the loss within the processing chain; they do not isolate the median filter from stereo lookup as its cause. Connections subsequently return, while the accumulated position error remains approximately 38 m.

The I-frames explain the observed absence of temporal vectors at particular transitions, not the complete sequence of errors. The catastrophic update itself occurs on a P-frame, and the shortage of usable observations extends beyond the I-frame. The containing six-frame window contributes 98.6% of the full sequence's sum of squared six-frame translation errors. Its RTE-6 is 37.8510 m, compared with 0.4066 m in the paired absolute-residual run. This episode exposes a strong difference in how the backends respond to the available observations; it does not show that every large PVO error has the same origin.

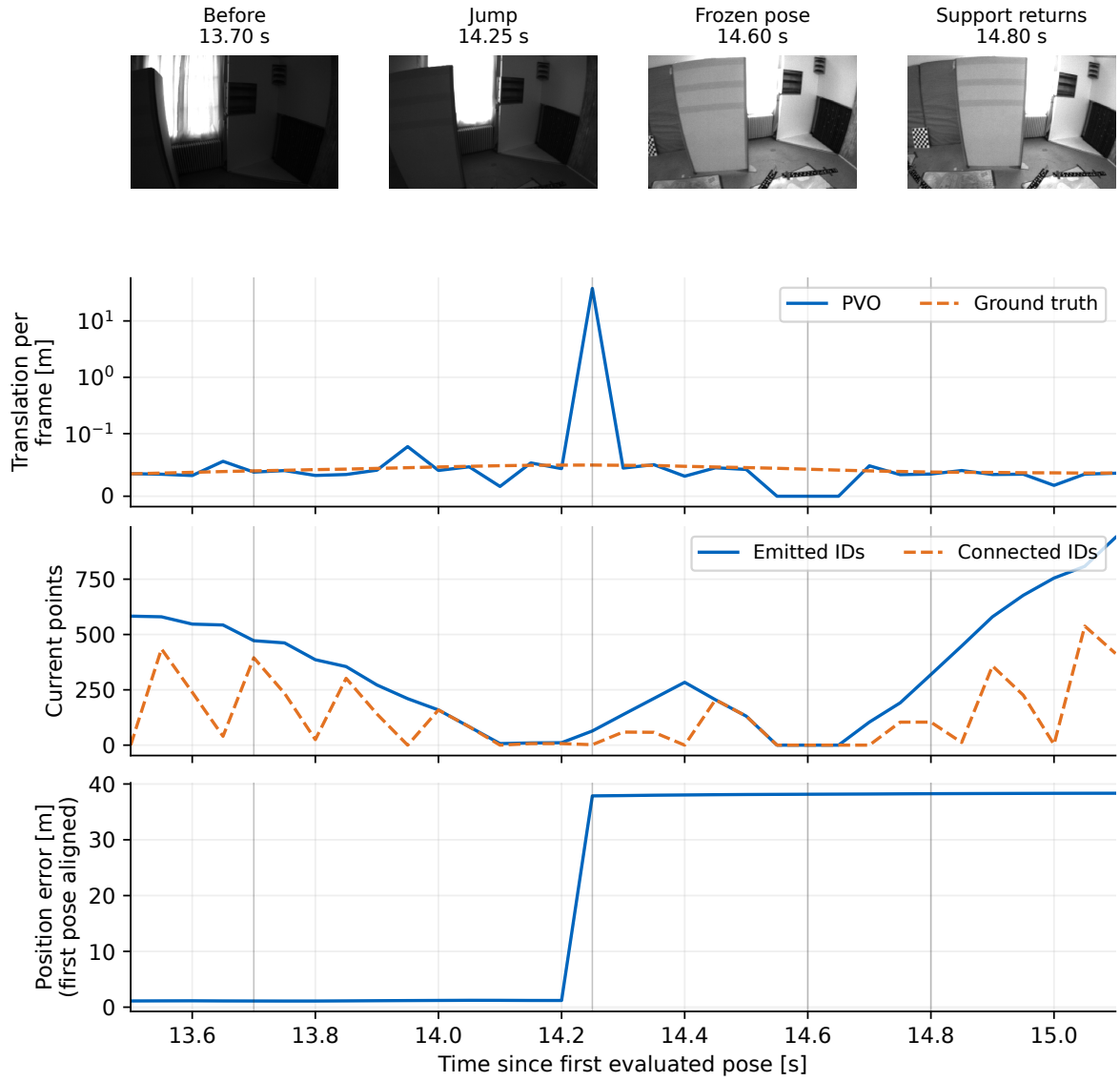


Figure 6.12.: EV103 around a catastrophic pose update. Per-frame translation uses a symmetric logarithmic axis so that ordinary motion, the large jump, and zero-motion outputs remain visible. The lower panels show the temporary loss of frontend support and the persistent position error after support returns. The sampled images are offline diagnostics.

6.8.4. Low-Motion Errors and a Textured Counterexample

EV202 contains consecutive estimated translations of 1.117 m and 0.895 m while ground truth moves only 0.013 m in each frame. The images change little and show fabric-covered blocks, a plain wall, and a small cart. Only three and two IDs, respectively, connect to existing landmarks. At the first jump, 2919 raw MV regions and 541 stereo-valid temporal candidates are still available, but only 25 reciprocal candidates and four candidates passing the fitted rigid-model gate remain. This is another case in which many initial observations do not translate into substantial temporal support.

A later EV202 event provides a counterexample to using image corner count as a sufficient indicator. A close, textured fabric surface dominates the view and 1640 corners are detected, yet the estimated translation is 0.752 m against a ground-truth step of 0.015 m. Only six IDs connect to existing landmarks. Conversely, a separate six-frame interval with an average of 7.4 connected IDs has RTE-6 of only 0.0376 m. Neither high corner count nor low connection count alone determines the outcome. Supplementary images and per-frame values are given in section B.1.

6.8.5. What the Paired Residual Comparison Establishes

Table 6.9 compares identical timestamp intervals from PVO, its absolute-residual ablation, and the frozen VOCA OF-first reference. The two PVO runs share encoded inputs and frontend settings, including the robust association procedure. EMH03's brief burst and EMH04's sustained error remain under absolute reprojection. EV103's catastrophic amplification is much smaller with absolute reprojection, as is EV202's textured-view error. However, the EV202 jump at 38.30 s becomes larger. The observation model thus changes the response to particular correspondence sets without providing a universal safeguard. The low OF-first errors on the same intervals show that the camera sequences still contain information from which the richer reference can recover motion.

Table 6.9.: RTE-6 in metres; lower is better. Entries report RMSE over the listed non-overlapping windows. Absolute changes PVO's observation model; OF-first is the frozen pixel-refined reference. The EV202 middle jump is at 38.30 s, distinct from the later textured-view event.

Episode	Windows	PVO [m]	Absolute [m]	OF-first [m]
EMH03 burst	1	0.4423	0.4397	0.0151
EMH04 dark view	26	0.3584	0.3719	0.0150
EMH04 recovery	23	0.0209	0.0270	0.0156
EV103 jump	1	37.8510	0.4066	0.0322
EV202 early jumps	1	1.3830	1.0851	0.0037
EV202 middle jump	1	1.1206	2.9941	0.0353
EV202 textured view	1	0.8402	0.2241	0.0019

7. Discussion

7.1. Answers to the Research Questions

RQ1: remaining trajectory accuracy. The accuracy difference from the closest higher-information reference is reported directly. For an error measure e, let

$$\Delta e_{\text{abs}} = e_{\text{PVO}} - e_{\text{MV-first}}, \qquad \Delta e_{\text{rel}} = 100\frac{e_{\text{PVO}} - e_{\text{MV-first}}}{e_{\text{MV-first}}}. \tag{7.1}$$

Positive values mean that PVO has greater error. For the easy group, the differences between group macro means are 0.334 m and 201.3% for ATE, and 0.0101 m and 49.6% for RTE-6. The small RTE-6 difference shows that local motion can remain close to the MV-first reference on easier motion even when the accumulated trajectory differs more strongly. Across the benchmark, the gap grows and is dominated by several difficult sequences; EV203 is the exception on which PVO has lower ATE and RTE-6 than MV-first. PVO nevertheless improves on the stationary baseline in both metrics on nine of eleven sequences. The relative differences must be read together with the absolute ones because division by a small reference error can produce a large percentage from a modest metric displacement.

Table 7.1.: Absolute and percentage differences between the group macro means of PVO and VOCA MV-first. Positive values indicate greater PVO error. Difficulty groups follow the EuRoC sequence labels.

Group	ΔATE [m]	ΔATE [%]	ΔRTE-6 [m]	ΔRTE-6 [%]
Easy	0.334	201.3	0.0101	49.6
Medium	1.091	796.6	0.0811	507.3
Difficult	4.341	165.3	0.2954	72.4
All sequences	1.998	189.7	0.1332	83.2

RQ2: stereo-temporal persistence and support. Persistent identities can be assembled without pixel descriptors, but their typical lifetime is short. Across the benchmark, PVO confirms 5,230,916 tracks with a median length of three frames and a 95th percentile of five frames. Only 9.92% reach five frames. Stereo support is likewise uneven: Between 13.48% and 19.88% of queries return a match on EMH01, EMH02, and EV101, whereas the rate is at most 1.99% on EV201 to EV203. In the separate structure diagnostic, the proportion of codec supports that contain a Shi-Tomasi corner changes across valid stereo, reciprocal

agreement, robust registration, and suppression: 30.59% for raw vectors, 37.45% with valid endpoint depth, 34.38% after reciprocal agreement, 39.57% after TEASER++, and 44.54% in the branched suppression-retained pool. Robust registration therefore concentrates its input near two-directional image structure, but most individual tracks still offer little multiview support.

RQ3: properties associated with failure. The episode analysis identifies several distinct events (section 6.8). EMH04 develops sustained drift while the view becomes dark and sparsely detailed, despite abundant stereo-valid candidates. EMH03 has a brief burst during rapid motion with substantially fewer retained connections than another, slightly faster interval with similar corner support. EV103 suffers a catastrophic pose update, temporary stationary outputs, and recovery of local motion without recovery of the accumulated trajectory. EV202 supplies both low-motion jumps and a failure in a corner-rich view. Visible structure, observation availability, and usable temporal geometry are therefore different quantities.

The paired residual comparison further limits the explanations. EMH03 and EMH04 retain their selected failures with absolute reprojection, while EV103's catastrophic error is much smaller. Another EV202 jump is worse with absolute reprojection. The recorded events establish when support is lost and how the estimated motion changes; they do not identify one cause shared by every sequence. Track lifetime and consecutive-depth summaries provide context but cannot explain an individual event without its temporal evidence.

7.2. Operating Envelope

The clearest feasibility evidence is the recovered trajectory geometry on the easy sequences, particularly the Machine Hall views in subsection 6.1.1. The reported PVO results on EMH01 and EV101 also show low accumulated error. Improvement over a stationary prediction on all four easy sequences provides a broader check that the estimates contain motion information. PVO also improves on all three medium sequences, although the quality varies substantially. The EV202 margin is small: 2.087 m ATE and 0.2078 m RTE-6 versus stationary values of 2.111 m and 0.2441 m. The difficult set is mixed rather than uniformly unsuccessful, with improvements on EMH04 and EMH05 but not on EV103 or EV203.

This grouping should not be read as a calibrated scale of codec difficulty. The EuRoC labels summarize the original sequence challenge, not the quality of H.264 prediction blocks or compressed-domain stereo. EV201 is labelled easy yet has substantially lower stereo yield than the other easy sequences. Conversely, local drift can remain moderate while small errors accumulate into a poor global trajectory. The evidence supports an operating envelope for the fixed EuRoC cameras, calibrations, and encoder configuration; it does not establish the same boundary for other cameras, frame rates, codecs, or encoder searches.

7.3. Track Persistence and Geometric Support

The three-observation median is only one observation longer than the provisional period required for confirmation. Consequently, most accepted landmarks supply little temporal baseline before disappearing. This limits the number of views that can jointly constrain a point and reduces the opportunity to average block quantization and stereo-depth errors. The small tail of longer tracks shows that identity persistence is possible, with a maximum of 204 frames, but that tail is too sparse to characterize the typical backend input.

Stereo availability is one bottleneck rather than a complete explanation. A continuation needs geometrically acceptable support, and a new identity also needs depth at both endpoints before it can enter the robust 3D registration. The low query yield on EV201 to EV203 therefore removes many potential associations early. Conditional consecutive-depth changes are small among points for which both depths exist, but this statistic says nothing about the larger set with missing depth. Track loss can additionally reduce the connected landmark ratio and request keyframes more frequently. Standard window optimization and marginalization preserve information from accepted factors, but they cannot recreate missing multiview observations.

The offline structure analysis provides a complementary result. Requiring valid stereo at both endpoints, reciprocal agreement, and TEASER++ consensus changes the fraction of regions containing a Shi-Tomasi corner at each stage. This indicates whether geometric selection concentrates observations near two-directional image structure without explicitly detecting it. It does not show that each codec support contains one uniquely localized physical point, and the diagnostic corner coordinates are never used by the estimator.

A quantitative lifetime comparison with the VOCA MV-first and OF-first configurations is deliberately omitted. The diagnostic build instrumented for track lifetimes did not reproduce their frozen trajectories, so its counters would not describe the same configurations as the ATE and RTE-6 tables. The qualitative expectation that pixel-refined tracking supplies more precisely localized and longer-lived observations remains plausible, but is not used here as a measured result.

7.4. Error Modes

The observed failures differ both in duration and in their relation to support. EMH04 continues estimating motion with many observations but accumulates a large orientation error. EV103's full-sequence RTE-6 is dominated by one catastrophic window, followed by a period without emitted observations. These require different explanations: the first calls into question the information supplied by the surviving geometry, whereas the second also exposes the estimator's response to an abrupt shortage of connections. Neither can be adequately described by the full-sequence mean track length.

The observation-model ablation adds the same qualification at the backend. Rigid flow lowers ATE on eight sequences, RTE-6 on seven, and both medians, but its EV103 failure makes both macro means worse than absolute reprojection (Table 6.6). Replacing eligible

absolute factors with displacement factors lowers the median errors in this experiment, while absolute initialization and fallback observations remain part of the estimator. The replacement does not by itself guard against an erroneous correspondence set or a poorly conditioned trajectory.

Two observation-level failures are directly visible before the later gates are applied. First, local codec support can disappear while the tracked image structure remains visible and raw temporal vectors remain elsewhere (Figure 4.8). The responsible encoder decision is unresolved in this example. Second, a codec block and its vector can remain available while a localized appearance inside the source block is no longer contained by the predicted destination block (Figure 4.7). The latter is possible because the encoder optimizes prediction of an extended block rather than persistence of a particular scene point. These cases explain why raw-vector availability alone does not establish a valid feature track.

In the separate EV103 episode, frame-type inspection identifies I-frames at the relevant endpoints in both encoding directions, explaining the absence of ordinary temporal vectors for that transition. Subsequent P-frames restore raw candidates before useful frontend support returns. This distinguishes an identified input interruption from the later losses whose individual causes are not recorded. Similarly, EMH04's dark views suggest ambiguous block predictions, but the saved logs cannot separate temporal-vector bias from incorrect depth or an unfavorable distribution of points.

TEASER++ success does not by itself resolve those uncertainties. Its role is to find a rigid transformation consistent with a subset of putative associations [16]; such consistency does not independently establish correspondence to the true scene. The implementation reports successful registration throughout EMH04 even while the trajectory diverges. Additional per-frame inlier geometry and residual diagnostics would be needed to assess conditioning or identify coherent incorrect associations. The present evidence does not establish that moving objects, calibration error, or a particular depth distribution caused these episodes.

The median filter targets isolated compression artefacts, reciprocal agreement tests temporal reversibility, TEASER++ rejects inconsistent 3D pairs, and age-prioritized suppression removes indistinguishable duplicate support. These stages address different symptoms. Their non-monotonic ablation results suggest that stricter rejection can also discard useful constraints. The case studies show where these distinctions matter, while identifying questions that the available diagnostics cannot answer.

7.5. Meaning of Pixel-Free Operation

The information restriction applies to the tracking frontend and pose estimator. They consume H.264 motion-vector side data, prediction-block geometry, timestamps, camera calibration, rectification transforms, and validity masks. They do not load decoded intensities, build image pyramids, detect corners, evaluate patches, compute descriptors, or apply a photometric residual. Shi-Tomasi and blur measurements are computed only after evaluation for diagnosis.

The restriction does not claim that the entire data-production process is image-free. Decoded stereo images are remapped during offline rectification and are then supplied to the video

encoder. This conventional preprocessing creates the H.264 streams on which the experiment operates. The resulting statement is therefore precise: after calibrated offline rectification and encoding, the frontend and pose estimator use codec side information and geometry without consulting image intensities. The extraction software used here decodes frames and reconstructs images internally, but does not forward those pixels to the tracking frontend.

7.6. Placement of Analysis and Transmission

The communication boundary can be placed at several stages of the codec-based frontend. Sending the required coded streams follows CTA: the capturing device encodes, and the receiver extracts motion vectors and estimates camera motion. Conventional CTA follows decoding with image-based feature extraction or tracking [3]; here the frontend uses codec metadata directly.

Alternatively, the capturing device could extract the vectors and perform some of the correspondence filtering before transmission. It could send temporal and stereo observations for the remaining geometric checks, or run the frontend locally and send confirmed observations to a remote backend. Analysis would then precede the compression and transmission of the geometric data, giving an ATC-like allocation of work while retaining codec-derived measurements. These options extend the processing–communication choice described in the CTA–ATC literature [4] to different stages of this frontend.

The transmitted representation must preserve the information needed after the chosen boundary. Early motion-vector observations require block support, reference information, timestamps, and calibrated geometry. A later split at the frontend output would require shared calibration, consistent track identities, per-camera coordinates, measured displacements, and predecessor timestamps, with frontend and backend state maintained on their respective devices. Filtering reduces the set of retained candidates, but the resulting bitrate also depends on their metadata and encoding. The experiments do not implement or measure these transport alternatives.

The evaluated input consists of two forward temporal streams, two reversed temporal streams, and twelve shifted stereo streams. Preparing those additional encodings has a cost. Two-pass temporal encoding and the current reversal of complete frame lists make preparation an offline procedure, despite the single-reference, B-frame-free prediction structure within each stream. A live deployment must establish how the observations are generated and delivered within its latency and throughput budget. Efficient metadata extraction must also account for the pixel reconstruction performed by the current decoder.

Passthrough+ motivates dedicated codec hardware with platform-specific power budgets [2]. The present software-encoded experiments establish geometric feasibility. Net energy savings depend on all preparation, communication, extraction, and estimation. Reusing motion from an already required stream has a different incremental cost from generating extra stereo encodings. Sending an unchanged stream also preserves its bitrate; filtering and transmitting geometric observations is a separate design choice.

7.7. Validity and Limitations

The evaluation covers all eleven EuRoC sequences, but only one benchmark, one stereo camera family, and the provided calibrations. The result is also tied to FFmpeg 4.4.2, `libx264`, H.264 High profile, the fixed temporal encoding, and the twelve shifted I-P-P stereo streams. Hardware encoders or other search, partition, bitrate, GOP, and colour-format choices may expose different motion vectors. Passthrough+'s hardware mosaic, densification, pose-assisted temporal depth propagation, and rendering stages are not reproduced. In particular, the original full proxy assumes headset poses from an independent SLAM system, which would violate the information restriction here. The stereo component is therefore an adaptation rather than a complete replication of that system.

The selected PVO parameters were chosen after a bounded twelve-candidate sweep on the same eleven sequences used for reporting. The final tables exclude abandoned searches, but the evaluation is therefore not an independently held-out benchmark study. The failure episodes are also selected retrospectively and do not estimate the prevalence of a failure mechanism. Their successful comparisons and counterexamples constrain the interpretation, but changing views, illumination, motion, and available geometry remain coupled. The paired residual ablation controls the observation model; image inspection and timestamped counters alone are not equivalent to a controlled intervention.

The static-scene model, two-camera depth construction, calibrated synchronization, and available ground truth further delimit the conclusions. The system has no explicit mechanism for rolling shutter, independently moving objects, calibration drift, or lost synchronization. A validity mask excludes undefined rectification regions, but no image-derived confidence is available inside valid regions.

7.8. Comparison with Pixel-Refined References

The VOCA MV-first configuration [1] uses codec motion to initialize KLT, followed by an unseeded optical-flow fallback. The VOCA OF-first configuration attempts unseeded KLT first and uses motion-vector initialization when tracks are lost. Both therefore receive information that the proposed estimator intentionally omits. Their median ATE values, 0.200 m and 0.118 m, are well below PVO's 1.082 m; their median RTE-6 values are 0.0163 m and 0.0155 m, compared with 0.0818 m. The absolute and percentage differences from MV-first are summarized in Table 7.1. This accuracy gap remains substantial even though the MV-first reference fails on EV203.

These references contextualize the accuracy available with pixel-based localization and refinement. PVO's nine improvements over the stationary baseline demonstrate useful motion information under the pixel-free restriction. The accuracy gap also includes differences in correspondence construction, filtering, and observation models, so it does not isolate the effect of pixel refinement.

8. Conclusion and Future Work

8.1. Conclusion

This thesis demonstrates that video encoding can supply the geometric observations needed for useful visual odometry. Motion estimation performed for compression can also support camera tracking without a separate pixel-based correspondence frontend. The successful trajectories, most clearly on the easy Machine Hall sequences, establish that this reuse is feasible. Their significance is that a representation produced for video coding can serve as the measurement source for geometric estimation.

This opens a route to repurposing widely available video-encoding accelerators for the odometry component of a SLAM system. The demonstrated method is naturally compatible with compress-then-analyze at its analysis interface: a capturing device can provide the prepared coded streams, and a receiver can estimate motion from their codec metadata and calibration. The receiving device need not repeat correspondence estimation on image intensities. For a small camera paired with a phone, this offers a way to build localisation around existing coding resources and the communication structure already used for compressed video.

The pipeline also permits other choices of where to place the communication boundary. Its filtering and geometric validation stages expose intermediate observations that could be transmitted instead of the video streams. A capturing device could extract motion vectors, perform some of that analysis locally, and then compress and send the retained correspondences for further estimation. Such a division resembles analyze-then-compress at the transmission boundary, while still obtaining its initial measurements through video compression. It therefore need not be assigned exclusively to either CTA or ATC: the same codec-based analysis principle supports a range of allocations between camera and receiver. The useful split would depend on local computation, link capacity, and the information required by the remaining estimation stages.

The contribution is a feasibility demonstration for this design direction. The present software-encoded experiments establish useful geometric information; their accuracy limits leave room for different correspondence constructions and estimators. Hardware execution and intermediate-data transport remain systems work to evaluate. What the thesis establishes is a basis for that work: visual odometry can be built from codec-generated motion information, making existing video-processing resources relevant to localisation as well as communication.

8.2. Future Work

A first priority is a causal hardware implementation that connects a constrained capturing device to a receiving phone. The current two-pass temporal encoding needs a streaming alternative. Offline reverse-temporal streams must be replaced by bounded buffering or another causal mechanism, and the shifted stereo inputs must be scheduled within the encoder's throughput and latency limits. A mosaic construction, fewer pre-shifts, or another means of obtaining depth could reduce input-preparation cost. Access to motion-vector metadata should also be measured directly, including any image reconstruction performed inside the extraction path.

The resulting system should be evaluated for energy per processed stereo pair, power, latency, transmission rate, and trajectory accuracy at the same operating point. Measurements should include rectification, every additional encode, memory movement, the radio link, metadata extraction, and estimation on both devices. Comparing image-based CTA, codec-based CTA, and capture-side feature extraction would test when the proposed allocation is beneficial. This follows the processing–communication trade-off demonstrated by Redondi et al. [4], rather than transferring a hardware power figure to a different complete workload. Sending an ordinary coded stream and sending a dedicated motion-vector payload are separate transport options to evaluate.

A second priority is to improve the geometric observations while preserving the pixel-free estimation boundary. The stereo examples show how grid phases refine spatial sampling and how pre-shifts can make depth available at different locations. Future work could select these transforms adaptively, compare hardware encoder motion fields, and account more explicitly for block support and measurement uncertainty. Observation models and association rules should be judged by their ability to retain geometrically valid connections as well as by their trajectory error. A longer assigned track alone does not establish that it follows the same scene point.

The preliminary experiments discussed in section B.4 also motivate maintaining surface or region identities while their supporting codec correspondences change. Such representations could reduce dependence on long-lived point tracks, provided that their association and geometry constrain camera motion.

Registered RGB-D measurements offer a complementary experiment. Replacing the codec-stereo stage with sensor depth would help separate temporal motion-vector limitations from missing stereo support, while keeping temporal tracking free of image-intensity access. This would be a different information setting from the stereo system studied here, and would test whether the same analysis principle is useful with another depth source.

Finally, evaluation should extend to other cameras, encoders, and motion patterns, including controlled exploration and mapping routes. Such tasks can make moderate, repeatable motion an operational choice, motivating study of where the favourable tracking seen here is useful. Tests under rapid rotation, blur, low texture, and changing illumination would establish the limits of that use. The present method provides a starting point for these investigations: codec motion can sustain useful camera tracking, and efficient systems built around that information are worth exploring further.

A. Reproducibility Material

The implementation is based on commit `29d09aa0b14ecefe7278d201fd31cd2a96dbf5d3`, identified locally by tag `x01-locked-20260906`. The selected PVO evaluation executable is identified by SHA-256 `bd9b6641fb7ba57df508cf188125838474fb29a689d96d7b271a99e2a1464659`. The SHA-256 values below identify the selected PVO configuration (internally p01), VOCA MV-first, VOCA OF-first, and EuRoC calibration JSON files, respectively.

```
8457ae6d36f8c137bdbf9f4dcaa29a23d480c9f2cad1bee41176c0578c27ee1b
9533b2a8f3e059bd0e46cfe6bf9fcd857b96ffe2d4839a6ff987e80e4fe11536
96d329823d5ad828e4ed43e3713324562aeb33bec6f1935899b40d66789b9bdf
ad8c5a18c48c55dacf61d18ebbc18cd7d4f3acccd5840bd5a5cd8646adb6271c
```

The direct submodule revisions are listed below. Their own nested submodules are fixed transitively by these revisions; `git submodule status -recursive` records the complete expansion.

```
CLI11 4af78beef777e313814b4daff70e2da9171a385a
Pangolin 73967b39ad2cf84245cf877ce0879dc0909d3be4
basalt-headers 1a519e990b2cbd62223a7cdab5e19a288412c5d9
magic_enum 3d1f6a5a2a3fbcba077e00ad0ccc2dd9fefc2ca7
opengv 91f4b19c73450833a40e463ad3648aae80b3a7f3
console_bridge f0b423c0c2d02651db1776c96887c0a314908063
ros_comm 194c776737134a429619f6851f1850734843e210
roscpp_core 1a74781ce707b1a2198d1bd6799b892b3c1fa195
```

The evaluation manifest records FFmpeg 4.4.2, Python 3.10.12, NumPy 2.2.6, pandas 2.3.1, and Matplotlib 3.10.3. The compiled environment uses GCC 11.4.0, OpenCV 4.5.4, Eigen 3.4.0 through the pinned `basalt-headers` tree, and a TEASER++ installation identified by revision `52a9c52`. A path-neutral release build is:

```
git submodule update --init --recursive
cmake -S . -B build-locked -G Ninja \
  -DCMAKE_BUILD_TYPE=Release \
  -DBUILD_EXAMPLES=OFF -DBUILD_TESTS=OFF \
  -DBASALT_ENABLE_TEASERPP=ON \
  -DBASALT_TEASERPP_ROOT=<teaser-prefix>
cmake --build build-locked --target basalt_vio -j2
```

The canonical dataset identifiers are `MH_01_easy` through `MH_05_difficult`, `V1_01_easy` through `V1_03_difficult`, and `V2_01_easy` through `V2_03_difficult`. These are the eleven EuRoC releases listed with their evaluated frame counts in Table 5.1. Temporal, reverse-temporal, rectified stereo, and mask products are generated with:

```
python3 .ci/get_dataset.py <dataset-path> <work-dir> \
  --config_path data/euroc/euroc_config_vo.json --device euroc
python3 .ci/get_dataset.py <prepared-dataset-path> <work-dir> \
  --config_path data/euroc/euroc_config_vo.json --device euroc \
  --reverse-temporal-only
```

Here the preprocessing configuration resolves `config.ffmpeg_stereo_mv_rectify_calib` relative to the configuration file. The generator checks synchronized timestamp lists, matching forward and reverse frame counts, twelve stereo files, I-P-P frame types, and both validity masks. A representative PVO invocation is:

```
build-locked/basalt_vio \
  --show-gui 0 --deterministic 1 --num-threads 1 \
  --dataset-path <dataset> --video-dataset-path <videos> \
  --dataset-type euroc --cam-calib <calibration.json> \
  --config-path <codec-only-config.json> \
  --use-mvs 1 --use-video-frames 1 --use-imu 0 \
  --save-trajectory euroc --save-trajectory-fn tracking.csv \
  --result-path result.json \
  --direct-mv-stereo-reprojection-gate 1.5 \
  --direct-mv-teaser-noise-bound-m 0.1 \
  --direct-mv-teaser-far-depth-m 5 \
  --direct-mv-teaser-far-reprojection-gate-px 1.5 \
  --temporal-reverse-policy hybrid \
  --temporal-reverse-cosine-threshold 0.98 \
  --temporal-reverse-min-iou 0.15 \
  --temporal-reverse-max-cycle-error 1.5 \
  --direct-mv-merge-indistinguishable \
  --direct-mv-merge-radius-px 1.125
```

The result directory follows the `xrtslam-metrics` layout:

```
runs/<configuration>/<sequence>/tracking.csv
```

The trajectory tables are regenerated with:

```
python3 <xrtslam-metrics>/batch.py <runs> <targets> \
  --metrics atec rtec --save_file batch-results.json --highlights md
python3 results/20260907-frozen-evaluation-matrix/scripts/generate_assets.py
```

The asset generator reads the selected PVO and frozen reference JSON files together with the frontend diagnostic CSVs. It produces the ATE and RTE-6 tables, component summary, empirical error distributions, track statistics, stereo-support plot, and feature-support tables. The failure-episode assets are generated separately from archived trajectories, timestamped diagnostics, and native image samples:

```
python3 results/20260912-failure-episodes/scripts/generate_emh_cases.py
python3 results/20260912-failure-episodes/scripts/generate_episode_evidence.py
```

Their manifests retain input hashes, selected intervals, metric definitions, and image timestamps. Results whose trajectory failed the selected metric check are omitted rather than substituted.

Full-sequence trajectory overlays and the confirmed-observation density figure are generated directly from the archived PVO trajectories and lifetime histograms:

```
python3 results/20260915-camera-trajectories/scripts/generate.py
python3 results/20260915-track-density/scripts/generate.py
```

The trajectory generator requires exact agreement with the archived ATE and RTE-6 before publishing a plot. Its manifest records the single alignment, associated timestamps, input hashes, and displayed bounds. For observation density, each lifetime-histogram count is multiplied by its lifetime before division by the number of input frames. This counts repeated observations of confirmed tracks rather than counting each track only once.

A.1. Configuration Listings

Table A.1 lists the switches and selected thresholds that differ across the reported configurations. The four component rows use the original operating point, whereas PVO uses the selected parameters from Table 5.2. All codec-only rows use two cameras, visual-only estimation, TEASER++, and the rigid-flow residual. For the VOCA MV-first and OF-first references, `-use-mvs 1` is retained but the frontend performs KLT refinement and uses ordinary absolute reprojection observations. The stationary x0 row is not an estimator configuration: it repeats the initial ground-truth pose for every timestamp.

Table A.1.: Parameters that differ between the reported estimated configurations.

Configuration	Reverse policy	Confirmation	Suppression	F-B gates	NMS [px]	Huber/std. [px]
TEASER++ only	best	off	off	n/a	n/a	1/8
F-B agreement	hybrid	on	off	.99/.25/1	n/a	1/8
Suppression	best	off	on	n/a	2.25	1/8
Joint frontend	hybrid	on	on	.99/.25/1	2.25	1/8
PVO	hybrid	on	on	.98/.15/1.5	1.125	.5/4
VOCA MV-first	not used	off	off	n/a	n/a	n/a/0.5
VOCA OF-first	not used	off	off	n/a	n/a	n/a/0.5

All estimated configurations use at most seven retained keyframes plus the current pose, square-root marginalization, seven nonlinear iterations, and no IMU. Although the configuration contains a three-state limit, that limit governs the inertial state container, which is empty in these visual-only runs. The codec-only configurations additionally use the 0.1-metre TEASER++ bound and the two-camera rigid-flow factors. The internal search label p01 is reported publicly as PVO; letter-coded development names are not used as method names.

B. Supplementary Results

The full per-sequence trajectory results are retained in Table 6.1 and Table 6.2 because the sequence-level variation is a principal result rather than supplementary material. All configurations in those tables completed all eleven sequences.

B.1. Failure Episode Details

Table B.1 records adjacent-pose translations at selected updates discussed in section 6.8. These are individual motion increments, not ATE or six-frame RTE. All times use the first ground-truth-associated pose as their origin. The analysis assets retain the corresponding native nanosecond timestamps, camera filenames, and complete before-and-after traces.

Table B.1.: Selected single-frame pose updates. Connected IDs count current observations linked to existing backend landmarks before new-landmark creation.

Sequence	Time [s]	GT step [m]	PVO step [m]	Connected IDs
EV103	14.25	0.050	37.716	2
EV202	3.45	0.013	1.117	3
EV202	3.50	0.013	0.895	2
EV202	38.30	0.096	1.376	2
EV202	112.60	0.015	0.752	6

For EV103, stream inspection identifies forward I-frames at relative times 14.10 s and 14.55 s. In the reverse stream, their predecessors in forward time are also I-frames. Neither direction therefore provides ordinary temporal MVs for these two connections. The first transition precedes the large pose jump by three frames; the second coincides with the first of three unchanged pose outputs. Subsequent frames are P-frames, so the continued shortage of observations cannot be attributed solely to I-frame coding. These checks use the encoded frame types and original timestamp order, including non-IDR I-frames rather than only frames marked as keyframes.

The candidate counters describe distinct populations. Raw regions are codec MV records. The logged stereo-valid temporal pool has already passed source, median, bounds, and mask checks and requires valid stereo at both temporal endpoints. A zero count at this stage does not identify which preceding check removed the candidates. The TEASER-stage count covers candidates passing the fitted rigid-model gate, whereas the suppression-retained pool can also include provisional births. Emitted IDs and IDs connected to existing backend

landmarks describe later populations. Their counts must not be interpreted as a sequence of interchangeable inlier ratios.

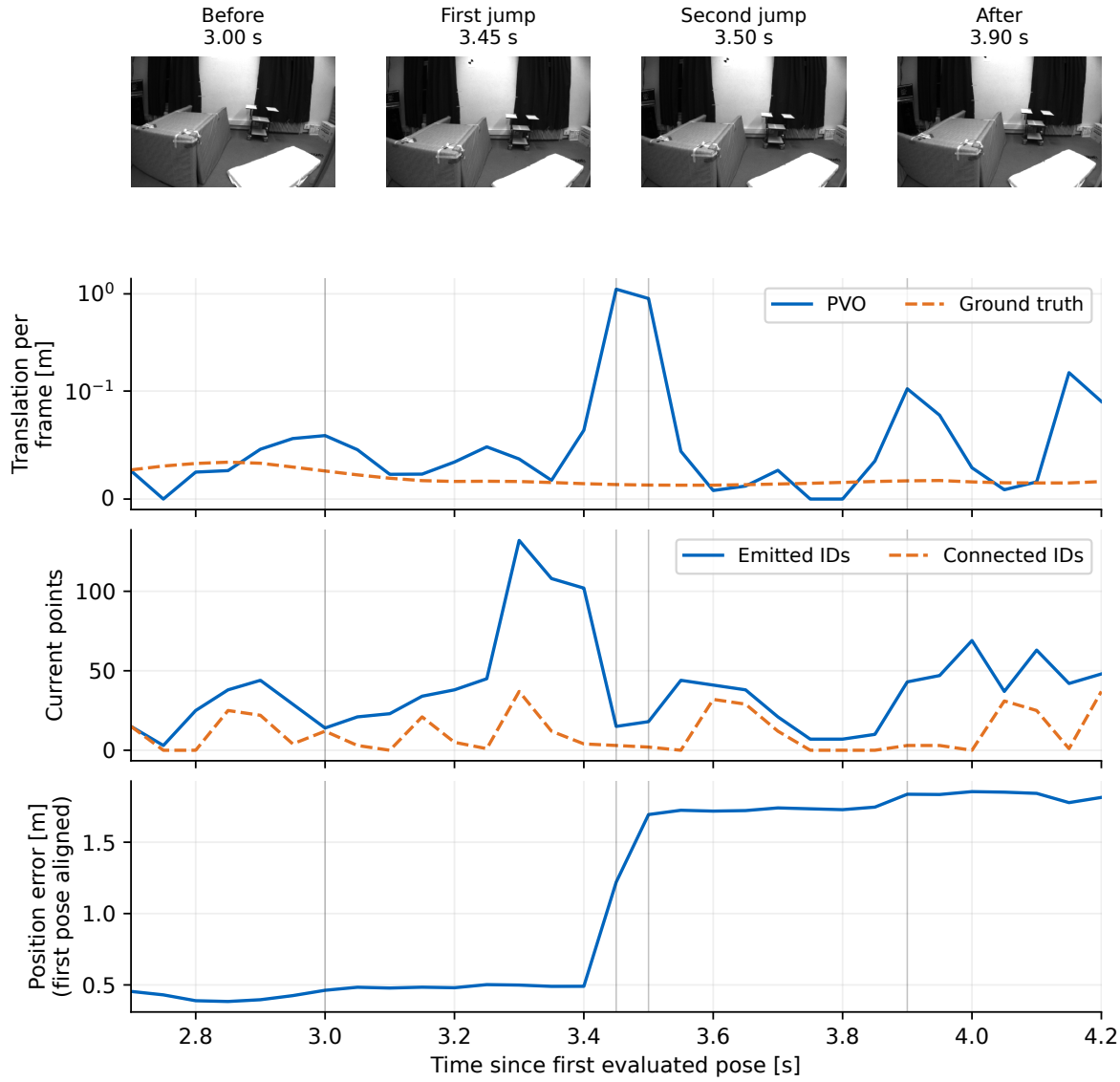


Figure B.1.: EV202's consecutive large translation estimates during small ground-truth motion. The native views change little across the updates, while only a few emitted IDs connect to existing landmarks. The position error uses first-pose alignment and is not globally aligned ATE.

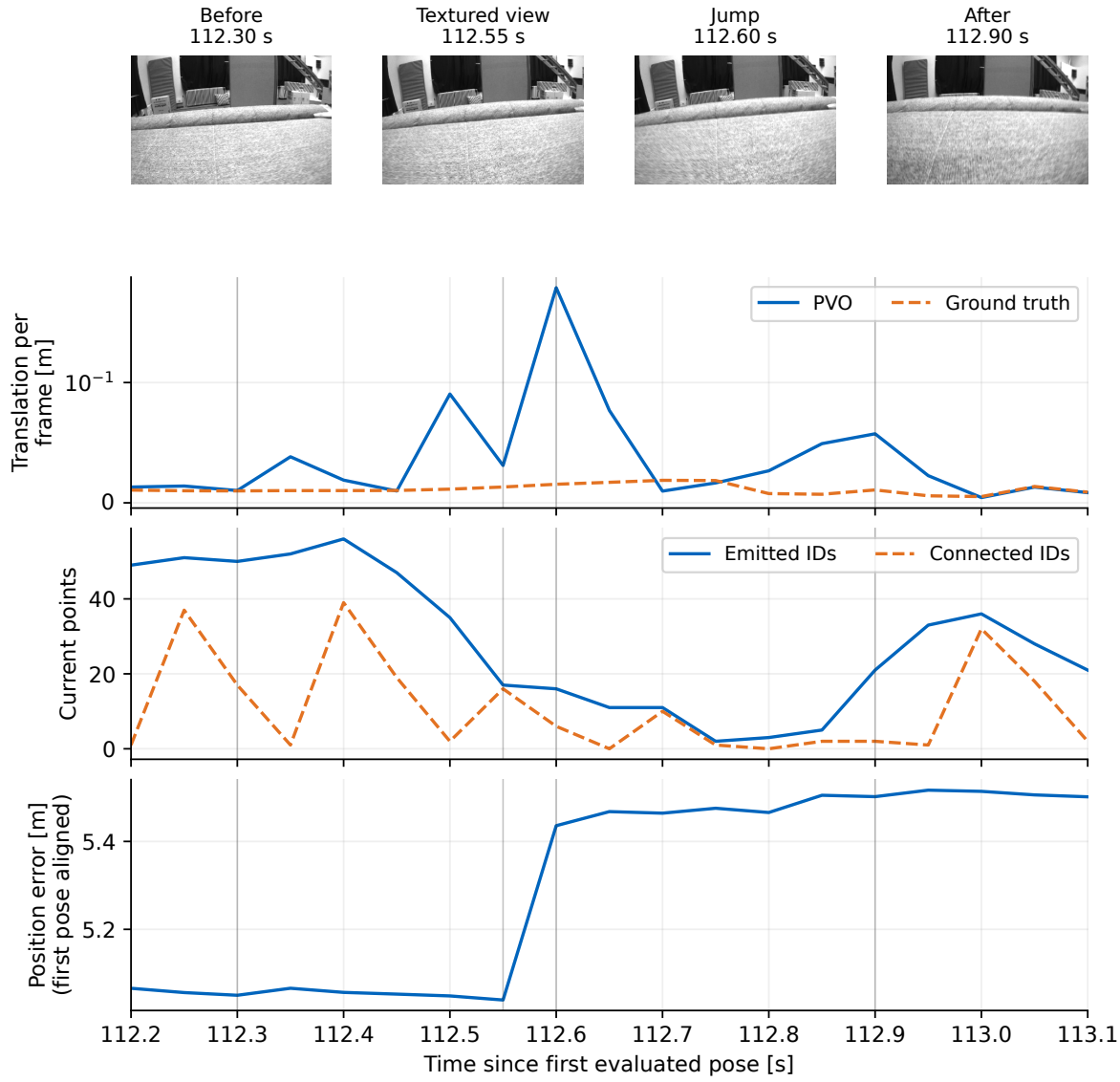


Figure B.2.: An EV202 failure despite abundant visible texture. The jump frame contains 1640 offline Shi-Tomasi corners but only six IDs connected to existing landmarks. Estimated translation is 0.752 m, compared with a ground-truth step of 0.015 m. High image corner count is therefore not sufficient to establish usable geometric support.

B.2. Full-Sequence Trajectories

The following plots show PVO on all eleven EuRoC sequences, complementing the examples in subsection 6.1.1. Each pair of projections shows every saved PVO position after the single whole-trajectory alignment used for ATE, together with the available reference path. No scale correction, smoothing, or removal of failure segments is applied. The axes use equal metric scale within a sequence and retain its full extent. These spatial views complement, rather than replace, the timestamped failure episodes.

EV202's estimated path is substantially distorted despite its small ATE improvement over the stationary baseline. EMH04 retains broad excursions but differs markedly in their geometry and vertical extent. EV103 contains a large discontinuity far outside the reference path. Because alignment is fitted over the complete sequence, this failure also shifts the earlier plotted estimates relative to ground truth; their offset must not be interpreted as an initial pose error.

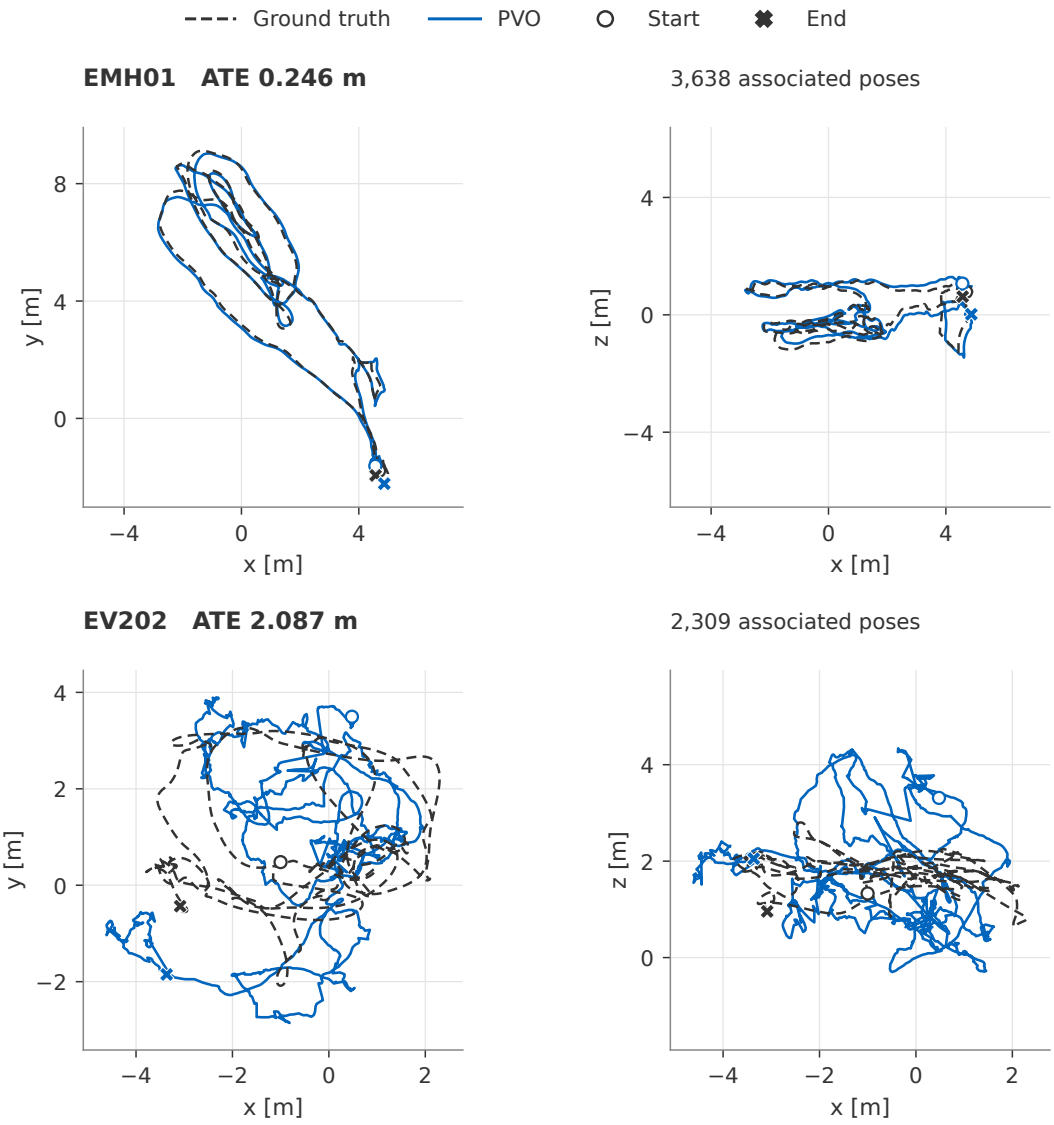


Figure B.3.: Full-sequence PVO trajectories and ground truth for EMH01 and EV202, viewed in the xy and xz planes. Both views use the same rigid alignment as ATE, without scale correction. All saved estimates are shown; the pose counts identify those associated with ground truth for the reported metric. Circles and crosses mark the start and end of each displayed path.

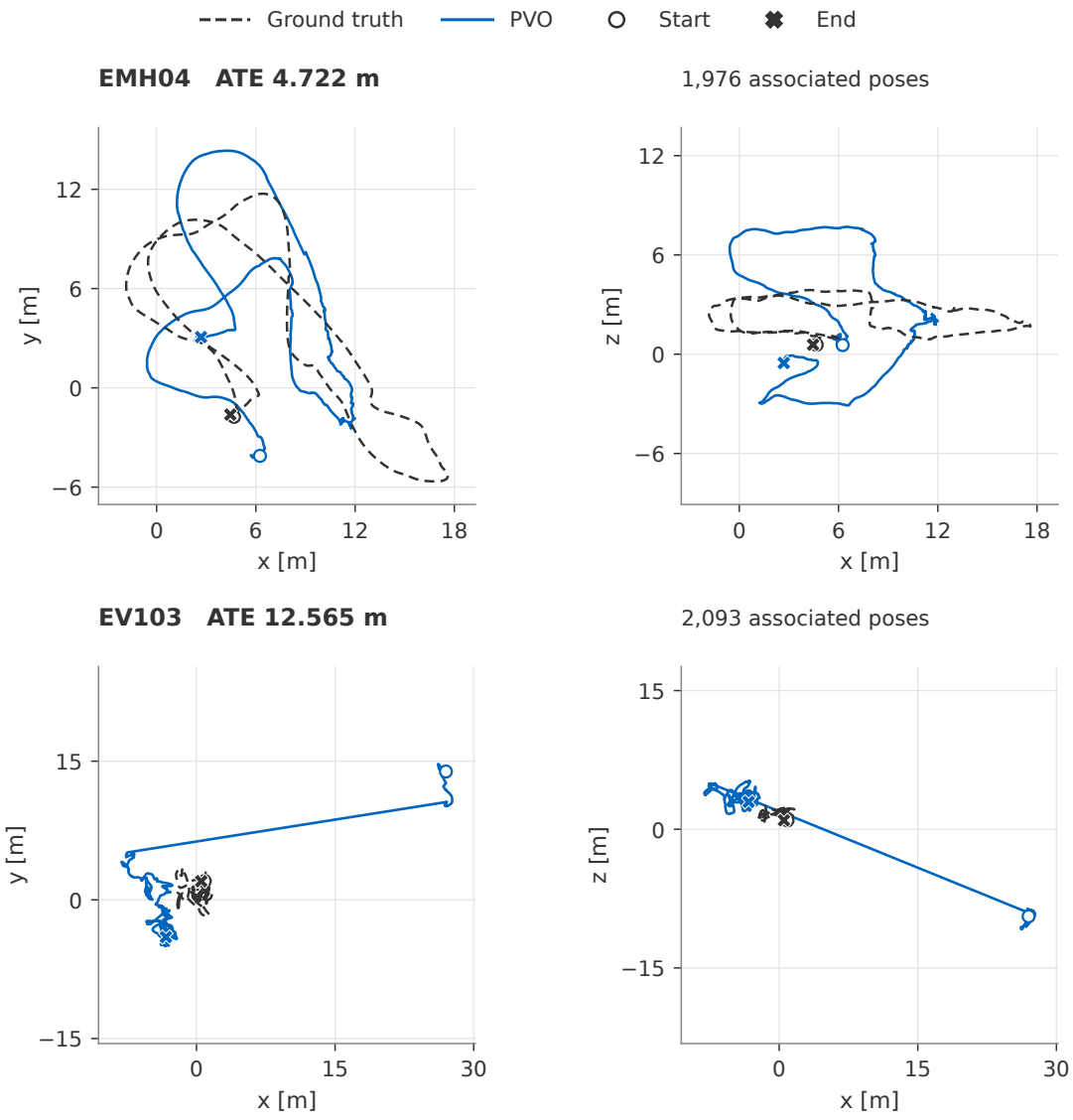


Figure B.4.: Full-sequence PVO trajectories for EMH04 and EV103, using the same convention as Figure B.3. The axes retain the full estimated extent, including EV103's catastrophic jump. Distances are equally scaled within each sequence, but axis ranges differ between sequences.

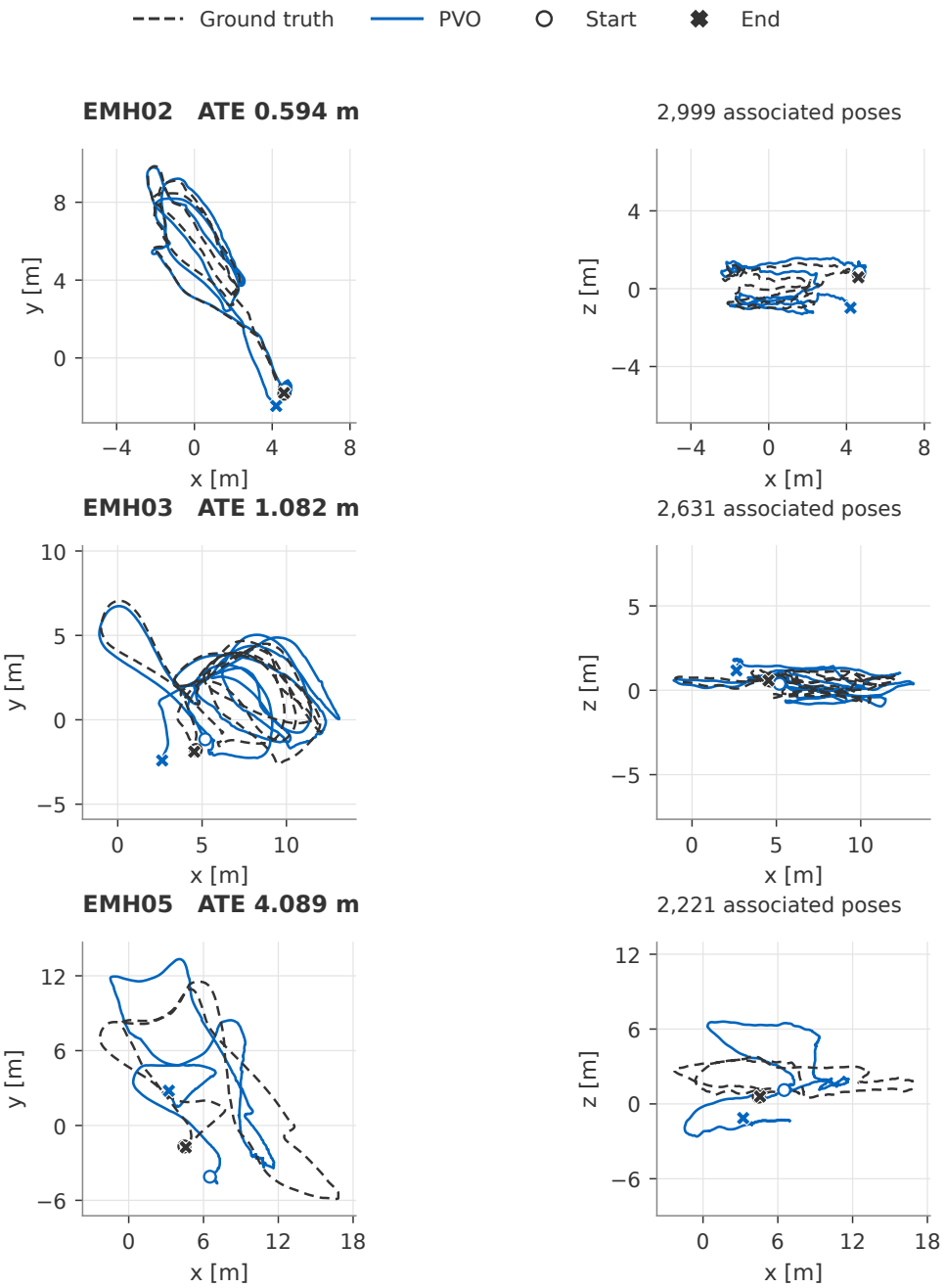


Figure B.5.: Full-sequence PVO trajectories for the Machine Hall sequences EMH02, EMH03, and EMH05. Left: xy projection; right: xz projection. Circles and crosses mark the start and end of each displayed path.

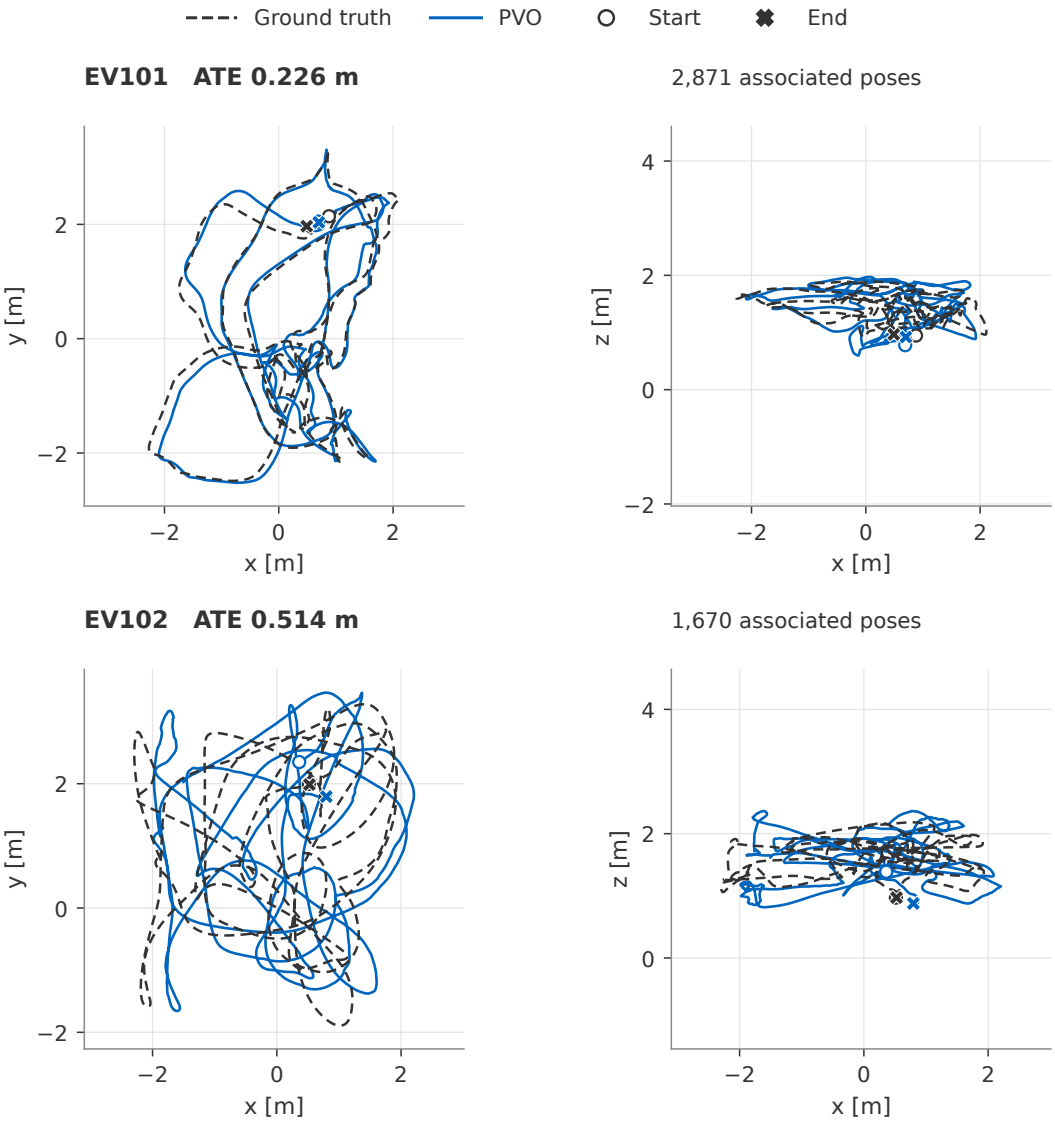


Figure B.6.: Full-sequence PVO trajectories for EV101 and EV102, using the same alignment and plotting convention as Figure B.3.

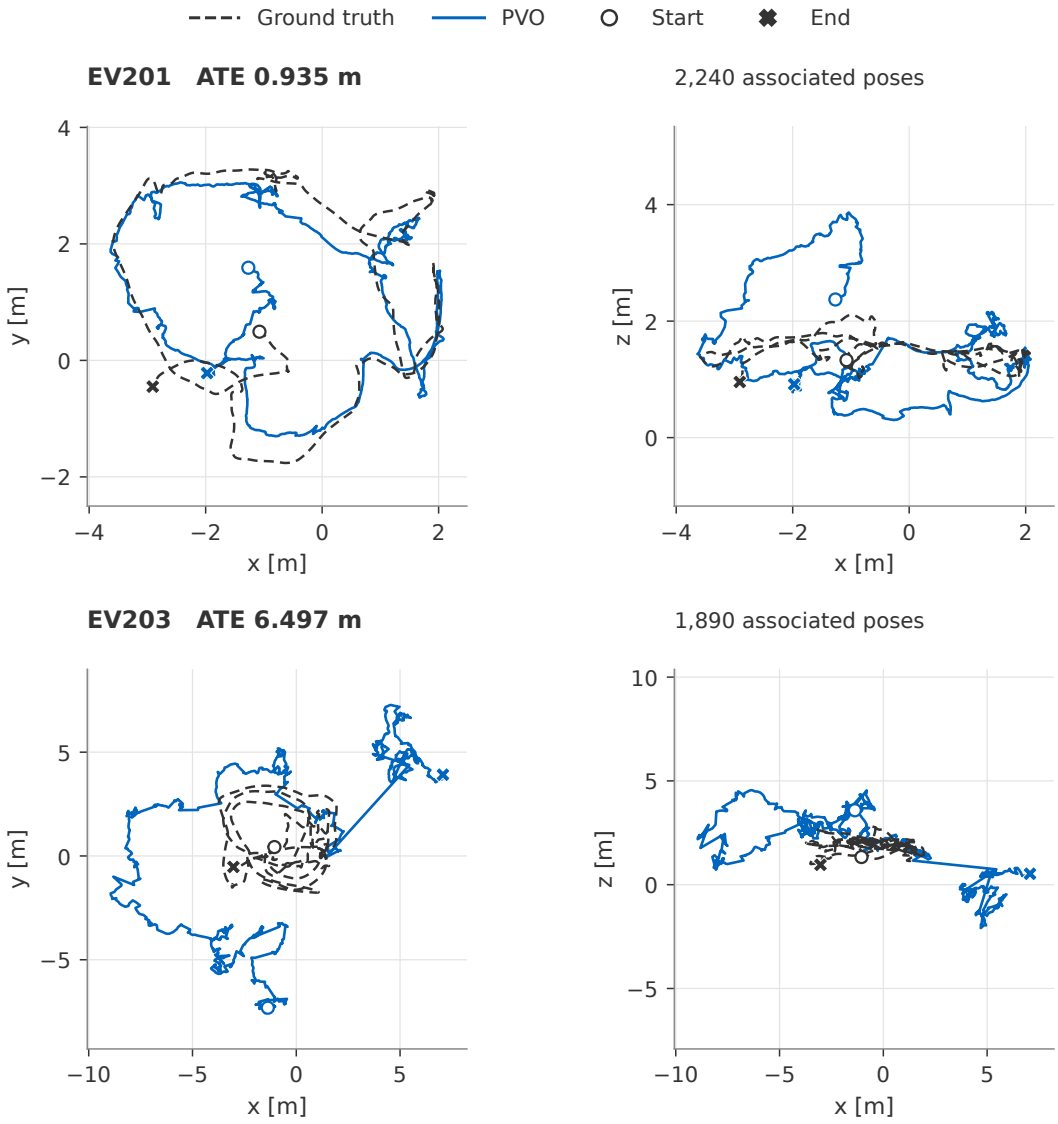


Figure B.7.: Full-sequence PVO trajectories for EV201 and EV203. The full extent of EV203's estimated motion is retained despite its much smaller reference path.

B.3. Stereo Transform Overview

The full set of twelve transformed encodings is illustrated on a later EMH02 stereo pair in Figure B.8. The rows vary disparity pre-shift and the columns vary grid phase. This overview complements the sampling construction and the location-specific recovery examples in Chapter 4.

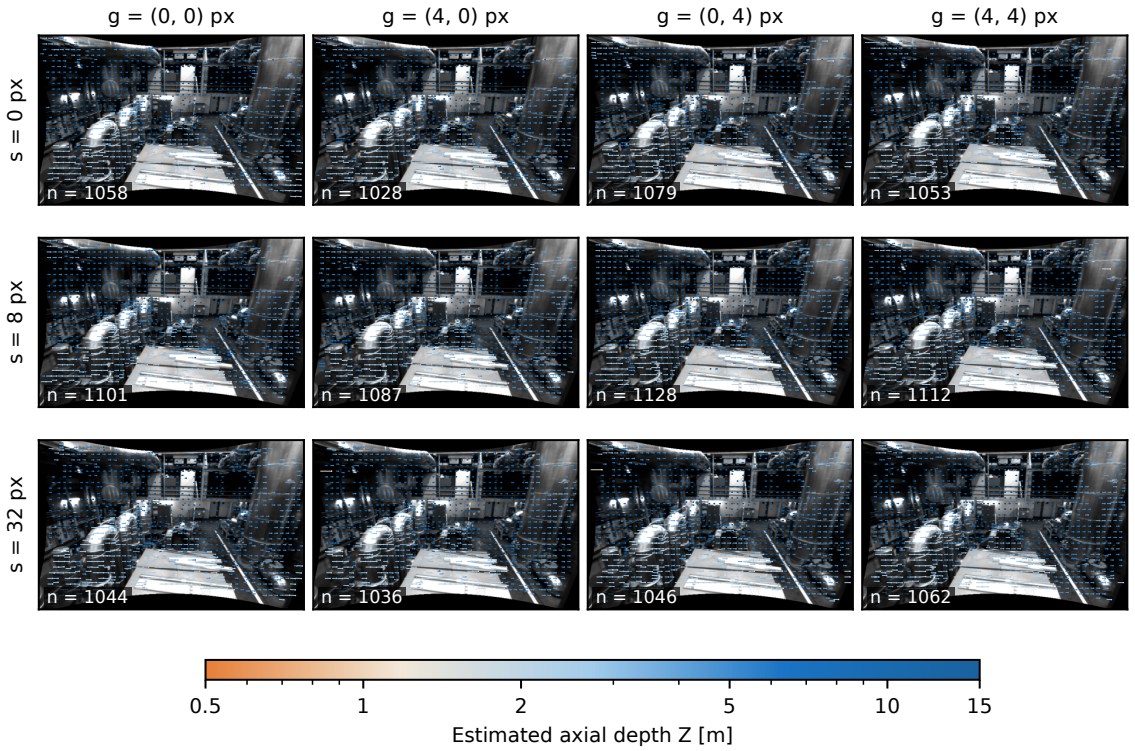


Figure B.8.: All twelve stereo transforms for EMH02 at 135 s (pair 2700, zero based). Rows vary pre-shift s and columns vary common grid phase g. Dots mark left-image locations; segments end at their right-image coordinates in the common rectified plane. Every positive-disparity observation surviving the diagnostic stereo checks is shown, with count n. Colour encodes estimated axial depth, clipped to the displayed 0.5 m to 15 m scale.

B.4. Lessons from Preliminary Experiments

Several approaches preceded the stereo-depth alignment method developed in this thesis. They explored filtering codec motion, using it to propagate existing image features, and changing the lifetime or placement of tracked samples. The examples below come from archived development experiments on Monado sequences [28]. Some used image features, inertial rotation, or ground-truth geometry to investigate particular limitations; they are separate from the final pixel-free EuRoC evaluation. Their common lesson is that useful local motion and persistent point identity are distinct requirements.

B.4.1. Frequency-Based Selection of Motion Vectors

An early idea was to distinguish coherent camera-induced motion from irregular codec predictions through their frequency content. Motion-vector fields were sampled on an eight-pixel grid and decomposed into spatial or spatiotemporal frequency bands. A complementary temporal diagnostic compared successive fields after alignment using the coded displacements. For two aligned motion samples $\mathbf{m}_a$ and $\mathbf{m}_b$, the normalized two-point Fourier transform has coefficients

$$\frac{\mathbf{m}_a + \mathbf{m}_b}{2} \quad \text{and} \quad \frac{\mathbf{m}_a - \mathbf{m}_b}{2}. \tag{B.1}$$

The second coefficient measures disagreement between the samples. A small value suggests temporal consistency, but can also occur when both samples share an incorrect displacement. A three-dimensional FFT over the two-frame volume additionally measures spatial variation.

The diagnostic assigned each vector to the frequency band with the largest reconstructed magnitude, provided that band's share of the summed magnitudes exceeded a chosen dominance threshold. Spatial and spatiotemporal bands used normalized radial cuts of 0.12 and 0.35.

Figure B.9 shows the resulting trade-off on MOO06. At a threshold of 0.45, the spatiotemporal low-frequency group retained 11.15% of the input vectors; 65.85% of those satisfied the one-pixel ground-truth epipolar check. Raising the threshold to 0.75 increased that fraction to 76.65%, but retained only 0.24% of the vectors. Frequency selection could therefore enrich a geometrically consistent subset at a substantial cost in coverage. Even a small epipolar residual does not establish the correct endpoint along the epipolar line or the identity of a physical scene point.

A later variant ranked blocks by the sum of their reconstructed band magnitudes and selected up to ten translation-support correspondences from low-ranked blocks containing active features, for an inertial-rotation-assisted epipolar filter. This score was called "energy" during development, although it was not squared Fourier energy. Without image refinement of these supports, global ranking produced 0.538 m ATE RMSE on the full 4,171-frame MOO06 run, compared with 0.177 m for image-refined translation support. Selecting supports within spatial windows improved the ranked variant to 0.299 m. These run-level results evaluate support sets, not the pose consistency of the single lowest-scoring vector; they therefore do not settle the latter question.

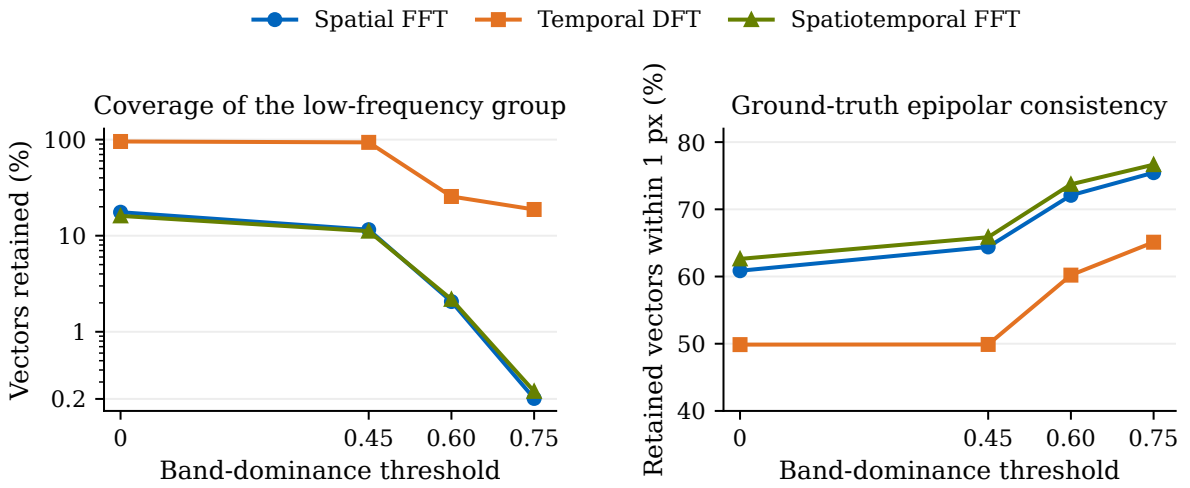


Figure B.9.: Low-frequency selection on MOO06. Coverage uses a logarithmic axis. Consistency measures the fraction of retained vectors with ground-truth square-root Sampson residual at most one pixel, evaluated in undistorted coordinates and scaled by $\sqrt{f_x f_y}$. Spatial grouping covers 7,108,462 vectors; the temporally aligned methods cover 7,017,517 because they require a preceding field.

B.4.2. Propagating Existing Features

Another approach retained the identities of existing image features and propagated their coordinates with codec displacements. At each transition, it selected a motion vector whose source support contained the current feature location, applied that vector, and stored the endpoint under the same track ID. The attraction was to reuse an established feature-based backend while replacing repeated temporal image matching. However, carrying an ID forward does not verify that its new coordinate still observes the same physical point.

Figure B.10 illustrates the difference between a promising short run and sustained tracking on MIO07. The archived direct-MV prototype reached approximately 0.096 m ATE over its first 800 frames, but approximately 30.27 m over the full 4,095-frame sequence. The image-based reference finished at approximately 0.120 m. The prototype still assigned vectors to 92.8% of attempted feature propagations. That count describes available assignments, not their geometric correctness or preservation of point identity.

This failure does not isolate identity drift as its sole cause: geometric support, stereo observations, and backend conditioning also matter. It shows why short-prefix accuracy and high assignment counts were insufficient validation. The feature-seeded diagnostic in Figure 4.7 shows a local mechanism: codec propagation can depart from an image track despite a common starting point and continuing vector support.

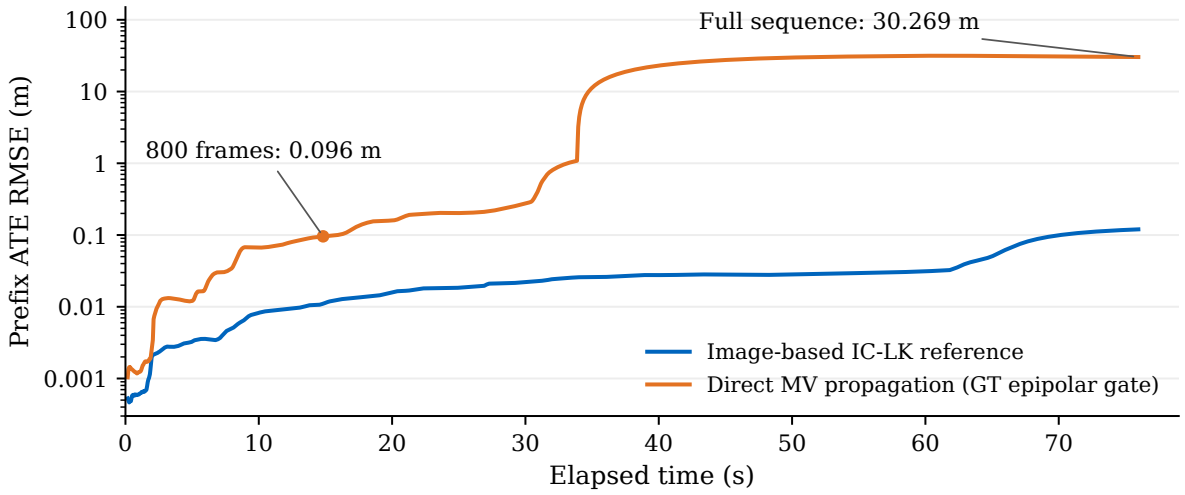


Figure B.10.: Archived MIO07 development run: direct temporal MV propagation and an image-based inverse-compositional Lucas–Kanade reference. Each plotted ATE is evaluated over the prefix ending at that time, with a new rigid alignment and fixed scale for that prefix. The vertical axis is logarithmic. The direct-MV prototype used a ground-truth epipolar gate with a loose 32-pixel threshold, retaining 99.6% of input vectors; it was not an oracle for correct point identity. Image-based feature detection and stereo matching were retained; codec vectors replaced temporal tracking only. The curves are redrawn from the archived per-prefix measurements and do not represent the final PVO configuration.

Further tests examined feature placement and temporal connectivity separately. In a stereo-inertial test with spatial refill opportunities controlled, using arbitrary codec block centres as starting locations reduced successful temporal KLT counts to 35.8% of the FAST-location reference on MIO07 and 37.1% on MOO07. This experiment retained image-based tracking and did not use codec displacements; it showed that block centres were not interchangeable with selected image features. In a separate visual-only stereo test on 800 MIO07 frames using KLT correspondences, restricting landmarks to adjacent frame pairs produced 0.521 m ATE. Making every frame a keyframe improved this to 0.035 m, while persistent tracks achieved 0.011 m. Shorter identities therefore also required changes to landmark admission and map connectivity; they were not a complete solution on their own.

B.4.3. Persistent Surfaces with Changing Point Support

These observations suggest a direction beyond improving a point-selection rule. The useful codec support on an extended surface can change from frame to frame. For example, observations on a tabletop might shift between its edge, a corner, and an interior textured region. Such observations may remain compatible with the motion of the same static surface even when they cannot be joined into one long-lived point track. This is a motivation for testing surface-level association, not evidence that the present experiments already recover object identities.

Li and Song [20] provide a relevant precedent under a planar scene model. They group codec correspondences using homography RANSAC, associate planes across frames, and jointly estimate plane geometry and camera motion in an extended Kalman filter. Their state can retain a plane while its supporting point correspondences change. The observation model still uses those correspondences through a plane-induced homography; the plane label alone does not supply camera motion.

A future codec-based estimator could similarly maintain a static plane or other geometric region while renewing its short-lived correspondence support. It would need reliable region association, enough spatially distributed observations, and geometry that constrains the camera motion. Merely seeing another part of the same object would not satisfy these requirements. Non-planar objects would require an appropriate shape or rigid-body model, and independently moving objects would need separate treatment. Preserving the pixel-free boundary would also require forming and associating these regions from codec metadata and geometry. The stereo-depth construction in this thesis supplies one possible source of that geometry; whether a surface representation makes better use of intermittent codec support remains an open experimental question.

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